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Improving Clinical Information Extraction with Public-Data Pretraining

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Abstract

Contents

Part I

Introduction

1 Artificial Intelligence and Healthcare

> *"The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin.* [...] *We should build in only the meta-methods that can find and capture this arbitrary complexity.* [...] *The eventual success is tinged with bitterness, and often incompletely digested."*
>
> Richard S. Sutton, *The Bitter Lesson*

Healthcare has been one of the earliest and most prominent application domains for artificial intelligence. Use cases range from diagnostic support to medical record management, cohort selection, report generation, and clinical information extraction.

As early as the 1970s, MYCIN, the iconic expert system developed at Stanford, diagnosed bacterial infections and recommended antibiotics. It already achieved performance comparable to human specialists in its narrow domain. INTERNIST-1, later known as CADUCEUS, tackled internal medicine diagnosis across hundreds of diseases. However, these systems relied on explicit rules and static knowledge bases, which limited their ability to handle the complexity and variability of real-world medical data. They were rigid and required constant maintenance to keep up with medical advances. Building such systems demanded expensive medical experts to annotate clinical data and write rules by hand.

With the rise of machine learning in the 1990s and 2000s, data-driven approaches began to dominate healthcare AI. Supervised and unsupervised learning algorithms trained models directly from data. The interpretability of rule-based systems was traded for the performance of deep neural networks capable of capturing complex patterns. R2 ImageChecker (1998) became the first FDA-approved computer-aided detection system for mammography, using neural networks to spot suspicious micro-calcifications. PAPNET applied neural networks to cytological screening for cervical cancer.

Then came the scaling era. The recipe became simple: pretrain a language model on billions of words of raw text, then fine-tune on the target task. This approach routinely outperformed systems built with years of domain expertise. The need for expensive annotation, hand-crafted rules, and curated knowledge bases faded, beaten by data and compute.

Despite healthcare offering many structured and curated knowledge bases such as UMLS, SNOMED-CT, and RxNorm, experience has shown that general-purpose language models pretrained on massive unstructured data outperform approaches based on domain-specific rules and knowledge bases. It seems our models cannot effectively integrate these structured resources, or perhaps these knowledge bases do not cover the full complexity and variability of medical knowledge as it is used in practice.

However, healthcare faces a limitation that other domains do not: the confidentiality and scarcity of annotated clinical data. Medical data is sensitive and difficult to obtain in large quantities for model training. Works adapting language models to the medical domain often rely on public data such as PubMed or patient forums, but these do not always reflect the real clinical language used in electronic health records. The clinical domain has unique jargon, abbreviations, and writing styles that differ from general medical texts. Moreover, while MIMIC exists in the United States, public clinical datasets remain scarce in other countries such as France.

This leads to a fundamental problem: the paradigm that has won is pretraining at scale, but in healthcare we lack massive public clinical data to pretrain language models. Can we find new ways to exploit public data for pretraining language models adapted to the clinical domain, without using sensitive clinical data?

2 Modern AI and New Opportunities

"Now go out and gather some data, and see what it can do."

Alon Halevy, Peter Norvig, and Fernando Pereira, *The Unreasonable Effectiveness of Data*

A concrete opportunity: the OncoLab project.

- TODO: describe OncoLab — goal, clinical setting, the information-extraction task (oncology reports / structuring).
- TODO: what today's models make newly feasible here, and the practical stakes (cohort selection, report structuring, decision support).

The challenge: opportunity without data.

- TODO: the wall — clinical text is personal data, not shareable; no large real French clinical corpus to train or even evaluate on.
- TODO: so the most direct lever, fine-tuning on in-domain clinical data, is unavailable; we must build from public text instead.
- TODO: this is the bridge to the next chapter — to use public text for the clinic we first need to know what biomedical text is.

3 What is Biomedical Text?

"Language was just difference. A thousand different ways of seeing, of moving through the world. No; a thousand worlds within one. And translation—a necessary endeavour, however futile, to move between them."

R. F. Kuang, *Babel, or the Necessity of Violence*

3.1 Clinical Text

FRANÇAIS

CENTRE HOSPITALIER DE SAINT-AUBIN
Dossier n^{o} 4471902
Cardiologie, Soins Intensifs
Séjour : 03/11–08/11/2024

Patient : CARON Julien, né le 14/03/1962 (62 ans). **Dest. :** D^{r} S. Lemaire.

Motif. Douleur thoracique constrictive rétrosternale, irradiation bras G, depuis 2h.

Antécédents. HTA. DT2 sous metformine. Tabac 30 PA. Dyslipidémie.

Examen. TA 148/92, FC 98, SpO_2 96% AA, EVA 7/10. Pas de signe d'IVG.

Paraclinique. ECG : sus-décalage ST V1–V4. Tropo T 4520 ng/L (N <14). Coro : occlusion IVA proximale, angioplastie + stent actif, TIMI 3. ETT : FEVG 40%.

Au total. SCA ST+ antérieur (I21.0).

Ttt sortie. Kardégic 75, ticagrélor 90x2, atorvastatine 80, bisoprolol 2,5, ramipril 5.

CAT. Réadaptation cardiaque. Consultation cardio à 1 mois. D^{r} C. Després

ENGLISH · TRANSLATION

SAINT-AUBIN HOSPITAL
Record no. 4471902
Cardiology, Intensive Care
Stay: 03/11–08/11/2024

Patient: CARON Julien, b. 14/03/1962 (62 y). **To:** Dr S. Lemaire.

Reason. Constrictive retrosternal chest pain, radiating to L arm, for 2h.

History. HTN. T2DM on metformin. Smoking 30 PY. Dyslipidaemia.

Exam. BP 148/92, HR 98, SpO_2 96% RA, pain 7/10. No sign of LVF.

Workup. ECG: ST elevation V1–V4. Trop T 4520 ng/L (N <14). Angiography: proximal LAD occlusion, angioplasty + drug-eluting stent, TIMI 3. Echo: LVEF 40%.

Summary. Anterior STEMI (I21.0).

Discharge tx. Aspirin 75, ticagrelor 90x2, atorvastatin 80, bisoprolol 2.5, ramipril 5.

Plan. Cardiac rehabilitation. Cardiology review at 1 month. Dr C. Després

Figure 3.1: A discharge summary for an acute myocardial infarction; French original, English translation.

Figure 3.1 is a letter from one cardiologist to a colleague. For an ordinary French reader, it is a text that is very hard to read and to understand. It has very few verbs, few adverbs, and sometimes neither. The sentences are telegraphic, the information very dense and simply separated by commas like a table. One might think this is just carelessness, that doctors have very little time and rush their letters without paying them much attention. There is some truth to that, but in reality, it is a style of communication specially tuned for the community it is written for. The writer and the reader share enough knowledge that the writer can give just the precise information he knows the other needs and go straight to what matters clinically.

This phenomenon has a precise name. Studying how specialised communities write, the linguist Zellig Harris observed in the 1960s that a technical field has a language of its own, governed not by a narrower vocabulary but by its own grammar, its own restricted rules for which words may combine (Harris, 1968). He called it a *sublanguage*: the text a community produces about a specialised domain, following its own rules for how words combine. Some of these rules are stricter than usual, and some ignore the usual rules entirely. Harris and his colleagues later made this concrete by reconstructing the grammar of a small body of immunology articles. They showed that its recurring sentence structures encoded the domain's entities and the relations between them, revealing how its knowledge developed over time (Harris et al., 1989).

The clinical note is the best-documented case of this. One of the earliest attempts to make computers read medical records, Naomi Sager's Linguistic String Project at New York University, set out to turn free-text clinical reports into structured data, and in the process found that the notes already followed a specific grammar (Sager, 1981; Sager et al., 1987). To read them, the program needed rules for the short, verbless phrases that notes use but everyday language does not, together with categories taken from medicine itself, such as patient, finding, and treatment, all visible in Figure 3.1. A line like *afebrile, chest clear* is not broken language; it is written this way for a reason. The style takes hold within a community that shares the same specialised knowledge and keeps up its patterns of usage (Kittredge, 1982). A grammar like this can live inside a hospital, kept up and extended by the people who use it.

But let us come back to the idea of *carelessness*. If this style was the result of a lack of care, its missing words would be scattered at random through the document, with no logic behind them. Instead, the omissions follow a pattern. Anderson et al. (1975) showed that clinical shorthand obeys a small set of deletion rules: the writer drops a subject here, a verb there, a tense elsewhere, but only when the reader can put it back. *Chest films normal* stands for *the chest films were normal*; *no bone pain* stands for *the patient has no bone pain*. What disappears is the part the reader already knows. A note is a form of *compression*.

This works because the reader is an expert of the same domain. Writing for a colleague who went through the same training, the clinician can leave out whatever the other already knows, and say only what is needed (Grice, 1975; Sperber and Wilson, 1995). This means that same words can behave differently depending on who reads them. *Afebrile, vitals stable, continue antibiotics* is complete for a physician and almost empty for anyone else.

That shared background is what linguists call *common ground*. The note records only what is new, leaving out what the writer and reader already know: medicine, the patient's history, and how the hospital works (Stalnaker, 2002). A clinical note is also performative. Written by a doctor, it can also declare or set off an action (Austin, 1962; Searle, 1969). A prescription lets the patient obtain the medication, and an order to transfer a patient sets the transfer in motion. The note is a document with real authority, signed and legally binding.

The text describes the illness by moving through the parts of the heart it affects. In Figure 3.1, the ECG first locates the problem in the anterior wall, the angiogram then finds a blockage in the LAD (IVA) artery, and the ultrasound shows damage to the left ventricle. Each test adds another location to the same picture of the disease. The patient comes across as a list of lesions, each at its site. This way of seeing has a history. When the modern hospital took shape, doctors learned to see illness as something placed in a specific tissue or organ, and to read the body as a map of local damage. Foucault called this trained way of looking the *clinical gaze*. He argued that it changed how doctors spoke and what could be said about disease at all (Foucault, 1963). The finding-to-site structure of the note is the daily trace of that gaze.

All of this makes the note worth studying, but it also makes it hard to obtain. The note rarely leaves the hospital that produced it, because it is full of personal data. A single report can hold the patient's name, age, and dates, a home address, a family situation, and a diagnosis rare enough to point to one person. Removing the obvious identifiers does not fix this. A pseudonymised note is still not anonymous. A malicious actor may combine the details it contains with other information to identify the patient. Confidentiality is inherent to this document type.

For French the problem is worse. There is no shareable corpus of real French hospital records, nothing like what exists for English (Névéol et al., 2018). Public data is an alternative: published case reports, drug leaflets, article abstracts. But a case report is written to teach. It reads like a short story, in full sentences and the past tense.

FRANÇAIS

Mme A. F..., 35 ans, vient consulter pour hématurie macroscopique intermittente avec lombalgies gauches, pollakiurie, brûlures mictionnelles évoluant depuis 10 ans. L'interrogatoire retrouve également un antécédent de tuberculose ostéo-articulaire. L'examen clinique est normal, le bilan biologique (numération sanguine, ionogramme) est normal. L'examen bactériologique des urines est stérile.

L'urographie intra-veineuse (UIV) montre une image lacunaire en battant de cloche au niveau de l'uretère gauche, s'étendant du tiers moyen de l'uretère lombaire à l'uretère iliaque, et dont l'extrémité inférieure est convexe vers le bas (Figure 1).

ENGLISH · TRANSLATION

Mrs A. F..., aged 35, presents with intermittent macroscopic haematuria, left-sided lower-back pain, frequent urination, and painful urination lasting for 10 years. The history also reveals previous osteoarticular tuberculosis. The clinical examination is normal, and the laboratory tests (blood count and electrolytes) are normal. The urine culture is sterile.

Intravenous urography (IVU) shows a bell-clapper-shaped filling defect in the left ureter, extending from the middle third of the lumbar ureter to the iliac ureter, with its lower end convex downwards (Figure 1).

Figure 3.2: A published clinical case from the French CAS corpus; French original, English translation.

In Figure 3.2, the patient is introduced, the symptoms are told in order, and the examinations are explained in complete sentences. The gap is easy to see when compared with the clipped lines in Figure 3.1.

Whether these stand-ins really match sufficiently hospital notes is still being argued. The team behind the case-report corpus is hopeful. They say the clinical part of a published case can be close enough to a real note that tools built on it still work on hospital text (Grabar et al., 2020). They also name the limits. Real notes are full of misspellings, carry administrative boilerplate, and lay out examinations and lab values in tables, while a published case rewrites all of this as clean sentences. A more recent project sees it differently. In building PARHAF, a corpus of 7,394 fictitious reports written by hand by 104 medical residents, its authors say plainly that "the style of case reports [...] is quite different from that of electronic health records" (Tannier et al., 2026). They wrote the reports from scratch so that they could be shared safely.

Taken together, these alternatives and the traits described above explain why our target text is so hard to obtain. Clinical notes are private by nature: they are written for a closed group of experts, contain personal data, and carry legal weight.

However, if we are willing to step slightly beyond the clinical domain, we find that medicine actually produces an enormous amount of text. This vast, freely available literature documents the state of medical knowledge. But how far is its language from the clinical text we ultimately want to model?

3.2 Scientific Text

FRANÇAIS

L'insuffisance mitrale (IM) ischémique est une valvulopathie fonctionnelle, restrictive, liée au remodelage ventriculaire gauche post-infarctus et qui complique environ 20% des cardiopathies ischémiques. Sa présence grève lourdement le pronostic de la maladie coronarienne, tant en termes de morbidité que de mortalité. Encore mal connue, elle suscite depuis quelques années de nombreux travaux visant à préciser ses mécanismes physiopathologiques, ses facteurs pronostiques et sa prise en charge thérapeutique.

ENGLISH · TRANSLATION

Ischaemic mitral regurgitation is a functional, restrictive valve disease, linked to post-infarction left-ventricular remodelling, which complicates about 20% of ischaemic heart diseases. Its presence weighs heavily on the prognosis of coronary disease, in terms of both morbidity and mortality. Still poorly understood, it has prompted in recent years a large body of work seeking to clarify its pathophysiological mechanisms, its prognostic factors and its therapeutic management.

Figure 3.3: A research article about myocardial infarction; French original, English translation.

Figure 3.3 is taken from a research article about a later complication of the heart attack described in the previous section. But the language has changed completely. The sentences are complete, the argument unfolds step by step, and the authors make clear where knowledge ends and uncertainty begins. Unlike the clinical note, this text belongs to a vast and freely accessible literature. It may seem like an expanded, more carefully written version of clinical language. It is not. The research article speaks a language of its own.

This difference is not simply a matter of writing style. Although the two texts concern the same disease, they are written to do different things. The rhetorician Carolyn Miller asked why texts settle into the forms they do. She answered that a genre is a recurring action in a community: to know a genre is to know what it is for and what it does (Miller, 1984). By that test the two are different genres because they answer different needs. The clinical note exists to record a patient and guide their care. The research article exists to share new findings with the scientific community so that others can examine and challenge them. This difference in purpose shapes how each one is written.

This purpose shapes how the article is written. John Swales found that research articles often begin by making three moves: they introduce an area of research, identify something that remains unknown, and explain how the present study addresses it (Swales, 1990). The passage in Figure 3.3 follows this pattern. It first explains that ischaemic mitral regurgitation is common and affects prognosis. It then identifies what remains *still poorly understood*, before presenting the research intended to explain it. In other words, the opening tells readers why the article needs to exist.

The body of the article follows the same logic. Its four parts, introduction, methods, results, and discussion, retrace the steps of a controlled study: a question, the steps used to answer it, what was observed, and what it means. The arrangement feels timeless, but it is recent. One survey looked at 1,297 original articles from four leading medical journals between 1935 and 1985. Not one used this structure in 1935. By the 1970s more than four in five did, and by the 1980s it was the only accepted form for an original paper (Sollaci and Pereira, 2004). A format only a few decades old now carries the shape of scientific reasoning in its headings alone.

The clinical note also gives form to a way of reasoning, but a different one. The problem-oriented record was designed to organise care around a patient's problems. Many clinical notes use a structure known as SOAP: what the patient reports, what the clinician observes, their assessment of the problem, and the plan for care. This structure follows the loop of diagnosis and management rather than experiment (Weed, 1968). The article asks whether a hypothesis holds. The note asks what is wrong with the patient and what to do next. Both are about the same medicine, but they reason in different ways. One is not simply a variation of the other.

These different purposes shape how the two texts are organised, but also how they make claims. Research articles constantly signal how certain each claim is. Ken Hyland's studies of academic writing call this running layer metadiscourse: the writer's own handling of the claim, including the words that tune how strongly it is made (Hyland, 2005; Hyland and Tse, 2004). Hedges such as *may* or *suggests* hold a statement back. Boosters such as *clearly* or *shows* push it forward. *These results suggest that X may contribute to Y* is a claim offered for debate. The plain *X causes Y* would be far bolder. The note hedges too, but rarely, and for another reason: its *rule out myocardial infarction* tells the clinician what to do.

These differences in language can be measured. Biber (1988) compared the features that commonly appear together across spoken and written texts. He found scientific prose to be one of the forms of written English most different from everyday conversation: it contains many nouns, long words, and prepositions, but few pronouns and few private verbs such as *think* or *believe*. The style also changes within an article. Methods sections tend to be impersonal, while discussions contain more hedging and expressions of uncertainty (Biber and Finegan, 1994).

The pattern reaches the smallest scale. Nwogu (1997) identified eleven recurring rhetorical moves across the sections of medical research articles. The same regularity appears at the phrase level: discontinuous frameworks such as *the … of* and *be … to* recur with a limited, discipline-specific set of words filling the gap (Luzón Marco, 2000). The research article is therefore highly structured at every level, from its sections to its individual phrases.

The wording of a scientific claim also changes as the claim gains acceptance over time. Latour and Woolgar (1979) observed that qualifiers such as *may* and *suggests* often disappear as other scientists repeat a claim. Over time, *X may contribute to Y* can become *X causes Y*. These words therefore show which claims are still open to debate. Ignoring these words may make an uncertain finding seem like an established fact.

These differences lead back to the idea introduced in the previous section and to a simple conclusion: there is no single biomedical language. Public research articles form the largest and most up-to-date record of humanity's medical knowledge in natural language, but they do not teach us how clinicians record patient care. Simply adding more articles will not close this gap. Beyond them lies an even larger body of medical text, written for patients and the public. It is even more different from clinical language, but too abundant to ignore. This is where we turn next.

3.3 Lay and Web Text

FRANÇAIS	ENGLISH · TRANSLATION
Bonjour, mon père a fait une crise cardiaque la semaine dernière. Il a eu une grosse douleur dans la poitrine qui descendait dans le bras gauche, il était tout pâle et en sueur, on a appelé le 15 tout de suite. Est-ce que quelqu'un sait combien de temps dure la convalescence après un infarctus ? Merci d'avance.	*Hello, my father had a heart attack last week. He had a strong pain in the chest that went down into the left arm, he was all pale and sweating, we called emergency services right away. Does anyone know how long recovery takes after a heart attack? Thanks in advance.*

Figure 3.4: A patient forum about myocardial infarction; French original, English translation.

Figure 3.4 shows the same heart attack once more, but this time in the words of the people who live through it. There is no technical vocabulary anymore. A worried relative writes *crise cardiaque* (*heart attack*) rather than *infarctus du myocarde* (*myocardial infarction*). The symptoms are described in plain, everyday words. For at least two decades the internet has made this kind of text widely available. By 2013, 72% of US internet users had looked online for health information (Fox and Duggan, 2013), creating a vast and freely accessible body of patient-facing medical text. This text is far from the language of the clinical domain, but it offers something research articles do not: an abundant and more varied representation of medical knowledge. It is therefore worth examining how medical language, and especially medical terms, change when they reach the public.

The idea that a fact changes as it travels is an old one. In 1935 the physician Ludwik Fleck argued that knowledge is first made inside a small circle of specialists, and is then changed as it moves out to a wider audience (Fleck, 1979). As knowledge reaches the public, cautious claims become more direct and technical terms are replaced by everyday words. Writing for patients therefore presents the same knowledge in a different form, adapted to a different audience.

Not all of this lay writing is unofficial. The most systematic patient-facing text of all is the drug leaflet, the *notice* folded inside every box of medicine. In the European Union, by law, every approved drug must carry one. It must be in the official language of each market where it is sold, and written "in clear and understandable terms for the users" (European Parliament and Council of the European Union, 2001). A shared template fixes the same six headings everywhere, from *what the medicine is and what it is used for* to *possible side effects*. A draft must also be tested on real patients for readability before it can be approved (European Medicines Agency, 2025). This makes the move from specialist to public explicit. The leaflet also shows how names vary. The rules require the brand name to sit beside the common one, *Doliprane* next to *paracétamol*. So a single substance carries several of its names side by side on one page.

EMEA · PATIENT-FACING MEDICINE SUMMARY

Ultratard est une insuline de substitution identique à l'insuline fabriquée par le pancréas. Le principe actif d'Ultratard, l'insuline humaine (ADNr), est produit grâce à une méthode dite de "technologie recombinante".

Figure 3.5: One medicine named as a brand, a treatment, and an active substance in the EMEA split of the MC-Bio corpus.

What changes most, for our purpose, is the naming. When terminology first emerged as a field, the Austrian engineer Eugen Wüster argued that each concept should have one precise term, preventing ambiguity in technical communication (Wüster, 1979). In practice, however, the same concept often has several names. María Teresa Cabré showed that the term people choose depends on the context and audience. Rather than being an exception, this variation is a normal part of specialised language (Cabré, 2003). François Gaudin's socioterminology studies this variation as a social phenomenon (Gaudin, 2003). The heart attack of the clinical note, *infarctus du myocarde*, coded *I21*, is the same event the patient calls a *crise cardiaque*. One concept, many names.

The variation goes beyond words. Medical concepts themselves can be understood in different ways. The heart, for example, can be described as a pump, leading to terms about pressure, output, and failure. It can also be described as an electrical system, leading to terms about rhythm, conduction, and arrhythmia. Temmerman (2000) showed that such ways of understanding a concept help shape the terminology used to describe it. Terminology must therefore account not only for different names, but also for the different views of a concept that those names express.

Medical terminology systems have evolved to represent this variation. The Unified Medical Language System brings together more than sixty medical vocabularies and links their terms to shared concepts (Bodenreider, 2004). These links form a many-to-many relationship: one concept can have several names, and one name can refer to several concepts. In 2004, the UMLS contained about 2.5 million names for 900,000 concepts, or almost three names per concept. It links these different names to shared concepts rather than selecting a single correct term.

Patient-facing language is itself diverse. A forum post is personal and spontaneous, while a drug leaflet is officially written, carefully structured, and directly addresses the reader. The CLEAR corpus makes this adaptation explicit by pairing specialised medical texts with versions written in plain language (Grabar and Cardon, 2018).

EXPERT VERSION · FRANÇAIS	PATIENT VERSION · FRANÇAIS
Chez les patients épileptiques ou à risque de convulsions, il est recommandé d'évaluer le rapport bénéfice-risque du thiocolchicoside et de renforcer la surveillance clinique. La survenue de crises convulsives impose l'arrêt du traitement.	*Utiliser avec précaution en cas d'antécédents d'épilepsie ou de convulsions. La survenue de crise convulsive impose l'arrêt du traitement.*
ENGLISH · TRANSLATION	ENGLISH · TRANSLATION
For patients with epilepsy or at risk of seizures, the benefit–risk balance of thiocolchicoside should be assessed and clinical monitoring should be increased. If seizures occur, treatment must be stopped.	*Use with caution if you have a history of epilepsy or seizures. If a seizure occurs, stop treatment.*

Figure 3.6: The same warning in expert and patient versions from CLEAR; French originals and English translations.

The expert version discusses risk assessment and clinical monitoring, while the patient version gives a direct instruction. Public medical text therefore does not form a single register, but several ways of adapting medical knowledge to different readers.

Biomedical text is not a single language. The same medical knowledge changes with who writes it, who reads it, and what the text is meant to do. Clinical notes, research articles, and patient-facing texts therefore offer different views of medicine, and only the latter two are widely available. The challenge for the rest of this thesis is to use this public text without losing sight of the clinical language we ultimately need.

Part II

Related Works

4 Language Models

"These semantic aspects of communication are irrelevant to the engineering problem."

Claude E. Shannon, *A Mathematical Theory of Communication*

4.1 Introduction

Long before computers, people knew you could guess the next symbol in a text from the ones before it. In 1913 the mathematician Andrei Markov tested this by hand. He took twenty thousand letters from Pushkin's *Eugene Onegin* and marked each one as a vowel or a consonant. Then he counted. A vowel was followed by another vowel 13% of the time and by a consonant 87% of the time (Markov, 2006). Letters do not stand alone. What comes before shapes what comes next, and you can measure it.

Claude Shannon turned this counting into a theory. He treated a text as a random source and defined its *entropy*, the average information each symbol carries, as

$$H = -\sum_i p_i \log_2 p_i,$$

measured in bits (Shannon, 1948). This number is highest when every symbol is equally likely. It gets smaller as some symbols become easier to guess.

Shannon showed this with a simple demonstration. Generate letters independently and the result is noise, *xfoml rxkhrjffjuj zlpwcfwkcyj*. Keep three-letter statistics and the output is already pronounceable and nearly English, *in no ist lat whey cratict froure birs*. Join whole words in pairs and the fragments begin to read like real sentences, *the head and in frontal attack on an english writer that the character of this point*. Each step uses a little more context, and each step looks more like the language.

How predictable is English, then? Shannon answered with an experiment. A person guessed the next letter of a text one symbol at a time (Shannon, 1951). Shown the text so far, the reader guessed correctly about two times in three, usually on the first try. From how the guesses fell, Shannon worked out bounds on the entropy. An alphabet of 27 symbols, the 26 letters and a space, could in principle carry $\log_2 27 \approx 4.76$ bits per character. But given enough preceding text, Shannon estimated the true entropy of English at about one bit per character, somewhere between roughly 0.6 and 1.3 (Figure 4.1). Nearly four of every five possible bits

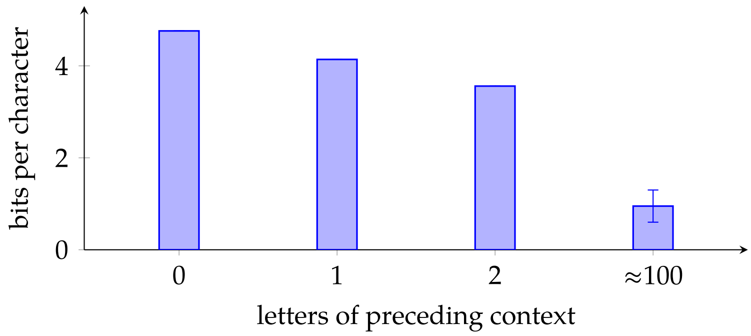

Figure 4.1: The entropy of printed English falls as more preceding context is used, from $\log_2 27 \approx 4.76$ bits per character with no context to about one bit (between 0.6 and 1.3) once roughly a hundred letters are taken into account (Shannon, 1951).

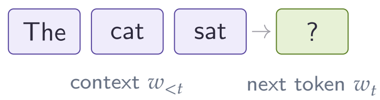

Figure 4.2: A language model predicts each token from the ones before it. The probability of a whole text is the product of these next-token predictions.

are redundant, already fixed by what came before. This redundancy is the whole opportunity. A model that predicts the next symbol well has learned how the language works.

This is what a language model is. By the chain rule, the probability of a whole text breaks down exactly into a product of next-symbol predictions,

$$P(w_1, \dots, w_T) = \prod_{t=1}^{T} P(w_t \mid w_{<t}),$$

so modeling language comes down to estimating a single conditional distribution: the probability of the next token given those before it (Figure 4.2). Every model in this chapter estimates that one quantity, and the history of the field is the history of estimating it better.

Estimating that conditional is harder than it sounds. The simplest idea is the one Markov and the early n-gram models used: count how often each short sequence is followed by each next word. But counting runs out almost at once. Most word sequences never appear twice, even in a large corpus, so the counts are zero exactly where we need them. Much of the history of language modeling is the search for estimators that get around this problem. The rest of this chapter follows that search, from n-gram counts to neural networks to the Transformer.

For this thesis the point is practical. Clinical information extraction rests entirely on these models. They are pretrained on large corpora and then adapted to a task. Adaptation is the hard part, because the target combines three scarcities at once, French, biomedical, and clinical. As the previous part showed, the text we can gather most easily is the least like the text we need. This chapter reviews the methods that make this adaptation possible: the architectures and pretraining objectives, how they scale, how they are specialised to a domain through continual pretraining, and the tokenization and long-context choices that matter for clinical documents. Continual pretraining, the mechanism at the heart of this thesis, comes last and in the most detail.

4.2 From Statistical to Neural Language Models

4.2.1 N-gram models

The first estimators keep only a short history. Markov assumed that the next token depends only on the last few. The full conditional $P(w_t \mid w_{<t})$ is then approximated by $P(w_t \mid w_{t-n+1}, \dots, w_{t-1})$, the probability of a token given the $n-1$ before it. These probabilities come straight from counts in a training corpus. The problem is the same as before, only sharper. The number of possible n-grams grows as $|\mathcal{V}|^n$, so for any real vocabulary most n-grams are never seen and their counts are zero. Smoothing methods help here. The best known is Kneser-Ney smoothing (Kneser and Ney, 1995). It moves some probability onto unseen sequences and falls back to shorter contexts when a longer one is missing. Smoothing made n-gram models good enough to be the state of the art for years. But these models cannot share what they learn between similar words. *Heart attack* and *cardiac arrest* are two separate entries in the table, with nothing in common.

4.2.2 Neural language models

The neural language model fixes exactly that. Bengio et al. (2003) replace the count table with a small neural network. Each word is first mapped to a vector, its *embedding*. The network reads the embeddings of the context and predicts the next word with a softmax over the vocabulary. It learns the embeddings at the same time as the prediction, so words used in similar ways end up with similar vectors and can pool their evidence. A sequence never seen in training is no longer a dead end, because its words resemble others the model has seen. This alone cut the perplexity of the best n-gram models by about a quarter. The embeddings turned out to be useful on their own. word2vec (Mikolov et al., 2013) showed that simple arithmetic on them captures regularities of meaning. The textbook example is that $king - man + woman$ lands near *queen* (Figure 4.3). These early vectors were still *static*: one vector per word, whatever the sentence. ELMo (Peters et al., 2018) made them contextual, giving a word a different vector in each sentence. It also showed that representations pretrained on plain text and then reused improve a wide range of tasks. This is an early form of the pretrain-then-adapt recipe the rest of this thesis depends on.

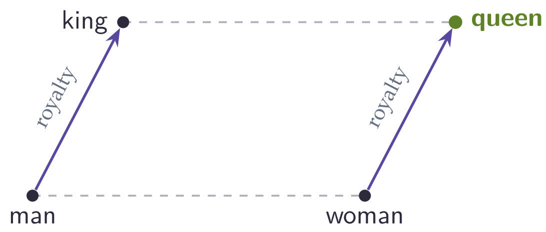

Figure 4.3: Word embeddings place the analogy king − man + woman ≈ queen as a parallelogram, with the "royalty" direction shared by both offset arrows (Mikolov et al., 2013).

4.2.3 Recurrent networks

All of the models so far read a fixed window of context. A *recurrent neural network* removes that limit. It reads the sequence one token at a time and carries a running summary forward. The idea goes back to Rumelhart and McClelland (1987) and was first made to work as a language model by Mikolov et al. (2010). At each step t the network takes the current token x_t and the previous hidden state h_{t-1}, and combines them into a new hidden state,

$$h_t = f(x_t, h_{t-1}),$$

where f is a small learned function and h_t is a fixed-size vector that summarises everything read up to position t. The next token is predicted from h_t with a softmax over the vocabulary, just as in the neural model above. The hidden state is passed from one step to the next (Figure 4.4), so a prediction can in principle depend on the whole history rather than the last $n - 1$ tokens.

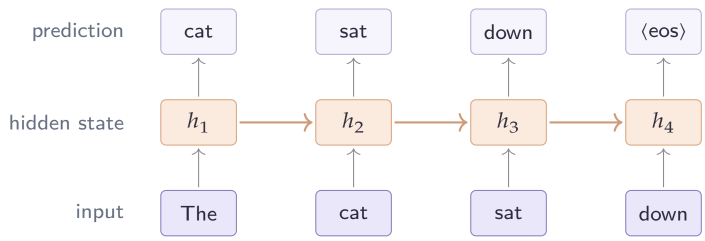

Figure 4.4: A recurrent language model, drawn one step per token. At each position the network combines the current input with the hidden state carried over from the previous step (the horizontal arrows) and predicts the next token, so the running state lets a prediction depend on the whole past rather than a fixed window.

In practice this memory is short. The network is trained by propagating the error back through every step. Pascanu et al. (2013) showed that this backward signal is scaled by a similar factor at each step, so it shrinks or grows geometrically as it travels

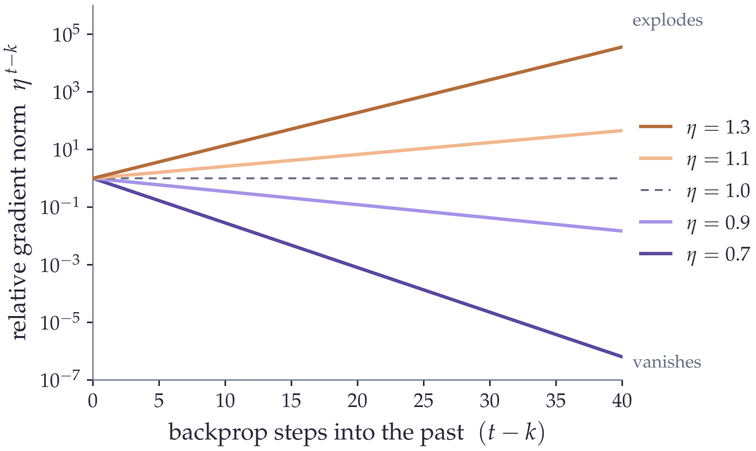

Figure 4.5: How the gradient of a recurrent network scales as it is propagated $t-k$ steps back in time, following the geometric bound η^{t-k} of Pascanu et al. (2013): below one it vanishes and above one it explodes, so links across long spans are hard to learn. The curves are this analytical bound, not measured gradients.

into the past (Figure 4.5). When the factor is below one the gradient vanishes and the model cannot learn links between distant tokens. When it is above one the gradient explodes and training becomes unstable. Bengio et al. (1994) had already identified this *vanishing* and *exploding gradient* problem as inherent to plain recurrence.

The *long short-term memory* network, or LSTM, was built to ease it (Hochreiter and Schmidhuber, 1997). Alongside the hidden state it keeps a separate memory cell that runs along the sequence. A set of *gates* controls it, deciding at each step what to write into the cell and what to read out. Information can travel along the cell almost unchanged, so useful facts survive over longer spans, and so does the gradient that carries them. A lighter variant, the gated recurrent unit, drops one of the gates and performs comparably (Chung et al., 2014). LSTMs became the standard language model of the early 2010s.

Two limits remained. Even with gates, very long dependencies are hard to capture. And the step-by-step processing cannot be parallelised, which makes training on large corpora slow. A related weakness showed up in translation. There an encoder had to compress a whole input sentence into its final hidden state before a decoder could read it. Bahdanau et al. (2015) relieved this bottleneck by letting the decoder look back at all of the encoder's states through an early form of attention. Long-range dependencies, parallelism, and this idea of direct access between positions are what the next architecture was built around.

4.3 Transformer

4.3.1 Self-attention

The Transformer (Vaswani et al., 2017) keeps the same goal, predicting the next token. It changes how the model reads its context. Its core operation is *self-attention*. It lets every token look directly at every other token. From the input representations X, the model computes three projections, a set of queries, keys, and values:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

Each token's query is compared with every key through a dot product. This measures how relevant the other tokens are to it. A softmax turns these scores into weights. The token's new representation is then the weighted average of the values:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V.$$

The division by $\sqrt{d_k}$, the size of the key vectors, keeps the dot products from growing too large. Large dot products would push the softmax to put almost all the weight on a single token.

This design has two effects. First, any two tokens now interact in a single step, however far apart they are. A recurrent network instead had to pass information through every position in between. Second, the scores for all positions are computed together as matrix products, so the whole sequence is processed at once. This is what made training on very large corpora practical. The cost is that every token is compared with every other one, so time and memory grow with the square of the sequence length, $O(L^2)$. In practice the operation runs in several *heads* side by side. Each head has its own projections, so different heads can follow different things, such as syntax, meaning, or position.

4.3.2 Positional encodings

Self-attention has one blind spot. It treats its input as a set, with no sense of order. *The cat sat* and *sat the cat* look the same to it, so position has to be supplied separately. The original Transformer added fixed sinusoidal signals to the embeddings (Vaswani et al., 2017). BERT instead learned a vector for each position (Devlin et al., 2019). Both encode an *absolute* position, and both cope badly with sequences longer than those seen in training. Later methods encode *relative* position. ALiBi adds a small penalty that grows with the distance between two tokens (Press et al., 2022). RoPE rotates the queries and keys by an angle that depends on their position, so that their dot product ends up depending only on how far apart they are (Su et al., 2024).

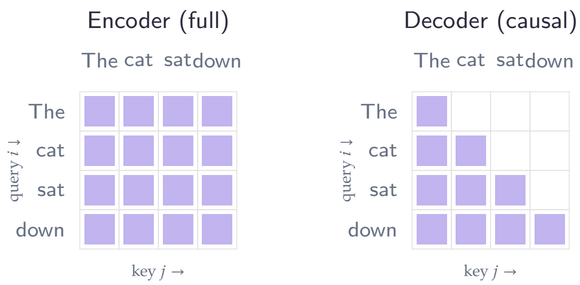

Figure 4.6: Attention masks as matrices. Row i is a query token and column j a key token; a filled cell means token i may attend to token j. The encoder allows every pair, so the matrix is full; the decoder allows only $j \leq i$, the lower triangle, so each token sees itself and the tokens before it but none of those that follow.

RoPE handles long and variable lengths well, and it is now the default in recent models such as LLaMA, Mistral, and ModernBERT.

4.3.3 Encoders and decoders

The same Transformer block can be wired in two ways. The only difference is which tokens each one is allowed to see (Figure 4.6). In an *encoder*, every token attends to the whole sequence, left and right. This two-sided view suits tasks that read a text and label it, such as classification or entity recognition. It underlies BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019). In a *decoder*, a token may attend only to the tokens before it. A causal mask imposes this restriction. It is what makes generation possible, one token at a time, as in GPT (Radford and Narasimhan, 2018) and LLaMA. A third design keeps both: an encoder to read an input and a decoder to write an output, as in T5 (Raffel et al., 2020). This design is natural for translation and other input-to-output tasks, but it is less common today. The two are built the same way. What really separates them is the training objective, the subject of the next section.

4.4 Pretraining Objectives

The architecture sets how a model reads its context. The pretraining objective sets what it must predict. That choice shapes the representations it learns. Two objectives dominate, and they differ in a way that matters for the rest of this thesis (Figure 4.7).

4.4.1 Causal language modeling

The most direct objective follows the chain rule: predict the next token. The model reads the sequence once. At each position t, it gives a probability distribution over the vocabulary for the token that should come next. Training lowers the negative

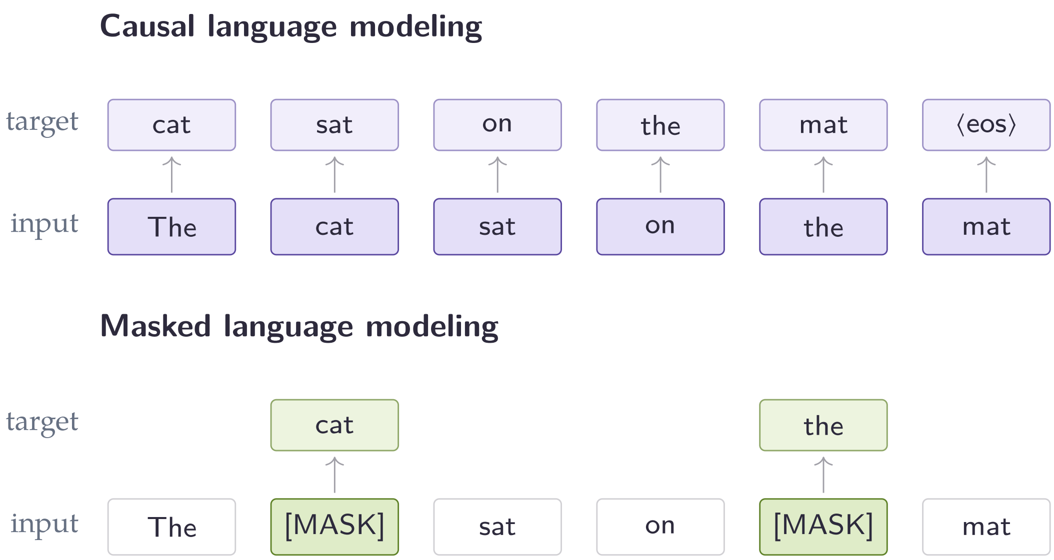

Figure 4.7: The same sentence under the two objectives. The bottom row is the input and the boxes above are the targets the model must predict, one arrow per predicted position. Causal language modeling predicts the next token at every position, so every token gives a training signal (dense). Masked language modeling hides about 15% of the tokens, here two, and predicts only those from the context on both sides, so most positions give no signal (sparse). This gap in how much signal each token carries is what the encoder pretraining of Chapter 11 exploits.

log-probability it assigns to the token that actually follows, summed over the whole sequence,

$$\mathcal{L}_{\text{CLM}} = -\sum_{t=1}^{L} \log P(w_t \mid w_{<t}),$$

where $w_{<t}$ is the tokens before position t. A causal mask lets this happen in a single pass. Each position may look only to its left, so the model can be scored at every position at once without seeing the token it must predict. Every position therefore gives a prediction and a gradient, so the supervision is dense. This is the objective behind GPT (Radford and Narasimhan, 2018) and the modern decoder models. It also fits generation well: at inference the model does exactly what it was trained to do, continue the text.

4.4.2 Masked language modeling

Understanding a text, rather than continuing it, calls for a different objective. BERT (Devlin et al., 2019) corrupts the input by replacing some tokens with a special `[MASK]` symbol. It then trains the model to put the originals back. At each masked position the model predicts the missing token from the context on both sides, and the loss counts only those positions,

$$\mathcal{L}_{\text{MLM}} = -\sum_{t \in \mathcal{M}} \log P(w_t \mid \mathbf{w}_{\setminus t}),$$

where $\mathcal{M}$ is the set of masked positions and $\mathbf{w}_{\setminus t}$ is the rest of the (corrupted) sequence. About 15% of the tokens are chosen. Of those, 80% become `[MASK]`, 10% are swapped for a random word, and 10% are left as they are. This mix stops the model from only ever reacting to the `[MASK]` symbol. Seeing the whole sequence at once makes the representations bidirectional, which is what makes encoders strong at classification and entity recognition. The price is that only the masked positions produce a gradient, about one token in seven, while the causal objective uses every token.

4.4.3 Mixing the two

Each objective has a strength and a cost. Causal training gives a signal at every token but only looks left. Masked training looks both ways but learns from few tokens. Recent work tries to get both at once. Some encoders are pretrained with a causal phase followed by a short masked phase, which beats masked pretraining alone (Gisserot-Boukhlef et al., 2025). Others alternate the two objectives during training (Team, 2024). This line is central to the thesis. Chapter 11 shows that a temporary detour through causal pretraining leaves a lasting and useful mark on an encoder. An earlier idea, ELECTRA's replaced-token detection (Clark et al., 2020), also sought a denser signal. It trained the model to spot tokens that a small generator had swapped in. Still, recent encoders such as ModernBERT (Warner et al., 2025) show that a good architecture with plain masking is already enough.

With both the architecture and the objective settled, the next question is how far this recipe scales.

4.5 Scaling Laws & LLMs

4.5.1 Scaling

The Transformer settled how a model reads its context, and the pretraining objective settled what it learns from raw text. The last lever is size. We can ask how far performance keeps improving as we add parameters and data, and whether anything new appears along the way. The answer, found around 2020, changed how the field builds models and how it uses them.

Until then, a pretrained model was put to work by *fine-tuning*. After pretraining, each new task needed a further round of supervised training on labelled examples. This produced a separate specialised copy of the model for each task. Brown et al.

(2020) showed another route. Their model, GPT-3, a decoder with 175 billion parameters, can carry out a task from nothing more than a description and a few examples written into its input, with no change to its weights. They call this *in-context learning*, and the abstract states the mechanism plainly: the model is "applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text" (Brown et al., 2020). The practical change is large. Instead of collecting labelled data and training a classifier for every task, one writes a prompt.

How much help the prompt carries defines three settings. In the *zero-shot* setting the model is given only the task, at most a short instruction. In *one-shot* it also sees one worked example. In *few-shot* it sees several, usually ten to a hundred, as many as fit in the 2048-token context window (the most text the model can read at once). Figure 4.8 places the three side by side on a simple arithmetic task. The examples are not training data. They are text the model reads before answering, and the same idea works across very different tasks: unscrambling letters (`skicts` → `sticks`), word analogies (*lull is to trust as cajole is to compliance*), or translating a sentence into French.

zero-shot

```
Q: What is 98 plus 45?
A:
```

one-shot

```
Q: What is 12 plus 7?
A: 19
Q: What is 98 plus 45?
A:
```

few-shot

```
Q: What is 12 plus 7?
A: 19
Q: What is 31 plus 6?
A: 37
Q: What is 98 plus 45?
A: 143
```

Figure 4.8: The same question put to GPT-3 in three ways. With no example (zero-shot) the model sees only the question; with one or a few solved examples in the prompt (one- and few-shot) it picks up the pattern and answers (in green). No weights are updated; the examples are just text. The format and the final example are from Brown et al. (2020).

The striking part is that more demonstrations help, and they help more as the model grows. On two-digit addition GPT-3 goes from 76.9% with no example to 99.6% with one and 100% with a few. On the TriviaQA question-answering benchmark it goes from 64.3% to 68.0% to 71.2% (Table 4.1). The same trend holds across many tasks, and Brown et al. (2020) note that "the gap between zero-, one-, and few-shot performance often grows with model capacity". They read this as a sign that larger models are better at learning from their context. A capability that is weak or absent in small models and grows stronger with scale is what later work called *emergent*.

These gains are not a fluke. They rest on something the model is trained to do directly. Below the downstream abilities sits the loss, the model's average surprise

Task	**Zero-shot**	**One-shot**	**Few-shot**
Two-digit addition	76.9	99.6	100.0
Two-digit subtraction	58.0	86.4	98.9
TriviaQA (open QA)	64.3	68.0	71.2

Table 4.1: GPT-3 (175B) accuracy as the number of in-context examples grows, with no weight updates. Numbers are from Brown et al. (2020).

on held-out text, and the loss falls with scale in a very regular way. Kaplan et al. (2020) showed that it follows a smooth *power law* in two quantities, the number of parameters N and the number of training tokens D. On logarithmic axes this is a straight line. The loss drops predictably as either grows, so model design becomes a forecast: pick a budget of computation and read off the loss to expect. Their estimate placed the optimum at large models trained on relatively little data.

Hoffmann et al. (2022) revisited this and found the trade-off was wrong. They fitted, on over 400 trained models, a loss law of the form

$$L(N, D) = E + \frac{A}{N^{\alpha}} + \frac{B}{D^{\beta}},$$

and estimated $E = 1.69$, $A = 406.4$, $B = 410.7$, $\alpha = 0.34$ and $\beta = 0.28$. Here E is the loss that would remain with unlimited data and parameters, a floor set by the language itself. The two fractions measure how far a finite model and a finite dataset still sit above it. The cost of training, counted in arithmetic operations (FLOPs), grows with both, roughly as $6ND$. So for a fixed budget the law gives the best split between model size and data, the *compute-optimal* point. The answer is that the two should grow together: each time the model doubles, the data should double with it (Figure 4.9). By the authors' own projection this means about twenty training tokens per parameter, far more data than earlier models had used.

To prove the point they trained Chinchilla, a 70-billion-parameter model, on 1.4 trillion tokens. It beat Gopher, a 280-billion model trained on the same compute, scoring 67.6% against 60.0% on MMLU, a multiple-choice exam spanning fifty-seven subjects. Most large models before it had simply been undertrained: built large, then fed too little data to fill them.

Chinchilla optimises the cost of *training* a model once. Touvron et al. (2023) pointed out that this ignores the cost of *using* it afterwards. A model served to many users is run countless times, and each run costs computation too, its *inference* cost. As they put it, "a smaller one trained longer will ultimately be cheaper at inference". They pushed small models well past the compute-optimal point and found that a 7-billion model keeps improving even after a trillion tokens. The resulting LLaMA models, trained only on publicly available data, make the case plainly: the 13-billion model outperforms GPT-3 (175B) on most benchmarks "despite being 10× smaller"

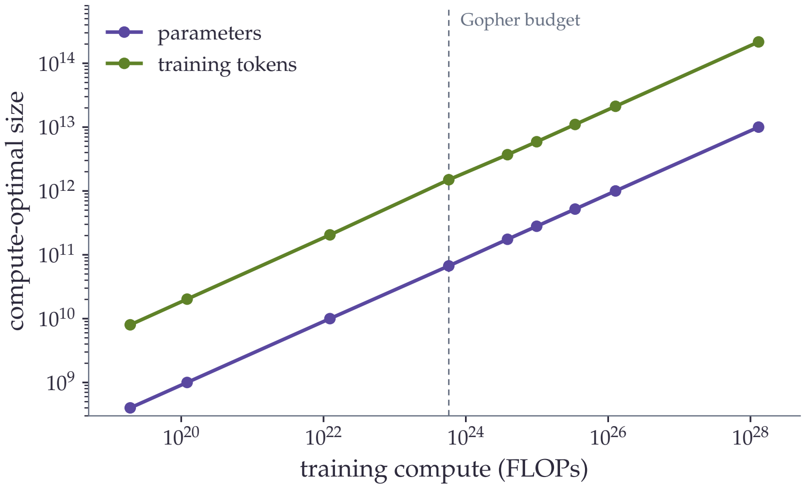

Figure 4.9: The compute-optimal model size and the compute-optimal number of training tokens rise together with the training budget, as the parallel lines show. Points are from Table 3 of Hoffmann et al. (2022).

(Figure 4.10), and the 65-billion model is competitive with Chinchilla and the 540-billion PaLM. Their weights were released, so they also brought capable models within reach of ordinary research budgets.

4.5.2 Instruction Tuning

The previous section showed that a large base model can already perform a task from a few examples placed in its prompt. That ability is fragile. It leans on well-chosen demonstrations and on phrasing the task as a text to be continued, and it is weakest in the zero-shot case, with no examples at all. A base model is trained only to predict the next token. So asked a plain question it may answer, or it may simply continue with another question. *Instruction tuning* closes this gap. The model is fine-tuned on many existing datasets, each rewritten as a natural-language instruction paired with its answer. After this it follows instructions phrased the same way, including instructions for tasks it never saw during tuning.

Wei et al. (2022a) introduced the idea and the name. They took a 137-billion-parameter model and tuned it on more than sixty datasets phrased as instructions. Its zero-shot accuracy then rose above that of the much larger GPT-3 on twenty of the twenty-five evaluation datasets they tried. The gain came from teaching the model to read instructions, not from more scale. Sanh et al. (2022) made the same case at a smaller size: an eleven-billion model tuned on a wide range of prompted tasks matched or beat GPT-3, about sixteen times larger, on nine of eleven held-out tasks. What matters most is the breadth of the tuning mixture. Adding more distinct tasks improves generalisation to unseen ones, with most of the benefit reached after a few

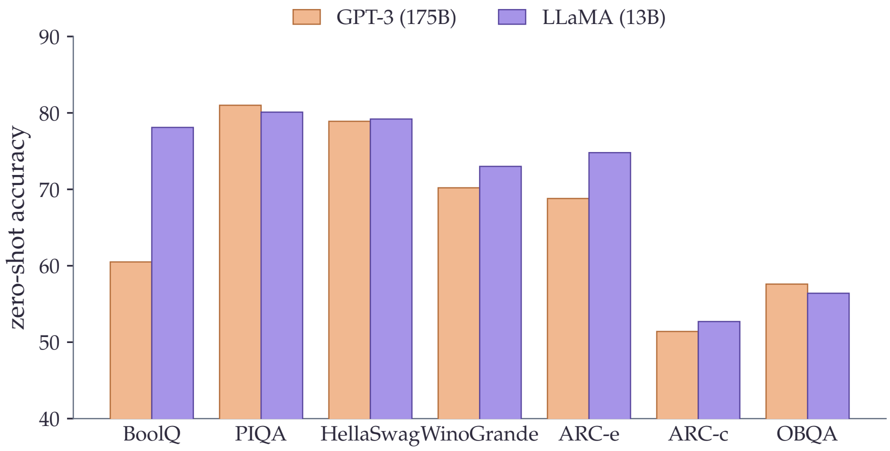

Figure 4.10: A 13-billion-parameter model trained on more tokens matches or beats GPT-3, which is more than ten times larger, on most of these zero-shot reasoning tasks. Numbers are from Table 3 of Touvron et al. (2023).

hundred tasks (Chung et al., 2024). Collections such as Super-NaturalInstructions were built to supply that breadth, gathering more than 1,600 tasks (Wang et al., 2022). Such instruction datasets are costly to assemble by hand, which opened a second route. Wang et al. (2023b) showed that a model can build its own instruction data, generating new instructions and answers from a small seed set and tuning on the filtered result. This approach seeded much of the open instruction-tuning that followed.

4.5.3 Aligning Assistants with Human Feedback

Instruction tuning teaches a model to attempt the task. It does not by itself make its answers the ones a person would prefer. A useful assistant should be, in an early statement of the goal, helpful, honest, and harmless (Askell et al., 2021). Reaching that goal from supervised data alone is hard: for an open-ended question there is no single correct target to imitate, only answers that are better or worse than others. The way around this is to learn from comparisons. Christiano et al. (2017) had shown, for control problems, that an agent can be trained from human judgments of which of two behaviours is better, with no hand-written reward. They fit a *reward model* to those judgments and then optimise against it.

Ouyang et al. (2022) brought this recipe to language in three steps: fine-tune the model on human-written demonstrations, collect human rankings of model outputs and fit a reward model to them, then optimise the model against that reward with reinforcement learning (Figure 4.11). The result, *reinforcement learning from human feedback* (RLHF), was striking: human raters preferred the answers of a 1.3-billion-parameter aligned model over those of the hundred-times-larger 175-billion GPT-3.

This is the recipe behind ChatGPT and the assistants that followed it. It is what let a non-expert state a request in plain language and get a useful answer. Bai et al. (2022a) built a helpful and harmless assistant the same way and reported that alignment training improved performance across most benchmarks while remaining compatible with specialised skills such as coding. The expensive part is the human labelling, so later work sought to reduce it. Constitutional AI replaces human judgments of harmful content with judgments the model itself makes against a short written list of principles, training from AI feedback instead of human feedback (Bai et al., 2022b).

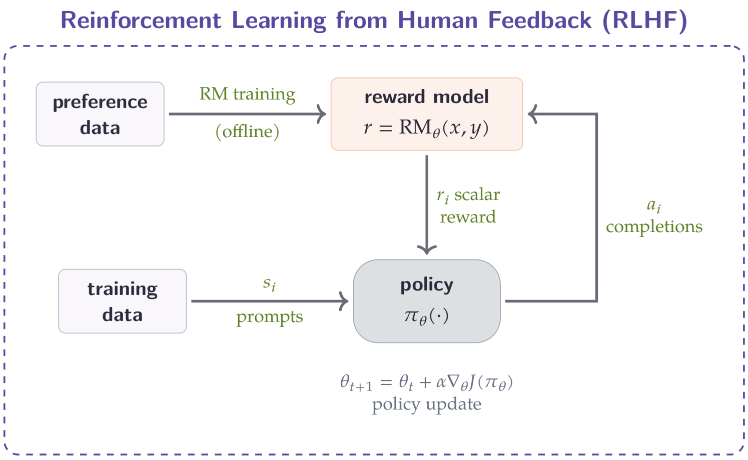

Figure 4.11: In reinforcement learning from human feedback the policy generates completions, a reward model learned offline from human preference data scores them, and the scalar reward updates the policy (Ouyang et al., 2022).

4.5.4 Reasoning and Verifiable Rewards

The most recent shift concerns problems that have a checkable answer. Wei et al. (2022b) observed that prompting a model to produce a *chain of thought*, a series of intermediate steps before its final answer, sharply improves its accuracy on multi-step problems, with no change to its weights. Sampling several such chains and taking the majority answer improves it further (Wang et al., 2023a). This made reasoning something one could first draw out and then train for. On tasks such as mathematics or programming the answer can be checked automatically, by matching the solution or running the code. This yields a reward computed by a rule rather than predicted by a learned model. Training against such a signal has been named *reinforcement learning with verifiable rewards* (Lambert et al., 2024). The reward is a correctness check, so it cannot be gamed the way a learned reward model can (Figure 4.12).

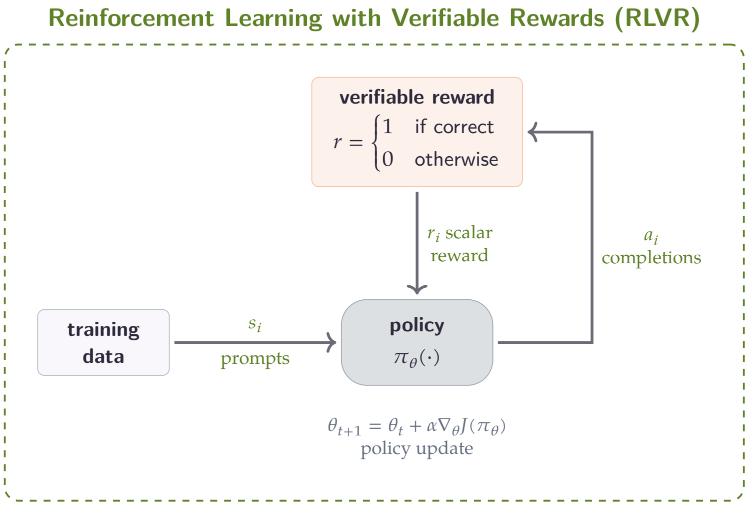

Figure 4.12: Reinforcement learning with verifiable rewards keeps the same loop as Figure 4.11 but replaces the learned reward model with a rule that checks the answer or runs the code, so the reward needs no human labelling (Shao et al., 2024).

The DeepSeekMath work introduced an efficient optimiser for this setting, Group Relative Policy Optimization. It drops the separate value network of standard policy optimisation and instead judges an answer against a group of sampled answers to the same question (Shao et al., 2024). DeepSeek-R1 then pushed the idea to its limit. A model trained by reinforcement learning alone, with no supervised fine-tuning first, developed reasoning behaviours such as self-checking and changing strategy from the reward signal, and this reasoning could afterwards be distilled into smaller models (Guo et al., 2025a). The strength of the approach is also its boundary. Verifiable rewards exist for mathematics and code, where correctness is a computable rule. They do not exist for open-ended writing, where there is no automatic check.

Two cautions close this picture. Models are tuned to follow instructions and increasingly trained on data shaped or generated by other models, so the line between training and evaluation blurs. This raises the problem of *benchmark contamination*: when test items, or text closely resembling them, leak into the training data, reported scores stop measuring genuine generalisation. And all of these advances, from instruction tuning to verifiable-reward reasoning, concern general-purpose models trained on broad web text. Turning such a model into one that performs in a narrow, low-resource domain is a separate problem, and the most direct lever for it is to keep pretraining on text from that domain.

4.6 Continual Pretraining

We just reached the idea of specialising a model to a domain by continuing its pretraining on in-domain text. This thesis relies on that idea, so we look at it closely. The section asks three questions: what continued pretraining is, why it is hard, and how recent work makes it reliable. The biomedical models that use it come in the next section.

4.6.1 A Second Phase of Pretraining

The idea is simple. Between the generic pretraining of Section 4.4 and fine-tuning on a labelled task, we add another phase of self-supervised training on text closer to the target. Gururangan et al. (2020) studied this across four domains, including biomedicine, and eight classification tasks, and named two variants. *Domain-adaptive pretraining* continues training on a large in-domain corpus. *Task-adaptive pretraining* continues it on the unlabelled text of the task itself. They found that both help in high- and low-resource settings, that task-adaptive pretraining helps even after the domain phase, and that it pays off even when the task corpus is small (Figure 4.13).

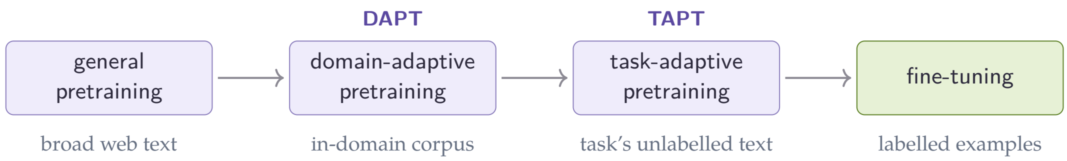

Figure 4.13: Continued pretraining inserts one or two self-supervised phases between general pretraining and fine-tuning: domain-adaptive pretraining on a large in-domain corpus, then task-adaptive pretraining on the task's own unlabelled text (Gururangan et al., 2020).

This second phase is one case of a broader topic. A recent survey divides continual learning for language models into three stages: continued pretraining, domain-adaptive pretraining, and continual fine-tuning. It sorts the methods that protect earlier knowledge into replay, regularisation, and architecture-based families (Shi et al., 2025). The rest of this section keeps to the first stage, continued pretraining, since it is the one this thesis uses.

4.6.2 Catastrophic Forgetting

Continuing to train a network has a cost. When a connectionist model is trained on new material, the weights that encoded the old material are overwritten. This

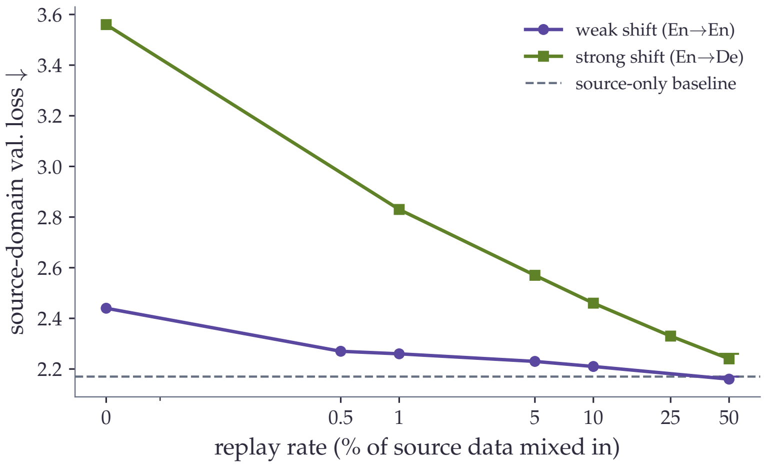

Figure 4.14: Catastrophic forgetting and its remedy: continuing to train a 405-million parameter model on new data raises its loss on the original domain, slightly when the shift is small (English to English) and sharply when it is large (English to German), and replaying a small fraction of the original data brings that loss back toward the source-only baseline (Ibrahim et al., 2024).

failure was named *catastrophic interference* when it was first described (McCloskey and Cohen, 1989). The standard remedy keeps the new training from straying too far. Elastic weight consolidation adds a penalty that, in the authors' words, works by "selectively slowing down learning on the weights important for" the earlier tasks (Kirkpatrick et al., 2017). For a model adapted to a domain the same risk applies in reverse: pushing it toward biomedical text can erode the general competence it began with. The danger is uneven. Yıldız et al. (2024) report that smaller models are the most sensitive, showing the highest rates of both learning and forgetting, so a small domain encoder is more exposed than a large one. The effect shows up directly as a rise in the model's loss on its original domain, sharp when the new data is distant from the old (Figure 4.14).

A drop in a benchmark score is not always true forgetting. Zheng et al. (2025) argue that the early steps of a new training phase often disturb a model's *alignment*, its learned way of exposing what it knows, more than its knowledge. In their words, "performance drops often reflect a decline in task alignment rather than knowledge loss", and much of the loss can be recovered. The lesson for what follows is to read a post-adaptation drop with care, and to design the adaptation so that it does not disturb more than it must.

4.6.3 Making Continual Pretraining Work

Recent work shows that, done carefully, continued pretraining can avoid forgetting and stay cheap. Ibrahim et al. (2024) reduce the recipe to three ingredients. The learning rate has decayed to almost nothing by the end of pretraining, so they re-warm it, then re-decay it, and they replay a small fraction of the original data alongside the new domain. This combination, they report, is "sufficient to match the performance of fully re-training from scratch on all available data" while "using only a fraction of the compute", and they verify it at both the 405-million and 10-billion parameter scales. So continued pretraining can reach the same place as starting over, far more cheaply.

Two refinements sharpen this. First, less data can work better than more. Choosing a small, relevant slice of the domain corpus rather than all of it can match or beat training on the whole. Xie et al. (2024) report data-selection strategies that "outperform vanilla continual pre-training's performance with just 10% of corpus size and cost", with no loss on general tasks. Second, the adaptation does not improve smoothly. Guo et al. (2025b) document a *stability gap*: in-domain performance falls for a time at the start of continued pretraining, then recovers. They show that training for several passes over a high-quality subset and mixing in pretraining-like data smooths it out, raising the average medical-task accuracy of a three-billion model from 36.2% to 40.7% while using only 40% of the training budget. The common thread is that continued pretraining rewards care in the learning-rate schedule, the amount of replay, and the choice of data.

4.6.4 From Scratch or Continue?

The alternative to continuing is to pretrain a specialised model from scratch on in-domain text. Each pole has a canonical biomedical example. BioBERT continued the pretraining of BERT on PubMed abstracts and full-text articles (Lee et al., 2020), the continued-pretraining route. PubMedBERT was instead pretrained from scratch on biomedical text, with a vocabulary built from that text rather than inherited (Gu et al., 2022). The trade-off is clear in principle. Training from scratch can fit a vocabulary to the domain and cannot forget what it never learned, but it is expensive and throws away the general competence of an existing model. Continuing is cheap and keeps that competence, but inherits a tokenizer and an initialisation built for general text (Figure 4.15).

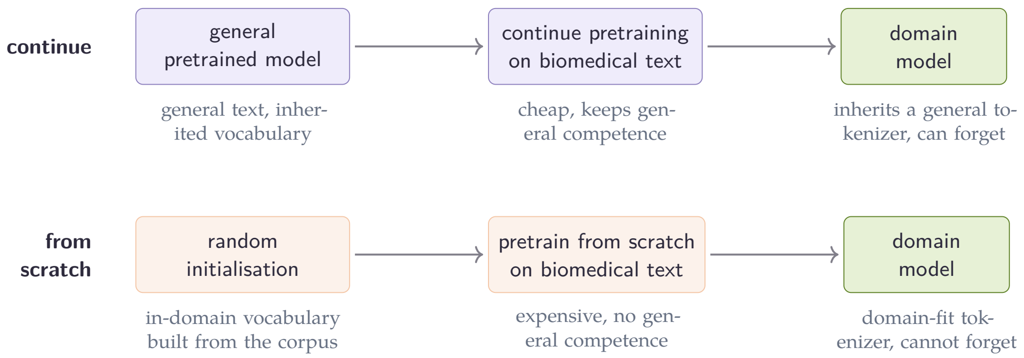

Figure 4.15: Two routes to a domain model: continuing from a general pretrained model is cheap and keeps general competence but inherits a general tokenizer and may forget, while pretraining from scratch fits the vocabulary to the domain and cannot forget but is expensive and starts with no general competence.

Which wins is contested. Gu et al. (2022) found that pretraining from scratch on in-domain text, with an in-domain vocabulary, did better than continued pretraining from a general model across a suite of biomedical tasks, and argued that the general-text vocabulary of a continued model holds it back. The general-purpose evidence above points the other way. With the right learning-rate schedule and a little replay, continued pretraining can match a from-scratch model at a fraction of the cost (Ibrahim et al., 2024). The disagreement is not settled, and it may depend on the architecture. Continued pretraining has often disappointed for generative decoder models. Dorfner et al. (2024) compared biomedical large language models with their general-purpose counterparts and found the specialised models mostly the weaker of the two, sometimes by a wide margin. One eight-billion biomedical model scored 30% against 64.3% for the general model it was built from on New England Journal of Medicine case challenges. The bidirectional, masked objective of encoders may behave differently. The open question is whether continued pretraining reliably specialises a model, and how the answer depends on whether the model is an encoder or a decoder. The next section takes it up through the biomedical models built to answer it.

4.7 Biomedical Language Models

The previous section left an open question. Does continued pretraining reliably specialise a model, and does the answer depend on the architecture? The biomedical models built over the past few years are the testbed. We look at them in three groups: the English encoders, the French models, and the generative decoders. The from-scratch-versus-continue question runs through all three.

4.7.1 English Biomedical Encoders

The first biomedical encoders took the continued-pretraining route. BioBERT continued the pretraining of BERT on PubMed abstracts and full-text articles, and reported clear gains over the general model: about half a point of F1 on biomedical named-entity recognition, nearly three on relation extraction, and twelve points of mean reciprocal rank on question answering (Lee et al., 2020). The gains were real and the recipe was cheap, since it reused the existing weights. This established the mixed-domain paradigm: start from a general model, then keep training on domain text.

A second line questioned that paradigm by training from scratch, with a vocabulary built from domain text rather than inherited. SciBERT measured what is at stake. Between BERT's general vocabulary and one learned from scientific text, only 42% of frequent tokens overlap, so a general tokenizer splits more than half of the common scientific words into pieces (Beltagy et al., 2019). Training from scratch on 1.14 million papers fixes this. PubMedBERT made the strongest version of the claim: pretraining from scratch on biomedical text, with an in-domain vocabulary, "substantially outperforms continual pretraining of generic language models", and they released the BLURB benchmark that became the standard yardstick for the field (Gu et al., 2022). BioLinkBERT added a twist. It pretrained from scratch on PubMed with its citation links so that related documents are seen together, for a further gain on biomedical question answering (Yasunaga et al., 2022b). Clinical text adds a constraint of its own. ClinicalBERT was trained on the notes of the MIMIC-III intensive-care database, and while its weights are public the underlying notes are not, since they sit behind credentialed access (Alsentzer et al., 2019). The current strongest encoder follows the continued route again, on a far larger and more varied corpus. BioClinical ModernBERT continues a modern long-context encoder on more than 53 billion tokens drawn from twenty datasets across institutions (Sounack et al., 2025), with the architectural changes that make it long-context deferred to Section 4.8.

4.7.2 French Language Models

French has its own lineage. CamemBERT adapted the RoBERTa recipe to French web text and made a point that matters for low-resource work: about four gigabytes of text gave results "as good as those obtained using larger datasets (130+GB)" (Martin et al., 2020). FlauBERT appeared in parallel, trained on a heterogeneous French corpus (Le et al., 2020). Later models refreshed the recipe. CamemBERTa adopted the DeBERTaV3 architecture and its replaced-token-detection objective for better data efficiency (Antoun et al., 2023), and CamemBERT 2.0 retrained the family on more recent data, with a longer context, an updated tokenizer, and new architectures, to counter the temporal concept drift caused by outdated training data (Antoun et al., 2024).

The two dedicated French biomedical encoders were both trained from scratch. DrBERT trained a RoBERTa model on NACHOS, a corpus of French biomedical and clinical text, and its experiments argued for the from-scratch route: models trained from scratch gave the best results, and continuing the pretraining of a French general model never reached the top on any of their tasks (Labrak et al., 2023). The picture is not one-sided, even in their own results. Continuing the pretraining of an *English* biomedical model on French data did work, which they read as cross-lingual transfer. They also found that publicly available biomedical text could match or beat private hospital data, a point that bears directly on the use of public corpora for the clinic. AliBERT, the other French biomedical encoder, was also trained from scratch, with a regularised Unigram tokenizer built for the domain (Berhe et al., 2023). On this evidence, whether continuing from a French general model can be made to match a from-scratch biomedical one is still an open question.

4.7.3 Biomedical Decoders

Generative models were specialised in the same way, by continuing the pretraining of a general large language model on medical text, but at a much larger scale. BioMistral continued Mistral 7B on PubMed Central and reported that the adapted model beat the general instruction-tuned Mistral on eight of ten medical question-answering tasks. To keep the general ability that specialisation can erode, the authors also merged the domain and general models, combining their strengths (Labrak et al., 2024b). Meditron continued Llama-2, at seven and seventy billion parameters, on a curated medical corpus of about 48 billion tokens that deliberately mixes in one percent of general data as replay, and its seventy-billion model outperformed GPT-3.5 and Med-PaLM on medical benchmarks (Chen et al., 2023). PMC-LLaMA continued LLaMA on biomedical papers and books (Wu et al., 2024).

Each of these reports gains on its own medical benchmarks, yet a broader picture is more sober. Dorfner et al. (2024) evaluated biomedical models against the general models they were built from, on tasks chosen to fall outside the fine-tuning data, and found the specialised models mostly the weaker of the two, especially away from pure medical knowledge. The contrast with the encoders is the lesson of this section. For encoders, continued pretraining on domain text is close to a default win on extraction tasks. For large decoders, specialising on a domain is a riskier trade, where in-domain gains can come at the cost of general ability and do not always survive an independent test. Why the two architectures should behave so differently is not settled, and it is one of the threads this thesis follows. These models, finally, are built on architectures that have moved on from the original Transformer, which is where we turn next.

4.8 Modern Architectures

The changes from the original Transformer are partly about quality and partly about cost. The same attention mechanism described in Section 4.3 becomes much cheaper to train and to run, and the context window grows from a few hundred tokens to several thousand. For clinical text the second change matters most, so this section ends on long documents. We first cover the encoder refinements and the attention engineering that made them possible.

4.8.1 Modernising the Encoder

Two encoder lines feed directly into the models used today. DeBERTa changed how a token's content and position are combined (He et al., 2021). The original Transformer adds a position vector to each word vector, so the two signals are mixed before attention sees them. DeBERTa keeps them apart, with one vector for content and one for position, and lets attention use the two separately. Its authors call this disentangled attention. The model was the first to pass the human baseline on the SuperGLUE benchmark, and the design influenced later encoders, including ModernBERT and CamemBERT 2.0.

A second line attacked the cost of pretraining. MosaicBERT is a recipe for training a BERT-style encoder cheaply (Portes et al., 2023). It raises the masking rate to 30%, against BERT's 15%, so each example teaches the model more, and it stacks several engineering changes: FlashAttention, ALiBi position information, gated linear units, and the removal of padding tokens from the computation. Together these let the authors train a BERT-base model in about 1.13 hours on eight A100 GPUs, for roughly twenty dollars. This recipe is the direct foundation of ModernBERT.

4.8.2 Efficient Attention

The encoder gains above rest on a handful of changes to attention and normalisation. FlashAttention reorganises the attention computation to fit the memory hierarchy of the GPU, using tiling and kernel fusion so that intermediate results stay in fast memory (Dao et al., 2022). The arithmetic cost is still the $O(L^2)$ of Section 4.3, but avoiding slow memory reads and writes makes it up to roughly three times faster in practice, and it is now a standard component. Grouped-query attention addresses memory at inference time by sharing key and value heads across several query heads, which shrinks the cache that the model must hold while generating (Ainslie et al., 2023); it is used in models such as Llama-2 and Mistral. RMSNorm simplifies layer normalisation by dropping the mean-centring step. This makes it cheaper to compute and is now common in large models (Zhang and Sennrich, 2019). These are engineering refinements, and together they made large-scale and long-context training practical.

4.8.3 Long Context

Context length is a concrete limit for clinical work. BERT accepts at most 512 tokens, while a hospital discharge summary often runs past 2000 tokens. Feeding such a document to BERT means cutting it short, and the part that is dropped may carry the diagnosis, the treatment, or the outcome. The model then never sees information that the task depends on.

Several architectures push the limit up. Longformer reaches 4096 tokens by replacing full attention with a sliding window, where each token attends to its neighbours, plus a small number of global tokens that attend to everything (Beltagy et al., 2020). This lowers the quadratic cost to something close to linear in the length. BigBird also reaches 4096 tokens, with a sparse pattern that combines local, global, and random connections (Zaheer et al., 2020). ModernBERT goes further, to 8192 tokens, sixteen times BERT's window (Warner et al., 2025). It combines FlashAttention, RoPE positional encoding (Section 4.3), an alternation of local and global attention layers, and the 30% masking rate from MosaicBERT. This is the long-context recipe that Section 4.7 pointed forward to.

BioClinical ModernBERT carries this long-context encoder to clinical text and is, on current evidence, the strongest biomedical and clinical encoder (Sounack et al., 2025). It inherits the 8192-token window, which lets it read a full clinical note rather than a truncated fragment. These architectures still inherit a vocabulary and a tokenizer, which is the next lever we examine.

4.9 Tokenization

We now turn to that tokenizer, which decides how a string of text becomes the discrete units a model reads. This choice shapes how well the model handles rare and technical words, and it carries forward into every later stage of training.

4.9.1 Subword Tokenization

A tokenizer first has to solve a coverage problem. A model with a fixed vocabulary of whole words can only represent the words it has seen, and any rare or unseen word becomes an out-of-vocabulary token that the model cannot read. Subword tokenization avoids this by splitting rare words into smaller pieces that are already in the vocabulary. A common word stays as one unit, while a rare word is broken into several known parts. Any word can then be represented as a sequence of subword units, so the out-of-vocabulary problem disappears.

The most widely used method is byte-pair encoding, often called BPE (Sennrich et al., 2016). It starts from individual characters and repeatedly merges the most frequent adjacent pair of units. Each merge adds a new, slightly longer piece to the vocabulary, so frequent sequences become single units while rare sequences stay split. After a fixed number of merges, the result is a vocabulary of common subword pieces that balances short frequent words against longer rare ones.

SentencePiece is a language-independent tokenizer that works directly on raw text (Kudo and Richardson, 2018). It does not need the text to be split into words beforehand, which helps for languages that do not mark word boundaries with spaces. It treats the input as a raw stream of characters and learns its subword units from that stream. CamemBERT used SentencePiece to build its vocabulary. A related method is the unigram language model (Kudo, 2018). It selects subword units with a probabilistic model rather than by greedy merging. It can also sample several different segmentations of the same text, which is useful as a form of regularization during training.

4.9.2 Domain Mismatch

A tokenizer trained on general text is a poor fit for a technical domain. SciBERT measured this gap directly (Beltagy et al., 2019). Between the general vocabulary of BERT and a vocabulary learned from scientific text, only 42% of the frequent tokens overlap. More than half of the common scientific words are therefore split into pieces by the general tokenizer. The model has to reassemble each technical term from generic fragments, which makes those terms harder to learn and to represent.

The same mismatch appears in French biomedical text. A general French tokenizer splits a technical term such as “échocardiographie” into several generic pieces, for example “écho”, “cardi”, and “ographie”. A tokenizer trained on biomedical text keeps the same term nearly whole, as one or two units. Over-splitting makes technical terms longer in tokens and harder for the model to handle, and biomedical text is full of such terms.

This connects back to the from-scratch-versus-continue question of Section 4.6. Training a model from scratch lets it build a vocabulary fitted to the domain, so technical terms stay compact. This route is expensive, because it needs a large corpus and a long training run. Continuing the pretraining of a general model is much cheaper, but the model is stuck with a general-purpose tokenizer that over-splits domain terms. The vocabulary is fixed once the general model is built, so continual pretraining cannot repair it. The choice between a fitted vocabulary and a cheap training path therefore stays open.

Tokenizer mismatch is one of several limits that matter for a clinical, French, and biomedical target, and we turn to those limits next.

4.10 Limits and Transition

This chapter followed language modeling from Shannon and n-grams to neural and recurrent models, then to the Transformer and its two main pretraining objectives.

It then covered scaling laws and large language models, with instruction tuning, alignment, and reasoning, and the use of continual pretraining to specialise a model to a domain. It closed on biomedical models, efficient attention, long context, and tokenization. One thread runs through all of it: the target of this thesis, text that is at once French, biomedical, and clinical. Seen from that target, the methods above share a set of limits, and each limit points past the models to the data they need.

The first limit is data rarity. Every method here assumes a large in-domain corpus, either to pretrain a model from scratch or to continue training an existing one. For clinical text that assumption does not hold. Clinical notes are private and regulated, in France under the CNIL, and a model trained inside one hospital cannot be shared with another institution. This is why public sources of biomedical text become attractive, and it raises a practical question that the models alone do not answer: how to build a corpus that is usable and shareable.

The second limit is a knowledge gap. Language models learn the statistical patterns of text, and they pick up many facts along the way. They do not absorb the explicit structure of medical ontologies, the hierarchies and relations that medical coding schemes such as ICD-10 depend on. That structured knowledge is needed for clinical tasks, and capturing it is a different problem from modeling running text.

The third limit is document length. Hospital reports are often long and exceed the input size of standard encoders. The long-context architectures of Section 4.8, such as Longformer, BigBird, and ModernBERT, raise this ceiling. Using them well on clinical text still calls for adapting them to the domain, which brings the discussion back to the question of in-domain data.

The chapter has described the models. The next chapter turns to what they need, the data: the corpora and the structured knowledge used to pretrain these models, and how such resources are collected, filtered, and prepared.

5 Pretraining Data: Corpora and Knowledge

5.1 Introduction

The previous chapter ended on a simple point: language models are only as good as the data they are trained on. The scale of that data is now very large. GPT-3 was trained on about 300 billion tokens (Brown et al., 2020), and the LLaMA models on one to 1.4 trillion tokens (Touvron et al., 2023). These numbers are not arbitrary. The Chinchilla scaling result showed that, for a fixed compute budget, the strongest models use far more data than earlier practice assumed, and that data and model size should grow together (Hoffmann et al., 2022). Good performance therefore depends on assembling corpora at this scale.

Scale alone does not settle the matter. The quality and composition of a pretraining corpus shape what a model can later do. A corpus drawn mostly from web pages teaches a model the language of the web, and a corpus rich in scientific writing teaches it the language of science. What to include, what to remove, and in what proportions are part of how the model is built. These choices receive less attention than model design, yet they account for much of the difference between models of similar size.

This chapter considers two kinds of resources used to pretrain language models. The first is raw text, such as web crawls, scientific articles, and books. The second is structured knowledge, such as medical ontologies and knowledge graphs, where facts are stored as explicit relations rather than as running prose. Both have been used to train and adapt biomedical models, and each raises different questions about how it is collected and prepared.

For a French clinical setting, the target data face a triple scarcity. The text must be French, it must be biomedical, and it must be clinical, all at the same time. Each of these constraints reduces the amount of usable text. French biomedical text is far less abundant than English biomedical text, and clinical text is the scarcest of all, because patient records are private and rarely leave the hospitals that produce them. This combination remains a standing difficulty for anyone building models for French clinical language.

The chapter follows the path of the data. It starts from general web corpora and the methods used to collect them, then turns to quality filtering and how it changes what a corpus contains. It then moves to biomedical and clinical sources, where this scarcity becomes concrete. Finally, it presents medical ontologies and discusses how such structured knowledge can be folded into the pretraining of language models.

5.2 Web-Scale Pretraining Corpora

Large language models are trained on text collected from the open web. Before describing how this text is cleaned and selected, we describe where it comes from and how the main public corpora were built.

5.2.1 Common Crawl

Almost every large pretraining corpus starts from Common Crawl, an open crawl of the public web that has been running since 2008. Common Crawl releases a new snapshot roughly every month. Each snapshot contains on the order of 2.5 billion web pages and reaches hundreds of terabytes once uncompressed. The content is very heterogeneous, mixing news articles, forum threads, spam, source code, advertising, and page boilerplate. For this reason the raw crawl cannot be used directly for training. Every corpus described below applies a heavy cleaning pipeline to Common Crawl before the text is suitable for a language model.

5.2.2 Early Large Corpora

The first widely used corpus of this kind was C4, the Colossal Clean Crawled Corpus. It contains about 750 GB of English text taken from a single Common Crawl snapshot from April 2019, cleaned with simple filters such as language detection, the removal of incomplete sentences, and deduplication. C4 was introduced together with the T5 model (Raffel et al., 2020). Dodge et al. (2021) later studied what C4 actually contains, and found patents, United States government and military sites, machine-generated text, and even examples drawn from evaluation benchmarks. They also showed that the blocklist used during cleaning removes a disproportionate amount of text written by and about minorities.

A different design choice is to assemble a corpus from many curated sources rather than from the web alone. The Pile follows this idea. It contains about 800 GB of English text from 22 sources, including PubMed, arXiv, GitHub, Stack Exchange, Wikipedia, and books. The motivation is that a diversity of sources matters more for a language model than the raw volume of any single source. The Pile was open-sourced by EleutherAI (Gao et al., 2020).

For languages other than English, OSCAR was an important early resource. It is produced by an efficient pipeline that classifies Common Crawl documents by language and groups them into per-language corpora. Its French portion, about 138 GB, is the corpus on which CamemBERT was trained (Ortiz Suárez et al., 2019).

5.2.3 Modern Corpora

Recent corpora are much larger and come with carefully documented pipelines. FineWeb contains about 15 trillion tokens drawn from 96 Common Crawl snapshots. Its pipeline combines text extraction, language filtering, quality heuristics, and deduplication based on URLs and MinHash. FineWeb also provides a subset of about 1.3 trillion tokens, called FineWeb-Edu, that is filtered for educational content, which we discuss in the next section (Penedo et al., 2024).

Other projects focus on making filtering easier for the people who reuse the corpus. RedPajama-V2 is an open dataset for training large language models, and it includes a French subset. It ships pre-computed quality signals, so that users can apply their own filters without recomputing those signals (Weber et al., 2024). TxT360 instead emphasises global deduplication. It deduplicates globally across 99 Common Crawl snapshots, leaving roughly 4.83 trillion tokens (LLM360, 2024).

Two further corpora were released mainly to support open research on language models: Dolma, about 3 trillion tokens (Soldaini et al., 2024), and ROOTS, a 1.6 TB multilingual composite built for the BLOOM model (Laurençon et al., 2022).

These corpora give us billions to trillions of tokens of web text, but their quality varies enormously from one document to the next, which raises the question of how to select the good documents.

5.3 Data Quality and Curation

The previous section showed that web text is abundant but uneven in quality. A monthly Common Crawl snapshot mixes clean articles with spam, boilerplate, and machine-generated pages. Training a model on this raw mixture wastes compute on text that teaches little. The question is therefore how to keep the good documents and remove the rest. Two families of methods address this question. The first uses simple rules to drop obviously bad text. The second uses learned classifiers to judge the quality of the content.

5.3.1 Heuristic Filtering

The first family is rule-based filtering, which has been standard practice since the Gopher pipeline (Rae et al., 2021). These rules look at surface properties of a document and discard it when a property falls outside an expected range. Common rules set minimum and maximum document lengths, limit the ratio of words to symbols, cap the proportion of uppercase characters, and remove documents with many duplicated lines. A second rule is perplexity filtering. A small n-gram model trained on Wikipedia scores each document, and documents with high perplexity are treated as noise and removed. Language identification with fastText keeps only documents in the target language. Almost every modern pipeline reuses these heuristics. They remove the obvious noise, but they do not judge whether the content is informative.

5.3.2 Quality Classifiers

The second family judges content quality with a learned classifier. The recent pattern has three steps. A large language model annotates a sample of documents, a small and fast classifier is distilled from those annotations, and that classifier is then run over the whole corpus. This keeps the judgement of a strong model while making large-scale filtering affordable. FineWeb-Edu is a clear example (Penedo et al., 2024). Its authors used Llama-3-70B-Instruct to score about 460,000 web pages for their educational value on a scale from 0 to 5, distilled a small regression classifier from these scores, and then classified the full 15 trillion tokens at a cost of about 6,000 H100 hours. The filter removes about 92% of the data, yet a model trained on the remaining fraction still beats one trained on the full corpus on benchmarks such as MMLU and ARC.

Other pipelines change the target of the classifier. DataComp-LM, known as DCLM, trains a fastText classifier to separate instruction-style text from general web text, which favours pages with a resolved question-and-answer structure (Li et al., 2024). Nemotron-CC uses a quality classifier with several criteria rather than a single educational score, which gives a more granular view of quality (Su et al., 2025).

5.3.3 Organising by Domain

Quality is not the only axis along which a corpus can be shaped. WebOrganizer organises the web into taxonomies by topic and by format, again using language model annotations distilled into efficient classifiers (Wettig et al., 2025). This view exposes what quality scores leave implicit. The authors show that FineWeb-Edu carries a hidden bias toward science, technology, and academic writing, and that part of its measured gains come from a domain mix that happens to match the benchmarks. They also find that combining domain organisation with quality filtering works better than quality filtering alone. What a corpus contains therefore depends on both how good its documents are and what they are about.

5.3.4 Deduplication

A separate problem is repetition. Web crawls contain the same text many times, because pages are copied, templated, and re-crawled across snapshots. Training on this material biases the model toward the most-copied text, such as boilerplate and templates, and wastes capacity on memorising it. Deduplication addresses this in two stages. The first removes exact duplicates at the level of URLs. The second finds near-duplicate documents using MinHash together with locality-sensitive hashing, which compares documents quickly without checking every pair. The scale of the problem is large. TxT360 deduplicated globally across Common Crawl snapshots to remove a large volume of near-duplicate documents (LLM360, 2024).

5.3.5 Contamination

Aggressive filtering toward benchmark-like text raises a further concern, benchmark contamination. Contamination happens when test items, or text that closely resembles them, leak into the training data. When this occurs, reported scores stop measuring true generalisation and instead reward memorisation. Deng et al. (2024) showed that strong models can guess a large share of masked benchmark options, which is evidence that the underlying question-and-answer pairs are already on the web. Tools such as Infini-gram help detect this by finding exact matches between benchmark text and training sets (Liu et al., 2024), and a survey reviews the broader set of contamination-detection techniques (Xu et al., 2024). This leaves a tension at the heart of curation. Filtering more aggressively raises benchmark scores, but some of that gain may come from implicit contamination rather than from real quality.

These methods work well for general web corpora, where text is abundant and the goal is to keep the best of a large pool. The biomedical setting is different. The corpora are far smaller, the domain is specific, and the criteria for good text are not the same, which raises the prior question of what biomedical sources exist in the first place.

5.4 Biomedical and Clinical Corpora

The data-quality criteria of the previous section were developed for general web text. The biomedical setting is different, and the corpora used to train biomedical models come from a small number of recurring sources. We review those sources here, separating the large English collections, the article-level curation built on top of them, and the more fragmented French resources. The models trained on these corpora were described earlier in Section 4.7; this section is about the corpora themselves.

5.4.1 Large English Sources

The most widely used source is PubMed, the database maintained by the United States National Library of Medicine. It contains about 39 million biomedical citations and abstracts, written in technical vocabulary and edited to a consistent standard. Its main limit is that it provides abstracts only, without the full text of the articles. PubMed Central (PMC) Open Access addresses this by providing full text, about 4.5 million articles (Chen et al., 2023). The biomedical text drawn from it is overwhelmingly English, about 98% in one large sample (Labrak et al., 2024b). The collection is heterogeneous, mixing research articles, reviews, case reports, editorials, and clinical guidelines. It is also hard to use at the article level, because a single research article can combine clinically relevant case material with methodological sections that are of little clinical use.

Clinical text is harder to obtain. The main reference is MIMIC (Johnson et al., 2016, 2023), a set of clinical notes from a single United States hospital, the Beth Israel Deaconess Medical Center. Access is restricted and requires a data use agreement and ethics training. MIMIC is treated as a gold standard for English clinical text, but it comes from one hospital in one country, which limits how far models trained on it generalise to other settings.

5.4.2 Article-Level Curation for Biomedicine

Some work tries to improve these corpora by scoring articles before training. The corpus used for Meditron (Chen et al., 2023) scores each article using its MeSH subject tags, its publication type, the reputation of the journal, its recency, and its citation count. High-scoring articles are upsampled and low-scoring ones are filtered out, which yields a corpus of about 46 billion tokens. The same limit returns here, because the scoring acts on whole articles and so still mixes relevant and non-relevant content within a single article.

A second example is the corpus used for BioClinical ModernBERT (Sounack et al., 2025), which combines PubMed, PMC, and around twenty clinical datasets into one large collection. Most of those clinical sources require their own data use agreements, so the assembled corpus is not fully public. The model built on it is discussed in Section 4.7; what matters here is that the access constraint sits in the clinical part of the corpus.

5.4.3 French Sources

Public French biomedical resources exist, but they are scattered across several projects. ISTEX is a base of about 27 million scientific references from which French biomedical documents can be extracted. CLEAR provides French drug leaflets and medical encyclopaedia articles in both technical and simplified versions (Grabar and Cardon, 2018). E3C is a multilingual corpus of clinical cases, drug leaflets, and PubMed clinical-case abstracts (Magnini et al., 2020). HAL, the French open repository, and its harvested version HALvest provide French and English scientific articles (Kulumba et al., 2024). Biomedical Wikipedia adds a free and multilingual source. Together these resources are useful, but each is partial, and none approaches the scale of PubMed or PMC.

The French biomedical models, including the ones trained from scratch and the ones obtained by continual pretraining, were described in Section 4.7; here we note only the corpora they used and the pattern of results. The continual-pretraining attempts on French biomedical text relied on small or loosely targeted corpora. One continued pretraining on about 31,000 abstracts and reported no improvement with the large model (Copara et al., 2020). Another assembled about 136 million words from PubMed, Cochrane, ISTEX, and Wikipedia, and obtained a small gain on drug leaflets and no gain on scientific titles (Le Clercq de Lannoy et al., 2022). A third used 21 million private hospital documents, which improved a private clinical benchmark but could not be shared with other researchers (Dura et al., 2022).

These French results are mixed, and their cause is not settled. We do not know whether the disappointing outcomes come from the corpora, which may be too small or not well targeted, or from the continual-pretraining method. These difficulties sharpen the clinical scarcity introduced earlier. Clinical text is private, and in France it is governed by the data-protection rules of the CNIL. Models trained inside one hospital cannot be shared between institutions. And the public French sources are fragmented and small next to PubMed and PMC.

Biomedical corpora help models learn the vocabulary and style of medical text. But language models capture statistical patterns and not the explicit structure that tasks such as medical coding require, so we now turn to the medical ontologies that provide it.

5.5 Medical Ontologies and Knowledge Graphs

5.5.1 Main Medical Ontologies

An ontology is a structured vocabulary. It records concepts, the names used to refer to them, and the relations between them, such as hierarchy, synonymy, and association. It therefore makes explicit the knowledge that a language model would otherwise have to learn on its own from running text. A diagnosis, the procedures that treat it, and the drugs prescribed for it are recorded as concepts and links, rather than left to be inferred from word statistics. Several large ontologies of this kind cover the biomedical domain, and a few of them are central to clinical work in France.

The Unified Medical Language System (UMLS) is the meta-ontology maintained by the United States National Library of Medicine (NLM) (Lindberg et al., 1993). It brings together about 189 source vocabularies, including SNOMED-CT, the Medical Subject Headings (MeSH), the International Classification of Diseases (ICD), and the Anatomical Therapeutic Chemical classification (ATC). In its current releases it contains roughly 3.5 million concepts and over 15 million unique concept names. For each concept it gives synonyms, hierarchical and associative relations, and definitions.

SNOMED-CT holds about 371,000 clinical concepts. They are connected by hierarchical relations, of the is-a kind, and by associative relations such as finding site and causative agent. It is an international standard for clinical information systems and covers diagnoses, procedures, anatomy, substances, and organisms.

The International Classification of Diseases, tenth revision (ICD-10, in French CIM-10), is the World Health Organization's (WHO) disease classification. It is organised as a hierarchy of chapters, blocks, categories, and subcategories. The French version, maintained by the Agence Technique de l'Information sur l'Hospitalisation (ATIH), adds inclusions, exclusions, and notes, and contains about 19,000 codes. It is used for hospital coding under the French case-mix system, the Programme de Médicalisation des Systèmes d'Information (PMSI).

The Classification Commune des Actes Médicaux (CCAM) is the French common classification of medical procedures. It is specific to France and arranges about 8,000 codes in a hierarchy.

The Anatomical Therapeutic Chemical classification (ATC) is the WHO classification of drugs over five levels, from the anatomical group down to the individual substance. It contains about 6,000 codes at the fifth level.

5.5.2 Formal Representations

Ontologies are published in a small set of standard formats defined by the World Wide Web Consortium (W3C). The Resource Description Framework (RDF) stores facts as subject-predicate-object triples. The Web Ontology Language (OWL) adds description logic, which allows automatic reasoning over classes and properties. The Simple Knowledge Organization System (SKOS) provides a lighter vocabulary for thesauri. It uses prefLabel and altLabel for a concept's preferred and alternative names, and broader, narrower, and related links for hierarchy and association.

These formats let an ontology state explicitly what a language model must otherwise infer from text. They record hierarchy, so that type 2 diabetes is marked as a kind of diabetes mellitus. They record exclusions, so that type 2 diabetes is kept distinct from type 1 and is not confused with it. They record synonymy, so that one concept carries several labels at once. They also record associative relations, such as the complications of a disease, its treatments, and links that cross from one ontology to another.

This structured knowledge is rich, but it is stored as graphs rather than as the running text that language models expect, which raises the question of how to bring it into pretraining.

5.6 Knowledge-Enhanced Pretraining

Several lines of work try to close this gap. They differ in how much of the graph they bring into the model and at what stage of training they do it.

5.6.1 Static Graph Embeddings

The first family turns a graph into fixed vectors, one per node, without any context. Node2Vec learns node representations by running random walks on a graph and feeding the resulting sequences to a skip-gram model, the same kind of model used for word embeddings (Grover and Leskovec, 2016). RDF2Vec adapts this walk-plus-skip-gram idea to RDF graphs, and it became the basis for later walk-based ontology methods (Ristoski and Paulheim, 2016). Snomed2Vec applies Node2Vec to SNOMED-CT and reports a large gain on concept similarity (Agarwal et al., 2019). OWL2Vec* extends the approach by running walks over OWL ontologies (Chen et al., 2021). These methods share one limit. The embeddings are static and live outside any transformer, so a concept gets a single fixed vector whatever the surrounding text says.

5.6.2 Injecting Knowledge into Transformers

A second family puts entities inside the model. ERNIE aligns entity embeddings with text representations during pretraining (Zhang et al., 2019). KnowBERT inserts entity linkers as extra attention layers within BERT (Peters et al., 2019). K-BERT injects knowledge-graph triples into the input using soft-position information, so that the added triples do not disturb the sentence order (Liu et al., 2020). A later generation pretrains on language and knowledge together. KEPLER trains masked language modeling and a knowledge-embedding objective at the same time on Wikidata, with one shared encoder (Wang et al., 2021). CoLAKE builds word-knowledge graphs that unify text and knowledge context, then pretrains a modified masked objective on them (Sun et al., 2020). DRAGON jointly pretrains, in both directions, masked language modeling and link prediction over PubMed text and the UMLS graph. It reports about 3% higher accuracy on a medical question-answering benchmark, with larger gains on questions that need several reasoning steps (Yasunaga et al., 2022a).

5.6.3 Medical Contrastive Approaches

A third family uses contrastive learning to pull related medical terms together in the embedding space. SapBERT does self-alignment pretraining on UMLS synonyms. It treats synonyms as positive pairs and unrelated terms as negatives, scales to millions of concepts, and reaches strong biomedical entity-linking results with a single model for all entity types (Liu et al., 2021). CODER extends this idea by adding relation triples from UMLS on top of synonyms, which helps with medical term normalization (Yuan et al., 2022b). Both methods read the graph through pairs of terms rather than through full sentences.

5.6.4 Graph-to-Text for Pretraining

A recent direction turns the graph into text that a language model can read directly. The "Textbooks Are All You Need" work showed that high-quality, textbook-like synthetic data lets small models perform well. This suggests that the structure and quality of the data can matter more than its raw volume (Gunasekar et al., 2023). EntiGraph follows this line with an entity-centric augmentation method. It breaks a corpus into entities and generates text about the relations between them, turning about 1.3 million real tokens into about 455 million synthetic tokens. Accuracy scales roughly log-linearly in the number of synthetic tokens. The method targets decoder models and has not yet been tested on encoders (Yang et al., 2025b).

Several gaps remain open across these families. Most of the work is in English and built on UMLS, so other languages and other knowledge sources stay unexplored. Graph-to-text for pretraining is recent, with the main results appearing between 2023 and 2025, and it has not been studied for bidirectional encoders, since DRAGON uses link prediction rather than generated text. National ontologies, such as the French ICD-10 maintained by the ATIH and the CCAM procedure coding, have not been used for pretraining. Contrastive methods such as SapBERT and CODER use UMLS through pairs of terms rather than by generating text from the graph. These open problems set up the limits we discuss next.

5.7 Limits and Transition

This chapter examined the data behind pretrained language models. We looked at web-scale corpora and how they are collected, at quality filtering and curation, at biomedical and clinical sources, at medical ontologies, and at ways to add structured knowledge to pretraining. Together, these threads leave several open problems for biomedical text in a lower-resource language such as French.

The first problem is scarcity outside English. The biomedical text drawn from PMC is about 98% English (Labrak et al., 2024b), and French biomedical corpora remain small and scattered. The main filtering pipelines that made web-scale curation work, such as FineWeb-Edu and DCLM, were built for English (Penedo et al., 2024; Li et al., 2024). How to collect and filter text for a specific domain in a language with little data is still an open question.

The second problem is the granularity of filtering. Existing biomedical curation works at the document or article level, as in Meditron (Chen et al., 2023), which leaves the mix of relevant and irrelevant content within an article untouched. Finer, paragraph-level filtering has received little attention.

The third problem is that quality criteria do not transfer across domains. The educational criterion used by FineWeb-Edu does not match biomedical text such as clinical cases, drug leaflets, and protocols. The domain biases identified by WebOrganizer (Wettig et al., 2025) may also take a different shape in the biomedical case. The right domain-specific quality criteria for biomedical text are still unknown.

Two further gaps concern knowledge. Research on medical ontologies is dominated by UMLS, but French hospital systems use ICD-10 as maintained by the ATIH, along with CCAM for procedures and ATC for drugs. No pretraining work has used these national ontologies, which is a clear gap for medical coding in French. At the same time, turning ontology graphs into text for pretraining is recent, and the existing work targets decoder models in English, as in EntiGraph (Yang et al., 2025b). Whether textbook-like text generated from a national ontology could help has not been tested for bidirectional encoders or for non-English ontologies.

This chapter covered the data used to pretrain and adapt language models. The next chapter turns to the task these models are ultimately built for, clinical information extraction, and to how pretrained models are put to work on named entity recognition, medical coding, and normalization.

6 Clinical Information Extraction

6.1 Introduction

The previous chapters built pretrained language models and asked what data they should learn from. This chapter turns to the task these models are built for. Clinical information extraction means turning unstructured clinical text into structured information that a computer can store, query, and analyse. A discharge summary, a referral letter, or a prescription is written for human readers, so the facts it contains are not directly available to software. The goal is to recover those facts, such as diagnoses, treatments, and their relations, and to record them in a form that supports research and care.

This task matters because hospitals increasingly store their records in clinical data warehouses, large databases that gather electronic health records for secondary use. Most of the clinically useful information in these records lives in free text rather than in structured fields. Most medical concepts appear only in the free text, which leaves reports, letters, and prescriptions as the main source (Raghavan et al., 2014). France adds a further reason to study the problem. The PMSI, the national system that describes hospital activity, requires every hospital stay to be coded with ICD-10 diagnosis codes and CCAM procedure codes, covering about 30 million stays a year. This coding is still done by hand by professional coders and reported to the ATIH, the national agency for hospital information. This makes automation a practical and economic priority.

Clinical information extraction covers a family of related tasks. Named entity recognition finds the spans of text that mention entities of interest, such as a drug name or a symptom. Relation extraction identifies how these entities are connected, for example which drug treats which condition. Medical coding assigns ontology codes to a document, such as ICD-10, CCAM, or ATC codes for medications. Entity linking, also called normalisation, maps a mention in the text to a single concept in a knowledge base such as the UMLS or SNOMED-CT. These tasks are often combined in a pipeline, since a coded record usually needs both the entities and the concepts they refer to.

These tasks are the right place to close the related work, because they are where the pretrained models of the earlier chapters are finally put to use and judged. Once a model has been built and adapted, its value shows in how well it supports named entity recognition, coding, and normalisation on real clinical text. The difficulties that recur across these tasks also frame the open problems of the field. Labelled clinical data are scarce, because the text is sensitive and annotation requires medical expertise. The label space can be very large, since coding systems contain tens of thousands of codes. Evaluation is not standardised, which makes results hard to compare across datasets and studies.

The chapter follows the structure of these tasks. It starts from the fundamentals of named entity recognition, then reviews biomedical and clinical benchmarks. It moves on to medical coding and to entity linking, before turning to zero-shot approaches and to the use of synthetic data. It closes with a discussion of how clinical information extraction is evaluated.

6.2 Named Entity Recognition

Named entity recognition (NER) is the task of finding entity mentions in text and classifying them into predefined types. It is often the first step in an extraction pipeline, since later tasks such as relation extraction and entity linking build on the entities that NER returns. The following subsections describe how the task has been formulated over time, from sequence labeling with handcrafted models to pretrained encoders and span-based methods.

6.2.1 Sequence Labeling

The classic way to formulate NER is as a sequence labeling problem with the BIO tagging scheme. Each token in the text receives one label: `B-TYPE` marks the beginning of an entity of a given type, `I-TYPE` marks its continuation, and `O` marks a token outside any entity. A common variant, BILOU, adds finer tags for the last token of an entity and for single-token entities. These labels turn the problem of marking spans into a problem of classifying tokens one by one.

Token labels are not independent, since some label sequences are not valid. An `I-DISEASE` label should not follow a `B-TREATMENT` label, for example, because the two belong to different entities. Conditional random fields (Lafferty et al., 2001) model the dependencies between consecutive labels and assign a score to the whole sequence, which lets them avoid these invalid transitions. They were the standard method for NER before deep learning. The BiLSTM-CRF model later combined two parts: a bidirectional long short-term memory network, which builds a contextual representation of each token, and a CRF decoding layer on top (Lample et al., 2016). It also used character-level embeddings to capture word morphology, and it was the state of the art before BERT.

6.2.2 BERT-based NER

With BERT (Devlin et al., 2019), NER becomes a pretrained encoder with a linear classification layer on each token, and the whole model is fine-tuned end to end. This approach needs no handcrafted features. It often needs no CRF either, though whether a CRF layer still helps is debated and the reported gain is usually small. The pretrained encoder brings general linguistic knowledge to the task through transfer learning, so the model starts from useful representations rather than from random weights.

The strength of this approach comes from the deep contextual representations that BERT produces. The same word takes a different vector in a different context, so the word "cancer" in "lung cancer" and in "breast cancer" is represented differently. This sensitivity to context helps the classifier separate entity types that depend on the surrounding words. The result is a simple and strong baseline that has become the default for general-domain NER.

6.2.3 Beyond Token-Level Labeling

Span-based NER takes a different view of the task. Instead of labeling each token, it enumerates candidate spans of text and classifies each span as an entity of some type or as no entity. This avoids the constraints of token-by-token tagging and handles nested entities, where one entity sits inside another. Nested entities are common in biomedicine. In "right lung cancer", an anatomy entity ("right lung") sits inside a disease entity ("right lung cancer"), and the BIO scheme cannot give a token two labels at once.

These methods work well on general-domain text, where entities such as people, places, and organizations follow fairly regular patterns. The biomedical domain raises its own difficulties, including a large and technical vocabulary, frequent nested entities, negation, and a telegraphic clinical style, which the next section takes up.

6.3 Biomedical and Clinical NER

6.3.1 English Benchmarks

The biomedical NER literature in English rests on a stable set of benchmarks. Scientific-text NER is mostly evaluated on PubMed abstracts. BC5CDR annotates chemical and disease entities in 1500 PubMed articles (Li et al., 2016), NCBI Disease marks disease mentions in 793 abstracts (Doğan et al., 2014), and JNLPBA tags genes and proteins in the GENIA corpus with five entity types (Collier et al., 2004). Other corpora target narrower entity sets, such as AnatEM for anatomical entities (Pyysalo and Ananiadou, 2014) and ChemProt for chemical-protein interactions, which is a relation-extraction task rather than plain NER (Krallinger et al., 2017). These datasets cover the language of scientific articles, where sentences are mostly complete and well edited.

Clinical-text NER is evaluated on a separate family of shared tasks built from hospital notes. The i2b2 challenges, later renamed n2c2, ran several editions on clinical text. The 2010 i2b2/VA challenge covered concepts, assertions, and relations in clinical records (Uzuner et al., 2011), and the 2018 n2c2 task addressed adverse drug events and medication extraction (Henry et al., 2020). A few benchmarks frame biomedical extraction as classification rather than span labelling, such as GAD for gene-disease associations as a binary task (Bravo et al., 2015) and the hallmarks-of-cancer multi-label classification (Baker et al., 2016). To make these scattered tasks comparable, BLURB collected thirteen biomedical datasets into one benchmark, and PubMedBERT was the first model evaluated on it (Gu et al., 2022).

6.3.2 French Benchmarks

French biomedical NER has fewer resources, but several are now standard. QUAERO is the main reference. It has an EMEA set of drug leaflets and a MEDLINE set of scientific titles, annotated with ten UMLS semantic groups and including nested entities (Névéol et al., 2014). CAS provides clinical cases drawn from scientific articles and was used as task 3 of DEFT 2020, with a two-class version (CAS1) and an eight-class version (CAS2) (Cardon et al., 2020). E3C is a multilingual corpus of clinical cases annotated with a single clinical-entity class (Magnini et al., 2020). The CLEF eHealth 2016 campaign combined NER and normalisation on QUAERO using the BRATeval tool, and added ICD-10 coding of death certificates (Névéol et al., 2016).

These corpora were long used in isolation, which made models hard to compare. DrBenchmark was the first unified French biomedical benchmark, gathering twenty tasks over twelve datasets so that models could be compared under one protocol (Labrak et al., 2024a). Its results show that biomedical pretraining helps on most of these tasks, though no single model is best on all of them.

6.3.3 Specifics of Clinical Text

Clinical text differs from both scientific text and general web text in ways that affect extraction. It is telegraphic, with incomplete sentences and lists of symptoms, and it is full of non-standard abbreviations, such as IDM for myocardial infarction and HTA for arterial hypertension in French. Negation is central, since a phrase like “no sign of pneumonia” mentions pneumonia while ruling it out, and the rule-based NegEx algorithm is still used as a baseline for detecting it (Chapman et al., 2001). Temporality also matters, since a model must separate medical history, current diagnosis, and hypothesis. Expressions of uncertainty such as “probable” or “to be ruled out" are common. Spelling errors are frequent as well, because clinicians type quickly under time pressure.

For these reasons, models trained on scientific or general text perform poorly on clinical text, which is why domain-pretrained models were developed (discussed in Section 4.7). These specialised models help, but they remain supervised, so each target class still needs annotated examples. One architectural alternative is ClinicalMamba, which adapts state-space models to clinical text with linear-time complexity in sequence length, useful for long clinical documents (Yang et al., 2024).

We now have benchmarks and models for biomedical NER, but medical coding is not simply NER, because it assigns codes from an ontology with thousands of possible labels to a whole document.

6.4 Medical Coding

6.4.1 The Problem

Medical coding assigns standardised codes to a whole clinical document. The codes come from controlled systems such as ICD-10 for diagnoses, CCAM for procedures, and ATC for drugs. In France, the PMSI requires that each hospital stay be coded with a principal diagnosis, a set of associated diagnoses, and the CCAM procedures performed. This covers about 30 million stays a year, and the coding is done by hand. The codes feed both hospital funding and national health statistics, so a coding error has financial and epidemiological consequences.

The task is hard for several reasons. The label space is very large, with about 17,000 codes in the French version of ICD-10, about 8,000 CCAM codes, and about 6,000 ATC codes. The documents are long, since a discharge summary runs from 1000 to 3000 tokens, well beyond the 512-token limit of BERT. The code distribution is very skewed, with a few codes that appear often and a long tail of codes that appear rarely. Finally, a single stay is multi-label, since one principal diagnosis and several associated diagnoses must be predicted at once.

6.4.2 Approaches

Early work treated coding as multi-label classification with attention over the document. Mullenbach et al. (2018) proposed CAML, a convolutional model with one attention head per label, so each code attends to its own passages of text. This per-label attention is also explainable, since it shows which words motivated each predicted code. Cao et al. (2020) extended this idea with HyperCore. It represents the ICD hierarchy with hyperbolic embeddings, since hyperbolic space fits tree structures better than Euclidean space, and it adds a graph of code co-occurrences.

Pretrained language models then entered the field. Huang et al. (2022) introduced PLM-ICD, the first to make pretrained encoders work competitively for ICD coding, with label attention over the model representations and segment pooling to handle long documents. Its main finding is that a better pretrained encoder leads to better coding, so pretraining quality matters directly for this task. Yuan et al. (2022a) took a different route with MSMN, a multiple-synonym matching network that uses UMLS synonyms for each code. This helps because clinical text uses varied expressions for one concept, and a synonym table lets the model match these variants to the same code.

Large language models do poorly on coding. Soroush et al. (2024) benchmarked large language models for medical coding and found them highly error-prone. Specialised approaches such as PLM-ICD and MSMN remain stronger, which reinforces the case for specialised encoders over generic large models.

6.4.3 Coding in French

French resources for coding are scarce. Pignat et al. (2025) released FRACCO, a French oncology corpus of 1301 synthetic clinical cases annotated with ICD-O codes. Zaghir et al. (2024) released FRASIMED, a French adaptation of Spanish coding corpora with about 2000 clinical documents. The CLEF eHealth 2016 shared task also provided ICD-10 coding of French death certificates. These corpora are small and cover narrow domains, so they support evaluation more than large-scale training.

Coding works on English benchmarks such as MIMIC, but the label space is huge, and annotating examples for every one of the thousands of codes is impossible. French resources are scarcer still. This motivates approaches that generalise to labels never seen in training, and before reaching those, we first ask how to connect a mention in the text to a concept.

6.5 Entity Linking and Normalization

6.5.1 The Problem

Entity linking maps a mention found in text to an entity in a knowledge base such as UMLS, SNOMED-CT, or ICD-10. The mention is a span of text, and the target is a single concept with a stable identifier. A related task, normalization, maps the many surface forms of a concept to one canonical entry. The phrases “myocardial infarction”, “MI”, and “heart attack” all describe the same condition, and normalization sends them to a single ICD-10 concept. Both tasks deal with the gap between how clinicians write and how knowledge bases store information.

Entity linking differs from medical coding in scope. Linking works at the level of a single mention, where each mention receives one concept. Coding works at the level of a whole document, where a report receives a set of codes that summarize the encounter. The two tasks are connected, and linking can be seen as a building block of coding. A typical pipeline first recognizes entities with named entity recognition, then links each mention to a concept, and finally aggregates these concepts into document-level codes. Strong linking is therefore useful well beyond the linking task on its own.

6.5.2 Approaches

Several biomedical encoders are used directly for this task. SapBERT gives strong entity linking with a single model that covers all entity types (Liu et al., 2021), and CODER adds relation information to improve term normalization (Yuan et al., 2022b). We described how both models are pretrained in Section 5.6, so here we note only their role as ready encoders for matching mentions to concepts.

BioSyn takes a more structured view of normalization (Sung et al., 2020). It represents a mention with both a sparse and a dense encoder and adds their scores to retrieve candidate concepts. Because one concept often has several synonyms, BioSyn aggregates scores across the synonyms of a candidate using a marginal objective during training, and it refreshes the hard candidates as training proceeds. This synonym marginalization is what improves the final results.

These methods share clear limits. Most of the work is done in English and built on UMLS. UMLS covers English vocabulary well, but it matches French clinical language poorly, with its informal abbreviations and local synonyms. The methods also treat each concept in isolation. They do not use the structure of the ontology, such as parent concepts, exclusions, and complications, and they ignore the RDF and SKOS relations that connect concepts, including broader, narrower, related, and exactMatch links.

These methods can normalize a mention once it has been found and once its label exists in training. But clinical extraction needs to recognize and normalize text across thousands of concepts never seen during training.

6.6 Zero-shot and Open NER

6.6.1 Why Zero-shot

Medical label sets are not fixed. New diagnostic codes are added, new clinical trials define new eligibility criteria, and new diseases appear, as the arrival of COVID-19 showed. A model that extracts information from clinical text must therefore handle labels that did not exist when it was trained. Collecting annotated examples for every possible label is not realistic. The ICD-10 classification alone contains about 17,000 codes, and annotating clinical text for each one would take more effort than any project can afford. Classic named entity recognition models do not help here, because a BERT model with a linear classification head has a fixed number of output classes chosen before training. This situation motivates models that can recognise entity types they have never seen during training.

6.6.2 GLiNER

GLiNER treats named entity recognition as a matching problem between text spans and natural-language descriptions of entity types (Zaratiana et al., 2024). A single bidirectional transformer encodes the input text and the candidate entity types together. The types are written as natural language, such as "disease" or "drug name", rather than as fixed numeric labels, so a user can ask for any type at inference time. The model places span representations and type representations in a shared space and matches them by similarity. GLiNER reaches higher zero-shot scores than ChatGPT while using up to 300 million parameters, which is far smaller than a large language model. GLiNER2 extends this design into a single framework that covers named entity recognition, classification, and structured extraction in one model (Zaratiana et al., 2025). A biomedical version, GLiNER-BioMed (Yazdani et al., 2026), fine-tunes the approach on biomedical entity recognition datasets and reports gains over the general GLiNER on biomedical benchmarks. This result shows that domain adaptation still helps even for a zero-shot model. The quality of such models depends heavily on their training data. UniversalNER builds its training set by asking ChatGPT to annotate text from the Pile, a large general corpus (Zhou et al., 2024). This yields about 45,000 instruction examples covering some 13,000 entity types, and a model tuned on this data beats ChatGPT on named entity recognition by several points.

6.6.3 Other Approaches

GoLLIE takes a generative route. A large language model reads natural-language annotation guidelines and produces structured annotations that follow them (Sainz et al., 2024). Because the guidelines describe the entity types in words, the model works zero-shot and can follow detailed definitions, at the cost of being larger and slower than an encoder. NuNER is a compact named entity recognition foundation model built on a RoBERTa-base encoder (Bogdanov et al., 2024). It is pretrained with contrastive learning on GPT-3.5 annotations of the C4 web corpus, and its authors find that the size and diversity of this training data matter more than any other choice. Large language models have also been applied directly to clinical named entity recognition. Early work was encouraging (Agrawal et al., 2022), but Lehman et al. (2023) argued that BERT-style models remain competitive on extractive clinical tasks, that large models are hard to run inside a hospital, and that sending clinical text to a proprietary API raises confidentiality problems.

6.6.4 Limits of Current Zero-shot

Several open problems remain. GLiNER and similar models are pretrained on general data such as the Pile, not on clinical text, so they carry the same domain gap that motivated specialised biomedical models. Entity types are given as short text descriptions, which do not capture the hierarchy of a medical ontology. The relations between codes, such as parent and child, mutual exclusion, or synonymy, are not used by the model. Performance also drops on the rare codes that make up the long tail of medical classifications, where examples are scarcest. Zero-shot named entity recognition is a promising answer to the problem of thousands of unseen labels. But the strongest models are trained on general web text, while clinical data are private and cannot be shared. This raises the question of how to train for the clinical domain without access to real clinical data.

6.7 Synthetic Data for Clinical NLP

6.7.1 The Scarcity Problem

Clinical text is hard to obtain because it is private. In France, the CNIL regulates the use of patient data, and in the United States HIPAA plays the same role. A model trained on the records of one hospital carries information about its patients, so it cannot be shared freely with another institution. This is the central obstacle for clinical information extraction: the data that matters most cannot leave the hospital where it was produced. Models and their training corpora fall under the same rules, which makes collaboration between hospitals difficult.

Public annotated French clinical resources exist, but they are small and not fully clinical. Datasets such as QUAERO, CAS, and E3C cover only a few thousand documents, and much of their text is scientific or drawn from drug leaflets rather than written by clinicians at the bedside. The larger accessible resources are English. MIMIC, the most widely used clinical corpus, comes from a single hospital in the United States and reflects its language and coding practices. For French clinical work there is no large public corpus that matches the style of real hospital notes, so researchers look for ways to train models without real clinical data.

6.7.2 LLM-as-Annotator

One response uses a large language model to annotate data and then trains a smaller model on those annotations. The idea appears in several places earlier in this related work. FineWeb-Edu uses a language model to score documents for educational quality (Penedo et al., 2024). UniversalNER distils annotations produced by ChatGPT into a smaller model for named entity recognition (Zhou et al., 2024). NuNER distils annotations of web text produced by GPT-3.5, again for named entity recognition (Bogdanov et al., 2024). In each case the large model supplies labels that would otherwise require human annotators.

For the clinical domain this pattern is attractive for a practical reason. The large model holds broad medical knowledge and can label text without supervision, while the small model it teaches is cheap to run and can be deployed inside a hospital where the data must stay. The expensive model does the labelling once, and the model that is deployed inherits much of its behaviour. The quality of the small model then depends on how reliable the large model is as an annotator, which is not guaranteed for technical clinical content.

6.7.3 Generating Synthetic Clinical Notes

A second route generates clinical notes directly with language models, instead of annotating existing text. Ive et al. (2020) generated synthetic psychiatric records and found that models trained on the synthetic notes reached performance comparable to models trained on real notes. This was an early sign that generated text can stand in for real text in clinical NLP. Chintagunta et al. (2021) used GPT-3 to augment medical-dialogue data. A small set of annotated examples together with generated variants matched the performance of a much larger annotated set, with an ensemble used to filter low-quality generations. Asclepius pushed the idea further by training a clinical language model entirely on synthetic notes. It used a large model to turn public case reports into realistic clinical notes and used no private clinical data. The result was competitive with models trained on real notes (Kweon et al., 2024).

Synthetic notes built this way still tend to read as too clean and complete, and they lack the terse, telegraphic style of real clinical writing. The generator can also produce stereotyped cases that under-represent rare presentations and unusual patients. When a large model supplies the labels rather than the text, those labels are not gold-standard expert annotations, and errors in them pass on to the model that learns from them. These caveats temper the otherwise encouraging finding that generated data can replace real data.

Synthetic data is one answer to the scarcity of clinical text, but it raises a further question: how do we evaluate a model trained on generated data correctly, and more generally how do we compare models that use different tokenizers on a fair footing?

6.8 Evaluation

Before any model can be compared with another, we need to agree on how its predictions are scored. The way we measure performance shapes which model looks best, and the field has not settled on one way to measure it. This section reviews the metrics used for clinical information extraction, shows how the choice of tokenizer leaks into the score, and explains why a shared evaluation protocol is still missing.

6.8.1 NER Metrics

The most common metric for named entity recognition is strict entity-level evaluation, implemented in the seqeval tool (Nakayama, 2018). Under this metric an entity counts as correct only when its type and its boundaries are both exact, read off from tags in the IOB2 scheme. Scores can then be aggregated in two ways. The micro-average pools all entities together, so frequent classes carry the most weight. The macro-average computes a score per class and then averages those scores, so every class counts equally, however often it appears. Strict entity-level seqeval has become the de facto standard for named entity recognition, and most recent French biomedical work reports it.

A second tool, BRATeval, was used in the CLEF eHealth 2016 shared task (Névéol et al., 2016). It scores predictions against exact character offsets in the original text. Because it works on character positions rather than on tokens, it does not depend on how the text was split or otherwise preprocessed. This makes it robust to tokenization. It is used far less often than seqeval, so it gives fewer points of comparison across papers.

A third approach scores each token on its own and reports a weighted F1 over all tokens. Here every token receives a label and is evaluated separately, instead of being grouped into entities. The problem is that the "O" label, which marks tokens that belong to no entity, is by far the most frequent label. It therefore dominates the average and inflates the reported score. This token-level weighted F1 was the metric used by DrBERT (Labrak et al., 2023), which makes its numbers hard to compare with models scored under entity-level seqeval.

6.8.2 The Effect of Tokenization

Token-level metrics carry a further problem. Different tokenizers split the same text into different numbers of pieces, so a score computed per token depends on the tokenizer used. The two vocabularies can diverge sharply: SciBERT found only a 42% overlap between the general BERT vocabulary and a vocabulary built from scientific text (Beltagy et al., 2019). A tokenizer that over-splits technical terms produces more tokens to predict, and therefore more chances to make an error. The word "échocardiographie" is a clear case. A general tokenizer breaks it into three pieces, while a specialised one breaks it into two. This changes the number of predictions for the same word, and so changes the score. Token-level metrics are biased by the choice of tokenizer, and that bias is hard to separate from genuine differences in model quality.

6.8.3 The Need for a Unified Benchmark

These problems are made worse by the absence of a shared protocol. Each paper tends to use its own datasets, its own metric, its own train and test splits, and its own hyperparameters. There is no standard protocol for French biomedical NLP that fixes these choices in advance. Many of the choices change the result on their own: a CRF layer versus a linear classification head, frozen versus full fine-tuning of the encoder, and micro versus macro F1 can each move the ranking of two models.

As a result, comparison across published systems is difficult and often unfair, since differences in the protocol are easily mistaken for differences in the models.

DrBenchmark was a step toward a shared protocol, gathering several French biomedical tasks under a common evaluation (Labrak et al., 2024a). Comparing very different architectures is harder still. Encoders, large language models, and GLiNER-style bi-encoders use different tokenizers and different fine-tuning procedures, so a token-level or even an entity-level score computed on one model's tokenization does not transfer cleanly to another. This points toward evaluation at the character level, which is independent of the tokenizer and so allows models of any architecture to be placed on the same scale. Such an evaluation is an open need rather than a solved problem, and the closing section takes up this missing piece of a fair, standardised comparison.

6.9 Limits and Transition

This chapter reviewed how clinical information extraction is done in French, from the tasks and corpora to the models that perform them. Across these topics, the same difficulties came back. They are not isolated problems with one method. They are structural features of the setting, and together they define the questions that the rest of this thesis takes up.

The first difficulty is the scarcity of clinical data. French hospital text is personal data, and the CNIL places strict conditions on its use. A model trained inside a hospital cannot be shared with other institutions, so work done on private corpora stays local and hard to reproduce. The public French resources that remain, such as QUAERO, CAS, and E3C, are small, and most of their content is scientific writing rather than clinical notes. This shortage of shareable clinical text pushes the field to look for alternatives to real hospital data.

The second difficulty is the size of the label space. Clinical coding relies on systems such as ICD-10 and CCAM, whose tens of thousands of codes (Section 6.4) make it infeasible to annotate examples for every label. A classifier with a fixed output head has one slot per label. It cannot handle codes it never saw in training, so the classic supervised recipe does not scale to this setting. The field therefore needs methods that generalise to labels for which no annotated example exists.

The third difficulty is that zero-shot extraction is not adapted to the clinic. Models such as GLiNER and UniversalNER recognise entities described by free-text labels, which removes the fixed head, but they are trained on general web text. On clinical writing, and especially on rare codes, their accuracy drops. Closing this gap requires adapting these models to the biomedical domain.

The fourth difficulty is that ontologies are underused. Coding systems are published as structured resources in formats such as RDF and SKOS, with a hierarchy of concepts, exclusion rules, and lists of synonyms. Zero-shot models reduce each label to a short description and ignore this structure. The field needs ways to bring ontological knowledge into the representation of labels.

The fifth difficulty is that evaluation is not standardised. There is no shared benchmark for French biomedical natural language processing. Token-level scores depend on how each model splits text, so the tokenizer biases the comparison, and protocols differ from one paper to the next. Fair comparison calls for an evaluation that does not depend on the tokenizer.

A final point runs underneath all of these. Much recent attention has gone to large decoder models, yet for extractive clinical tasks encoder models remain competitive. They are also small enough to run on the hardware available inside a hospital, which matters when the data cannot be moved off-site. This keeps encoders at the centre of the work that follows.

These three survey chapters covered the language models used in biomedical natural language processing, the data used to pretrain them, and the clinical tasks they are built for. The gaps above set the agenda for the contributions of this thesis. The next part begins with the first of them, the construction of a biomedical corpus for French from public sources.

Part III

Building a Biomedical Corpus

7 Collecting Biomedical Text

7.1 Introduction

Hospitals produce millions of clinical documents every year. These can be used to train language models internally, but the resulting models inherit the legal constraints of their training data: they cannot be released publicly or exchanged between institutions. For a model that is truly open, the training data must come from somewhere else.

Yet hospital records remain the richest source of medical information. Clinical data warehouses have made them increasingly accessible for research purposes. It is estimated that up to 80% of entities are missing from structured modalities (Raghavan et al., 2014), making clinical reports the richest source of medical information. Yet extracting this information requires high-performing language models adapted to the biomedical domain.

BERT-based models (Devlin et al., 2019) consistently demonstrate state-of-the-art results for a wide range of natural language processing tasks. The adaptation of BERT to the French language, particularly with the CamemBERT model (Martin et al., 2020), has replicated these performances in French natural language processing. However, the results of this model on biomedical data are disappointing (Cardon et al., 2020), as it exhibits lower performance compared to heuristic models on certain evaluation datasets. These results are predictable because CamemBERT is trained on plain language, often sourced from web pages such as forums. Biomedical data, especially clinical data, are significantly different. They contain technical terms that are very rare or absent in everyday language, and they have a radically distinct style, often telegraphic, rarely consisting of complete sentences, with varying abbreviations.

In this chapter, we address this gap by introducing *biomed-fr*, a public French biomedical corpus of 413 million words assembled from three complementary sources. We describe the corpus construction, analyze the vocabulary mismatch between general and biomedical French tokenizers, and situate our effort among existing French biomedical corpora. The use of this corpus for continual pretraining of CamemBERT, yielding CamemBERT-bio, is presented in Chapter 10.

Authors	Source	Size	Public
Copara et al. (2020)	PubMed	31K docs	Yes
Le Clercq de Lannoy et al. (2022)	Misc.	136M words	Partial
Dura et al. (2022)	APHP CDW	21M docs	No
Labrak et al. (2023)	Misc.	7.4 GB	Partial

Table 7.1: French biomedical corpora used for language model adaptation. Size units vary across works as reported by their respective authors.

7.2 Existing French Biomedical Corpora

Several works have attempted to adapt CamemBERT to the biomedical domain, each relying on different training data. Table 7.1 summarizes these corpora.

Copara et al. (2020) explored a second phase of pre-training on 31,000 French biomedical scientific articles. However, they did not observe a significant improvement on a clinical named entity recognition task using the large version of CamemBERT. This could be explained by the combination of a relatively small corpus (31K documents compared to the 18 billion words in BioBERT) and the large version of the CamemBERT model.

Le Clercq de Lannoy et al. (2022) also adapted CamemBERT to the biomedical domain. They aggregated documents from various sources, including PubMed, Cochrane, ISTEX, and Wikipedia, forming a larger partially public corpus of approximately 136 million words. They observed a 2-point improvement in F-measure on a named entity recognition evaluation set composed of drug notices (EMEA), but no significant improvement on a set composed of scientific article titles (MEDLINE).

Dura et al. (2022) continued the pre-training of CamemBERT on 21 million clinical documents from the APHP (Assistance Publique – Hôpitaux de Paris) clinical data warehouse. They observed a significant 3% improvement on APMed, a private clinical named entity recognition dataset owned by APHP. They also achieved similar scores to CamemBERT on EMEA and MEDLINE. Their new model performs better on clinical data, yet it obtains scores similar to CamemBERT in other biomedical domains.

Labrak et al. (2023) introduced a public French biomedical model named DrBERT. Through their experiments, they explored both continual-pretraining and from-scratch training strategies. Their findings indicated a superior performance when training from scratch, suggesting that continual-pretraining with CamemBERT for French biomedical data may not be as effective. However, when applied to PubMedBERT, continual-pretraining yielded results nearly on par with the from-scratch approach.

Finally, Berhe et al. (2023) introduced AliBERT, which was trained on a French biomedical corpus primarily comprising articles from ScienceDirect and theses collected through Sudoc. The model leverages a new regularized Unigram-based

Source	Size (M words)	Content
ISTEX	276	Scientific articles
CLEAR	73	Drug leaflets
E3C	64	Clinical cases, leaflets
Total	**413**	

Table 7.2: Composition of the *biomed-fr* corpus (in millions of words).

tokenizer and underwent extensive training on 48 GPUs for a total duration of 20 hours. Their pretrained model outperforms notable French non-domain-specific models, such as CamemBERT and FlauBERT, in two biomedical downstream tasks. Unfortunately, the model is not currently available.

7.3 The biomed-fr Corpus

We built a French biomedical corpus composed exclusively of public documents to minimize the usage constraints mentioned earlier. The documents come from three different sources (see Table 7.2), the main one being ISTEX. This new corpus, named *biomed-fr*, consists of 413 million words, equivalent to 2.7 GB of data. Martin et al. (2020) have shown that with only 4 GB of data, it is possible to achieve performance almost comparable to the model trained with the 138 GB OSCAR dataset (Ortiz Suárez et al., 2019). For an adaptation of CamemBERT to the biomedical domain, this amount of data can be considered sufficient.

ISTEX. The ISTEX database contains references to 27 million scientific publications. We extracted 108,183 French documents published in a biology or medical journal since 1990. Articles published before this date often contain numerous typographical errors, as they are often scanned articles that require optical character recognition algorithms, resulting in a certain number of errors. Such errors are found to a lesser extent in articles published after 1990. Some documents, although in French, contain passages in English. Therefore, there is an indeterminate amount of English in this corpus. However, it is unlikely that this will significantly impact pre-training. Typographical errors and the presence of other languages are aspects that can be addressed in future versions of *biomed-fr*.

CLEAR. The CLEAR corpus (Grabar and Cardon, 2018) consists of encyclopedia articles, drug leaflets, and abstracts of scientific articles. Each document is available in two versions: one in technical language and the other in simplified language. We retrieved all of these documents in both versions. Regarding the drug leaflets, we removed redundant sentences at the beginning and end of each document, such as

Term	General	Specialized
échocardiographie	écho-cardi-ographie	échocardiographi-e
transthoracique	trans-thorac-ique	trans-thoracique
glimépiride	g-lim-épi-ride	gli-m-épi-ride
cardiopathie	cardio-pathie	cardiopathie
diastoliques	dia-s-tol-iques	diastolique-s

Table 7.3: Comparison of tokenization between a general-purpose tokenizer and a specialized tokenizer for some biomedical technical terms. Hyphens indicate subword boundaries.

the website navigation bar from which the documents were extracted or information about the company selling the documents.

E3C. This corpus (Magnini et al., 2020) is composed of three layers. The first two layers are annotated or semi-annotated and will be used for evaluation in Chapter 10. The last layer is not annotated, and that is the one we retrieved for *biomed-fr*. It consists of medical specialty admission competitions, drug leaflets, and medical thesis abstracts. There may be duplicates of some leaflets found in the CLEAR corpus.

biomed-fr-small. By randomly selecting 10% of content from *biomed-fr*, we created a smaller corpus called *biomed-fr-small*. This subset allows us to study the impact of corpus size on downstream performance. Results of this comparison are presented in Chapter 10.

7.4 Vocabulary Analysis

CamemBERT-bio is a biomedical-adapted model based on CamemBERT. Unlike a new model trained from scratch, it shares the same vocabulary. The vocabulary of CamemBERT was constructed using SentencePiece (Kudo and Richardson, 2018) on an OSCAR sample. Therefore, it is a general-purpose vocabulary designed for everyday language. We can hypothesize that the tokenization of CamemBERT may result in oversplitting of biomedical technical terms.

To investigate this possibility, we trained a specialized tokenizer on *biomed-fr-small* and calculated the intersection of the two vocabularies.

We find a 45% intersection between the two vocabularies (Table 7.3), which is quite close to the 42% intersection found by Beltagy et al. (2019) between the vocabulary of BERT and SciBERT. Therefore, there is a significant difference in the most frequent terms.

The relatively modest performance gain of PubMedBERT compared to BioBERT and the similar performance of DrBERT compared to CamemBERT-bio despite their specialized vocabulary, together with the over-segmentation of technical terms and the low intersection rate, demonstrate the value of investigating vocabulary adaptation. However, taking into account the comparable performance levels and the significant difference in computational cost (see Chapter 10), we advocate for the continual-pretraining method as the preferred adaptation approach.

7.5 Limitations

It is important to discuss some limitations of our corpus.

Firstly, *biomed-fr* is limited in its diversity. This is because we chose to only use publicly available materials, which tend to lean towards scientific content. As a result, models trained on this corpus may not fully represent performances on real private clinical data. The gap between scientific and clinical text is a recurring challenge that we address in Chapter 8.

Furthermore, we did not explore any potential biases in our data and some parts of the dataset would benefit from further cleaning. These aspects could impact the outputs of models trained on this corpus and should be investigated.

Third, all documents from the selected sources were included indiscriminately, without assessing whether individual passages were genuinely useful for pretraining. Whether such filtering would improve downstream performance is an open question that we explore in Chapters 8 and 9.

Conclusion

More than half of the most frequent biomedical subwords are absent from CamemBERT's general vocabulary. A mismatch this large raises a natural question: should we abandon the general tokenizer entirely and train a new model from scratch? The answer, perhaps surprisingly, is no, as we show in Chapter 10.

Before we get there, however, a more immediate question deserves attention. We included all 413 million words of *biomed-fr* without filtering. Not all of this text is equally useful. In the next chapter, we investigate whether selecting data by quality can improve upon including everything indiscriminately.

8 Detecting Content Types

8.1 Introduction

For web data, the question of what to keep and what to discard has a straightforward answer. FineWeb-Edu (Penedo et al., 2024) showed that filtering entire documents by educational quality, discarding 92% of the data, actually improves downstream performance. A low-quality web page is low-quality throughout: if the source is bad, every paragraph is bad. But a scientific article is not a web page.

Figure 8.1 illustrates the difference. A web page about astronomy scores low throughout and can be safely discarded. A PMC article on pulmonary hypertension, by contrast, contains a textbook-quality review paragraph, a boilerplate committee note, and a clinical case report with real diagnostic values. Document-level filtering would either keep all three or discard all three. Neither is the right answer.

In the previous chapter, we assembled *biomed-fr* by including all available documents from three public sources. This approach treats every paragraph equally. How much of this text actually helps, what types of content does it contain, and can we be smarter about what we include?

In this chapter, we introduce a paragraph-level annotation framework that distinguishes content types, domains, and educational quality within scientific articles. We show that selective pretraining on the most relevant paragraphs matches the performance of training on the full corpus using one-third of the tokens. We also extract 2 million clinical case paragraphs from PubMed Central, providing an openly available alternative to restricted hospital records. Finally, we investigate the risks of neural quality filtering: not all selection strategies are equally safe, and some can amplify benchmark contamination by a factor of 20.

8.2 Content Heterogeneity in Scientific Articles

Large language models for biomedicine are typically pretrained on PubMed Central (PMC) Open Access articles. While valuable, this corpus presents challenges including heterogeneous quality, predominance of English content (approximately 98%), uneven representation of clinical cases, and variable educational value. Existing filtering approaches operate at the article level: BioMistral (Labrak et al., 2024b) upsampled non-English articles, while Meditron (Chen et al., 2023) scored articles

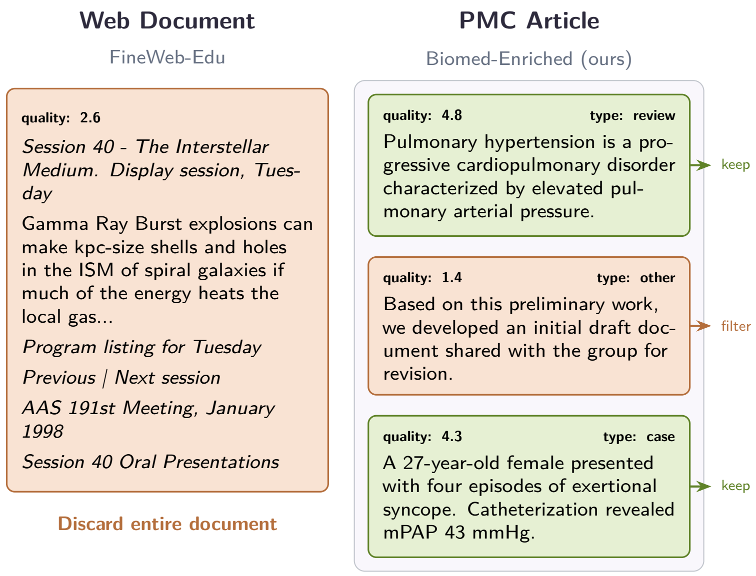

Figure 8.1: **Document-level vs. paragraph-level filtering.** A low-quality web page can be discarded entirely (left). A scientific article mixes high-value content (clinical cases, reviews) with boilerplate (right). Paragraph-level annotation allows keeping the valuable parts.

using MeSH tags, publication type, journal reputation, recency, and citation count. These methods assume that entire documents are either valuable or not.

This assumption does not hold for curated scientific corpora. Unlike web-scraped data where low-quality sources justify discarding entire documents, scientific articles are all potentially valuable. The real issue is variance *within* articles. To quantify this, we analyzed 10,000 PMC articles (378,000 paragraphs) and found that 91.6% of articles contain multiple document types (clinical case, study, review, other). Educational scores vary by 2.6 points on average within articles, spanning more than half of the 1–5 scale. Some paragraphs contain excellent educational or clinical content, while others add little domain knowledge but remain difficult to detect with simple heuristics.

8.3 Method

8.3.1 Data Collection and Preprocessing

We extracted text from the PubMed Central (PMC) Open Access Subset, containing approximately 4.5 million full-text scientific articles. Using a custom pipeline, we processed the raw XML files to extract article content, segment articles into 133 million individual paragraphs, filter out non-textual elements, and retain only paragraphs containing a minimum of 64 tokens.

	Single	Mixed
Document Type	8.4%	91.6%
Domain	44.7%	55.3%
Educational Score (1–5 scale)		
	Mean	**Median**
Intra-article std	0.68	0.70
Intra-article range	2.61	2.86

Table 8.1: Intra-document heterogeneity in 10K PMC articles (378K paragraphs). 91.6% of articles contain multiple document types. Educational scores vary by 2.6 points on average within articles.

Dimension	**Agreement**
Educational score	$r = 0.876$, MAE $= 0.326$
Domain	$F_1 = 0.954$
Document type	$F_1 = 0.925$, $\kappa = 0.821$

Table 8.2: Distillation quality: agreement between the XLM-RoBERTa student and Llama-3.1-70B teacher on held-out paragraphs.

8.3.2 Two-Stage Annotation Framework

We follow a two-stage annotation process on a single 8×A100 node: 160 GPU-hours for the first stage and 80 GPU-hours for the second.

Stage 1: LLM annotation. We used Llama-3.1-70B-Instruct (Grattafiori et al., 2024) to annotate a diverse subset of 400,000 paragraphs (approximately 0.33% of the corpus) across multiple dimensions: type classification (clinical case, study, review, other), domain categorization (clinical, biomedical, other), and educational quality (1–5 scale). Language was identified separately using langdetect.

Stage 2: Distillation. To scale annotation to the full corpus, we distilled the LLM annotations into a smaller XLM-RoBERTa-base model (Conneau et al., 2020) trained to jointly predict all annotation dimensions. On 39,800 held-out paragraphs, the student closely matches the Llama-3.1-70B labels (Table 8.2). It is important to note that these annotations are not gold-standard labels verified by medical experts, but neural heuristics for data curation. Similar to FineWeb-Edu (Penedo et al., 2024), they guide which content to prioritize during pretraining.

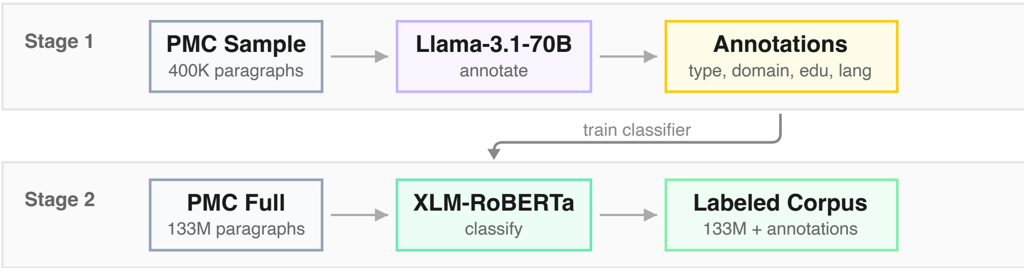

Figure 8.2: **Two-stage annotation pipeline.** Stage 1: Llama-3.1-70B annotates 0.33% of PMC paragraphs across four dimensions. Stage 2: a fine-tuned XLM-RoBERTa distills these annotations to label the full corpus, enabling flexible filtering strategies.

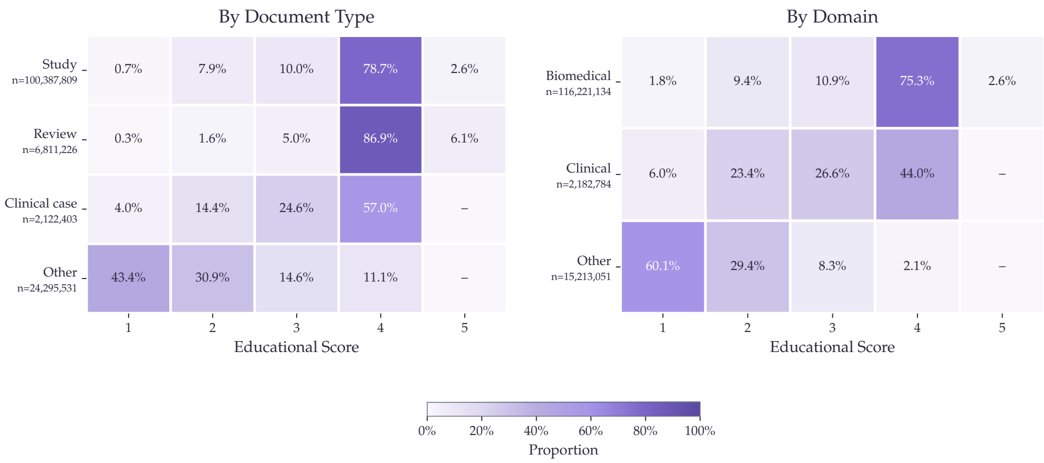

Figure 8.3: **Distribution of educational quality scores by document type and domain.** Reviews and studies show the highest proportion of high scores, while clinical texts display more variance. Content labeled "other" rarely reaches high scores.

8.3.3 Data Analysis

Most PMC paragraphs receive an educational score of 4, with a mean of 3.48 and median of 4.00. This distribution varies by content type: reviews and studies contain the highest proportion of educational content (86.9% and 78.7% scoring 4, respectively), while clinical cases show more variance (57.0% rated 4). Domain-wise, biomedical paragraphs score higher (75.3% at score 4) than clinical text (44.0%), and content labeled "other" rarely reaches high scores (2.1%). These patterns support the score threshold (≥ 3) used in BE-Educational and BE-All, and explain why combining domain and quality filters yields consistent gains across tasks.

Variant	Strategy
BE-Base	Complete unmodified PMC Open Access Subset (baseline).
BE-Educational	Removes paragraphs with educational quality scores below 3.
BE-Clinical	Upsamples 10× articles with predominantly clinical domain content.
BE-ClinicalCase	Upsamples 10× articles containing at least one clinical case paragraph.
BE-French	Upsamples 10× articles containing French text.
BE-All	Combines quality filtering (score ≥ 3), upsampling of clinical content, French text, and clinical cases, plus metadata prefixing.

Table 8.3: Biomed-Enriched dataset variants. All variants preserve the original article structure and use an 8K context window during pre-training.

8.3.4 Dataset Construction

Using the distilled model, we annotated the full PMC corpus and constructed several dataset variants through strategic filtering and upsampling, summarized in Table 8.3. Upsampling refers to duplicating articles in the training corpus, proportionally increasing their sampling probability.

For all variants, we preserved the original structure of the article and employed an 8K context window during pre-training. This ensures that models can process complete scientific articles, capturing dependencies where information presented in earlier paragraphs is essential for understanding later content.

8.4 Experimental Setup

Continual pre-training served as a method to evaluate the relevance and utility of our annotations. Our evaluation focuses on isolating the effects of data curation rather than pursuing state-of-the-art scores on benchmarks. A more powerful foundation model would likely yield higher absolute scores, but would probably obscure the precise impact of our dataset.

8.4.1 Decoder Setup

We selected OLMo2-7B-stage1 (Team et al., 2025) as our foundation model for continual pre-training, strategically choosing this intermediate checkpoint to more

Parameter	Value
Base model	OLMo2-7B-stage1
Peak LR	6.15e-5
Min LR	6.15e-6
LR Decay	Linear
Batch size	1024
Weight decay	0.1
Context	8,192 tokens
Hardware	128 MI250X
Tokens	33.6B / variant

(a) Decoder

Parameter	Value
Base model	ModernBERT-base
Peak LR	5e-4
Min LR	5e-7
LR Decay	one_minus_sqrt
Batch size	576
Weight decay	1e-5
Context	8,192 tokens
Hardware	4× H100
Tokens	70B

(b) Encoder

Table 8.4: Hyperparameters for continual pre-training.

clearly attribute performance changes to our data curation. Although stage 1 has already developed strong language modeling capabilities, it precedes the knowledge-intensive tuning of stage 2, providing an ideal balance of baseline capabilities without the risk of catastrophic forgetting of instruction-following abilities during domain adaptation. Notably, the data mix used in stage 1 includes DCLM (Li et al., 2024), which is a dataset obtained by filtering web-data using a classifier trained on instruct-data. Hence, OLMo2-7B already has relatively strong question-answering capabilities after stage 1.

Each Biomed-Enriched variant was trained with exactly 33.6 billion tokens, using identical hyperparameters. We follow the annealing strategy of OLMo2 used in the mid-training phase. By maintaining strict parameter parity across experiments, we created a controlled environment focused solely on measuring the effectiveness of our different data curation strategies.

8.4.2 Encoder Setup

We also conducted continual-pretraining experiments on encoder models. Starting from ModernBERT-base (Warner et al., 2025) after its context extension phase, we followed exactly the same hyperparameters as BioClinical-ModernBERT phase 1. Our baseline reproduces their standard continual pretraining setup on public PubMed/PMC. Our clinical-upsample recipe additionally applies educational filtering (score ≥ 3) and upsamples clinical cases 100× and clinical domain content 10×.

Table 8.4 summarizes the hyperparameters for both setups.

8.4.3 Evaluation

For decoders, our evaluation consists of zero-shot testing on MMLU medical subcategories, MedQA (Jin et al., 2021), MedMCQA (Pal et al., 2022), and PubMedQA (Jin et al., 2019). For the French adaptation assessment, we use 5-shot evaluation on

FrenchMedMCQA (Labrak et al., 2022). We compare our models against three baselines: OLMo2-7B-stage1, Llama-3.1-8B, and Meditron-70B.

For the encoder models, we evaluate on 11 tasks spanning clinical and biomedical domains. Clinical tasks include ChemProt, Phenotype, COS, SocialHistory, and DEID. Biomedical tasks include AnatEM, BC5CDR, JNLPBA, NCBI Disease, GAD, and Hallmarks of Cancer. Following the evaluation recipe of BioClinical-ModernBERT, we finetune for 10 epochs on clinical tasks and 20 epochs on biomedical tasks, with early stopping patience of 3 epochs for all. We compare against BioClinical-ModernBERT (Sounack et al., 2025), which was trained on 169B tokens from PubMed, PMC, and 20 clinical datasets including MIMIC-III, MIMIC-IV, CheXpert Plus radiology reports, and various clinical notes from multiple institutions, most requiring data use agreements.

8.5 Results

8.5.1 Decoder Results

Tables 8.5 and 8.6 present decoder results on medical QA and MMLU benchmarks. The combined strategy BE-All achieves the best average performance (61.08%), with the strongest gains on MedQA (+2.4 pts) and MMLU Clinical Knowledge (+1.5 pts). Different enrichment strategies show complementary strengths: clinical upsampling yields a +4 point improvement on MMLU Professional Medicine, while educational filtering improves Medical Genetics by +2 points and MedMCQA by +1.2 points. BE-All reaches target performance using roughly one-third of the training tokens compared to BE-Base, with stable improvements visible from early checkpoints. BE-French also demonstrates successful adaptation to French medical QA (40.5% vs 38.3% baseline, Figure 8.4), showing that language-specific upsampling generalizes beyond English.

8.5.2 Encoder Results

Tables 8.7 and 8.8 present encoder results across 11 clinical and biomedical benchmarks. Our BE-Clinical recipe achieves 77.08% average F_1, matching BioClinical-ModernBERT (77.05%) while using 2.5× fewer tokens and only public PubMed/PMC data. Adding a decay phase with high-quality educational content brings slight additional gains: the Biomedical decay variant (edu ≥ 4) achieves 77.32%.

Different decay targets show task-specific strengths: Study decay yields the best results on AnatEM (79.7%) and NCBI Disease (81.2%), Review decay excels on GAD (80.2%), and Clinical Case decay achieves the highest SocialHistory score (57.0%). These results suggest that public PMC data has more potential for clinical NLP than commonly exploited, and that targeted curation can close the gap with models trained on restricted datasets.

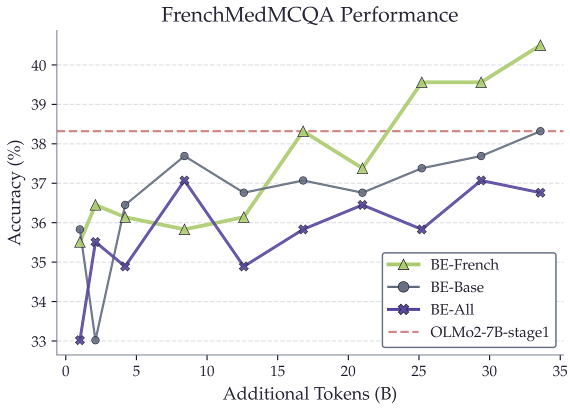

Figure 8.4: **FrenchMedMCQA performance.** BE-French outperforms other variants on French medical QA, demonstrating effective language-specific improvement through targeted upsampling.

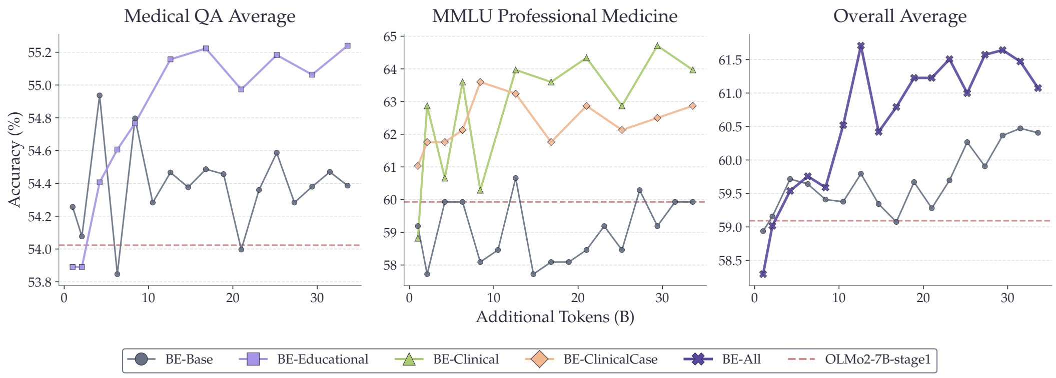

Figure 8.5: **Decoder training progression.** BE-All reaches target performance with approximately one-third of the training tokens required by BE-Base.

8.6 Discussion

The core advantage of paragraph-level annotation lies in its ability to identify valuable content that would be missed by document-level approaches. Clinical case descriptions, for instance, often appear in isolated sections of broader scientific articles. By operating at the paragraph level, we extracted 2 million clinical case paragraphs from PMC Open Access. This content is otherwise difficult to obtain at scale due to privacy restrictions on hospital records.

For decoders, our results reveal clear task-specific benefits from different enrichment strategies. Educational filtering improves knowledge-intensive QA tasks, clinical upsampling enhances clinical reasoning benchmarks, and the combination of both yields the best overall performance with faster convergence. The successful French adaptation through targeted upsampling further demonstrates that this framework

Model	MedQA	MedMCQA	PubMedQA	Avg
Reference Models				
Llama-3-8B	59.70	57.47	74.80	63.99
Meditron-70B	57.10	46.80	76.60	60.17
Our Variants				
OLMo2-7B-stage1	45.33	41.14	75.60	54.02
+ BE-Base	44.85	41.91	76.40	54.39
+ BE-Clinical	41.95	39.35	76.60	52.63
+ BE-ClinicalCase	42.11	39.52	76.60	52.74
+ BE-Educational	45.64	**43.08**	77.00	55.24
+ BE-All (33.6B)	**47.21**	42.79	76.60	55.53
+ BE-All (12.6B)	**47.21**	42.94	76.40	**55.52**

Bold indicates best result per column. Underline: second-best.

Table 8.5: **Decoder results: Medical QA benchmarks.** BE-All achieves the best QA average with both 33.6B and 12.6B tokens.

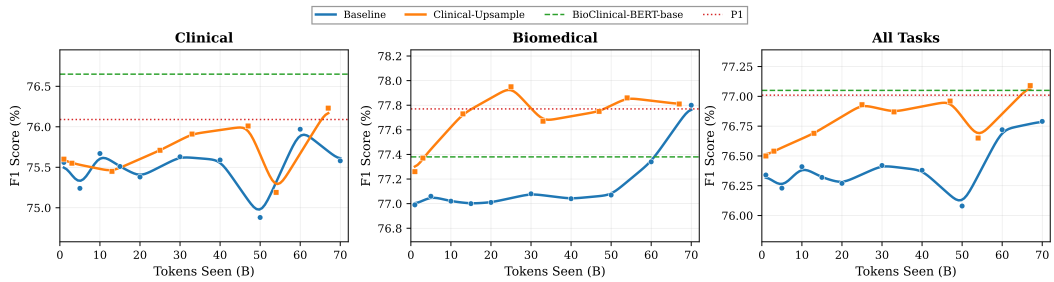

Figure 8.6: **Encoder training curves** on clinical, biomedical, and combined benchmarks. Our clinical-upsample recipe (orange) reaches higher performance faster on clinical tasks. At 67B tokens, we match BioClinical-ModernBERT (green dashed) which required 169B tokens and 20 clinical datasets.

generalizes beyond English. However, aggressive clinical upsampling can reduce performance on general biomedical tasks like College Biology, suggesting a trade-off between specialization and breadth.

For encoders, the decay phase experiments reveal that different paragraph types benefit different downstream tasks: Study paragraphs improve NER performance, Clinical Case paragraphs improve social history extraction, and Review paragraphs excel on relation extraction. The distinct task-specific strengths of each decay variant suggest a promising direction: model merging or checkpoint averaging could combine the benefits of specialized training runs without requiring additional compute. Rather than training general-purpose models on massive undifferentiated corpora, these findings support a more modular approach where data composition is tailored to downstream requirements.

Model	Anat	Clin	Bio	Med	Gen	Prof	Avg
Reference Models							
Llama-3-8B	68.89	74.72	78.47	61.85	83.00	70.22	72.86
Meditron-70B	53.30	66.70	76.30	63.00	69.00	71.60	66.65
Our Variants							
OLMo2-7B-stage1	54.81	63.40	69.44	53.18	69.00	59.93	61.63
+ BE-Base	57.04	64.15	**70.83**	**59.54**	69.00	59.93	63.42
+ BE-Clinical	53.33	63.40	65.28	58.38	66.00	**63.97**	61.73
+ BE-ClinicalCase	57.04	64.91	66.67	**59.54**	69.00	62.87	63.34
+ BE-Educational	57.04	65.28	68.06	56.65	**71.00**	58.82	62.81
+ BE-All (33.6B)	60.00	**65.66**	68.06	58.96	69.00	61.40	63.85
+ BE-All (12.6B)	**64.44**	64.53	68.75	57.23	**71.00**	62.87	**64.80**

Anat=Anatomy, Clin=Clinical Knowledge, Bio=College Biology, Med=College Medicine, Gen=Medical Genetics, Prof=Professional Medicine.

Table 8.6: **Decoder results: MMLU Medical subcategories.** BE-All (12.6B) achieves the best MMLU average despite using one-third of the tokens.

8.7 The Risks of Neural Quality Filtering

[1]

The results above show that selecting content by type and educational quality can improve pretraining efficiency. A natural next step is to apply neural quality classifiers, as done by FineWeb-Edu (Penedo et al., 2024) and DCLM (Li et al., 2024), to filter pretraining data more aggressively. However, this raises a question: do these classifiers treat all content equally, or do they have blind spots?

To investigate this, we designed Benchmark-In-A-HayStack (BiaHS), an experiment to test how different text-quality classifiers rank benchmark-like content within a large web corpus. We sampled 35 benchmark instances from three datasets: 15 samples from MMLU (3 samples each from anatomy, computer security, high school geography, moral scenarios, and college physics), 10 from GSM8K (Cobbe et al., 2021), and 10 from GPQA (Rein et al., 2023). Each benchmark sample was formatted as a standalone document containing the question, answer choices, and reference solution. These 35 benchmark documents were inserted as independent entries into a corpus of 99,965 documents sampled from FineWeb, yielding a final evaluation set of 100,000 documents where benchmarks constituted 0.035% of the total.

We scored all documents using four quality classifiers: DCLM (FastText-based, used to filter OLMo 2 training data), Textbook (FastText-based, used to filter OLMo 1 training data), FineWeb-Edu (transformer-based), and the GAPeron classifier (transformer-based). Each classifier produced a full ranking of all documents based on their predicted quality score.

[1]This section is based on our contribution to Godey et al. (2026).

Model	ChemPr	Pheno	COS	Social	DEID	Avg
Baselines						
ModernBERT-base	89.5	48.4	94.0	53.1	78.3	72.66
BioClinical-MdnBERT†	90.0	**60.7**	94.8	56.0	**81.8**	**76.66**
Our Models (Public Data Only)						
Baseline	89.8	59.6	94.5	55.5	78.5	75.58
BE-Clinical	89.9	59.9	94.8	56.8	79.7	76.22
With Decay Phase						
+ Review	**90.3**	59.7	94.6	55.4	79.6	75.92
+ Study	**90.3**	**60.3**	94.7	55.6	78.6	75.90
+ Clinical Case	89.5	59.4	94.5	**57.0**	79.3	75.94
+ Biomedical	90.1	59.9	**94.9**	56.4	79.6	<u>76.18</u>

† Trained on 169B tokens including 20 clinical datasets (MIMIC-III/IV, CheXpert, etc.). Our models use only public PubMed/PMC.
ChemPr=ChemProt, Pheno=Phenotype, Social=SocialHistory.

Table 8.7: **Encoder results: Clinical tasks.** Our BE-Clinical recipe matches BioClinical-ModernBERT on clinical tasks using only public data.

The results reveal a striking pattern. The DCLM classifier ranks all MMLU and GSM8K samples in the top-5 percentiles. If only the top 5% of documents are selected through filtering, the probability of encountering a benchmark sample in a training batch increases by a factor of approximately 20. The FineWeb-Edu classifier also ranks benchmark samples consistently higher, likely because its annotation prompt explicitly asks for documents with high educational value, which naturally favors exam-style items and step-by-step solutions.

The DCLM classifier pushes benchmark samples even higher and more consistently. Its FastText model is trained to separate synthetically generated instructions from high-scoring Reddit posts, from general web text. This training objective naturally favors solved Q&A structures with a short question, a direct answer, and brief reasoning, which aligns with the format of common benchmark datasets.

In contrast, the GAPeron classifier does not significantly push benchmark samples. It was trained on annotations focused on general content quality across multiple dimensions: accuracy, clarity, coherence, grammar, depth of information, and overall usefulness. This broader framing does not specifically reward the structured question-answer format typical of benchmark samples.

Implications. This experiment suggests that the way quality classifiers are trained can strongly influence contamination risks. As neural quality filtering becomes a standard step in data curation, these design choices deserve closer examination. For the Biomed-Enriched work presented earlier in this chapter, we rely on content-type

Model	AnatEM	BC5CDR	JNLPBA	NCBI	GAD	HoC	Avg
Baselines							
ModernBERT-base	77.2	87.9	74.3	77.7	76.8	66.6	76.75
BioClinical-MdnBERT†	**79.2**	88.7	74.8	78.7	75.8	67.0	77.37
Our Models (Public Data Only)							
Baseline	78.8	88.7	74.8	**81.3**	76.2	67.0	77.80
BE-Clinical	78.5	88.1	74.6	79.4	77.7	68.5	77.80
With Decay Phase							
+ Review	78.5	88.5	74.7	79.6	**80.2**	68.8	<u>78.38</u>
+ Study	**79.7**	88.4	**74.8**	81.2	78.7	67.7	**78.42**
+ Clinical Case	78.7	88.3	74.7	80.0	77.9	68.1	77.95
+ Biomedical	79.0	**88.6**	74.7	79.9	78.4	**69.0**	78.27

† Same model as Table 8.7. Trained on 169B tokens with 20 clinical datasets.

Table 8.8: **Encoder results: Biomedical tasks.** Study decay achieves 78.42%, outperforming BioClinical-ModernBERT (77.37%) with only public data.

classification and educational quality scoring rather than general quality filtering, which avoids this particular risk.

8.8 Limitations

Our decoder experiments used 7B parameter models. Larger models may respond differently to data curation choices. The two-stage annotation pipeline relies on XLM-RoBERTa-base, which may have limited capacity for nuanced paragraph classification. Our clinical case paragraphs come from published case reports, which may differ stylistically from actual hospital records. The educational quality scores are neural heuristics rather than gold-standard pedagogical assessments.

For the BiaHS experiment, the evaluation was conducted on a relatively small number of benchmark samples (35) inserted into a single web corpus (FineWeb). While the results are directionally clear, a larger-scale study across multiple corpora would strengthen the conclusions.

Conclusion

Two findings emerge from this chapter. First, paragraph-level annotation reveals that scientific articles are far more heterogeneous than their metadata suggests: 91.6% mix content types, and educational quality varies by 2.6 points within a single article. Selecting the right paragraphs allows matching full-corpus performance with a third of the tokens, and extracting 2 million clinical case paragraphs provides public clinical text without data use agreements.

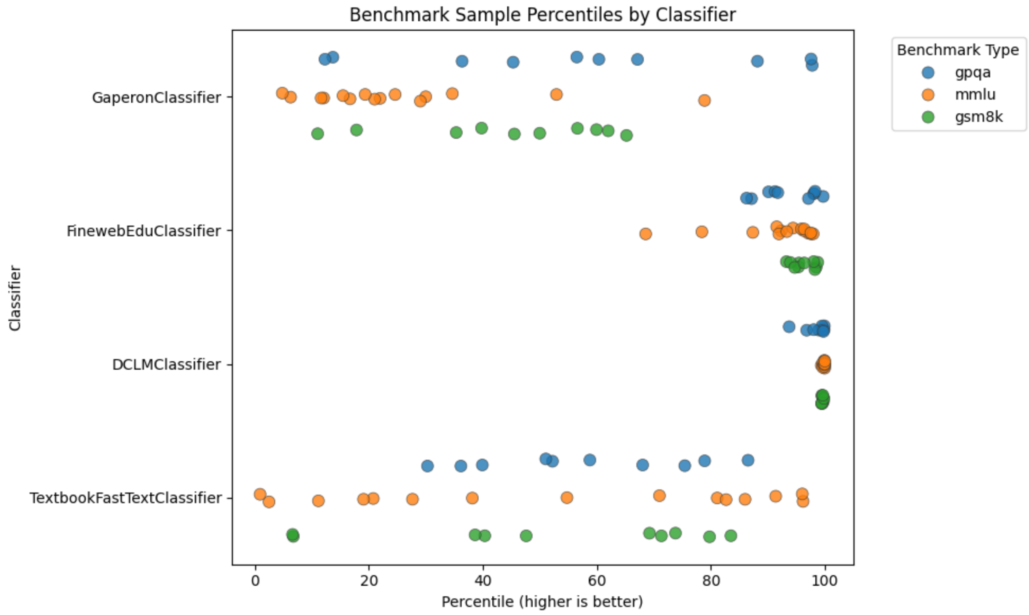

Figure 8.7: **BiaHS: Benchmark-In-A-HayStack results.** Percentile rank of 35 benchmark samples (MMLU, GSM8K, GPQA) within 100K FineWeb documents, as scored by four quality classifiers. DCLM ranks all benchmark samples in the top-5 percentiles.

Second, not all filtering is created equal. Neural quality classifiers trained on instructional or educational content can amplify benchmark contamination by a factor of 20, effectively turning a data curation step into a memorization accelerator. Content-type classification, by contrast, selects for relevance rather than format, and does not carry this risk.

We now have both a general biomedical corpus (*biomed-fr*, Chapter 7) and a method for selecting the most relevant paragraphs within it. The next chapter addresses how to combine these resources into a curated training mix, and how quality signals can be used at scale to build a better corpus for pretraining ModernCamemBERT-bio.

9 Which Quality Signals Matter?

9.1 Introduction

Here is one passage as it came out of an ISTEX PDF:

```
… vise ~ remonter l&apos;organe prolab6 (correction de la pt6se) et
h le soutenir dans sa position id6ale plus qu&apos;/~ le fixer (fixation).
Ce sout~nement est assur6 par le renforcement du p6rin6e grace aux
tissus naturels que l&apos;on r6pare (rapphie) ou que l&apos;on consolide
avec du mat6riel local (plastie, ligamentopexie)…
```

And here is what it should read:

> … la chirurgie vise à remonter l'organe prolapsé (correction de la ptose) et à le soutenir dans sa position idéale plus qu'à le fixer (fixation). Ce soutien est assuré par le renforcement du périnée grâce aux tissus naturels que l'on répare (raphie) ou que l'on consolide avec du matériel local (plastie, ligamentopexie)…

OCR had read every *é* as a *6*, apostrophes came as `l'`, and tildes had drifted into the middle of words. Multiply this by the hundreds of thousands of similar PDFs in HAL and ISTEX, and assembling a 10-billion-token biomedical training mixture stops being a collection problem and becomes a repair problem.

Chapter 7 introduced *biomed-fr*, a 413-million-word public French biomedical corpus. Chapter 8 showed that paragraph-level annotation by a large language model can identify content types and educational quality more precisely than document-level filters, and that neural quality classifiers carry contamination risks. This chapter applies these lessons at scale. We assemble *biomed-fr-v2*, a 10-billion-token mixture from six French sources, and ask which quality signals actually improve downstream performance once the dust settles. The corpus is the training data for ModernCamemBERT-bio, presented in Chapter 11.

The contributions of this chapter are: (1) a multi-source French biomedical corpus 25 times larger than *biomed-fr*, including a curation pipeline that handles the OCR artifacts, boilerplate, and structural noise of HAL and ISTEX; (2) an LLM-based annotation of 2.16 million paragraphs along four quality signals and nine content types; (3) ablation studies that quantify the effect of each signal and each type on downstream performance, with one notable surprise: writing-quality and terminological-precision filters *degrade* performance.

9.2 Sources

We assemble *biomed-fr-v2* from six complementary French-language sources. HAL, ISTEX, and FineWiki-bio cover scientific biomedical literature. EMEA contributes drug package inserts. E3C contributes clinical case reports. Synthetic textbooks generated from medical ontologies complete the mixture with the vocabulary of French administrative nomenclatures. The final composition is summarized in Table 9.4.

- **HAL.** Medical theses, research articles, and reports deposited on the HAL open archive, filtered to the biomedicine section.
- **ISTEX.** French scientific articles from the ISTEX corpus, filtered to biology and medicine.
- **FineWiki-bio.** Biomedical subset of FineWiki, the Wikipedia counterpart to FineWeb (Penedo et al., 2024), restricted to French and filtered for the biomedical domain.
- **EMEA** (Névéol et al., 2014). Drug package inserts published in French by the European Medicines Agency.
- **E3C** (Magnini et al., 2020). Multilingual clinical corpus (layer 3, French partition), composed of clinical cases extracted from medical journals and theses.
- **Synthetic ontology textbooks.** Educational text generated from three French medical ontologies published by the Agence du Numérique en Santé in RDF/OWL format: CIM-10 (diagnoses, 19,161 codes), CCAM (procedures, 38,191 codes), and ATC (medications, 6,950 codes). Random walks through the ontology graph (1.3 million walks total) traverse hierarchical relations and, for CIM-10, causal and exclusion relations. A large language model (Qwen3-235B-A22B-Instruct) reformulates each walk into a paragraph of medical prose while preserving the original codes and relations. The full pipeline is described in Chapter 12.

HAL and ISTEX are distributed in heterogeneous formats: some documents come with structured XML, others only as PDFs, sometimes scans of printed material. For documents without clean XML, we apply the extraction and cleaning pipeline described in Section 9.3. FineWiki-bio, EMEA, E3C, and the synthetic textbooks are already of high quality and are integrated directly into the final mixture, bypassing this pipeline.

9.3 Curation Pipeline

A portion of HAL and ISTEX documents come with clean XML and proceed directly to annotation. The other portion is available only as PDFs, sometimes scans of printed material (particularly for articles published before the 1990s). Once converted to text, these PDFs contain OCR artifacts (incorrect ligatures, character substitutions, fragmented words), bibliographic references, repeated headers, and irrelevant content. We therefore apply an LLM-based extraction and cleaning pipeline, described below. The other sources (FineWiki-bio, EMEA, E3C, synthetic textbooks) are already clean and bypass this pipeline entirely.

Extraction and cleaning. Each document is split into segments of approximately 5,000 tokens at paragraph boundaries. A large language model (Qwen3-Next-80B-A3B-Instruct) receives each segment and produces structured JSON output containing semantically complete passages, with references, metadata, headers, and captions removed, and OCR errors corrected (ligatures, encoding). Extracted passages must contain between 30 and 600 words and at least 60% alphabetic characters. We then apply Jaccard-similarity deduplication (threshold 0.80, sliding window of 5 passages), followed by the FineWeb-2 filter chain for French: encoding correction, repetition filters, and Gopher and FineWeb quality filters.

Fidelity verification. To ensure the LLM does not introduce content absent from the source, we measure character-level overlap between each extracted passage and its source document on 300 ISTEX passages (after typographic normalization). 94.4% of passages have $\geq$ 90% overlap and 84.1% have $\geq$ 99% overlap; the median rate of words absent from the source is zero (mean: 3.4%). Manual inspection of edge cases shows that divergences come from OCR corrections (such as the surgical passage shown at the start of this chapter) or from converting tables into prose. The opposite case is also common: a clean 1,143-character passage on cancer-related depression is extracted with 100% character overlap and no additional words introduced. These two cases bracket the typical behavior of the pipeline: it cleans aggressively when the source is corrupted by OCR, and copies verbatim when the source is already clean.

LLM quality annotation. The same LLM annotates each passage along four quality signals scored from 1 to 10: educational value (`edu`, mean 6.5), content richness (`cont`, 6.8), writing quality (`writ`, 7.6), and terminological precision (`term`, 7.6). The codes in parentheses are the abbreviations used in ablation tables. A content type is also assigned among nine categories: `research_findings` (526k passages), `medical_knowledge` (388k), `clinical_guidance` (218k), `research_methodology`,

Type removed	Avg F_1	Δ
Removal helps		
`drug_information`	64.35	+0.67pp
`research_findings`	64.24	+0.56pp
`policy_administrative`	64.07	+0.39pp
Controls		
Random (same volume)	63.72	+0.04pp
Full corpus (reference)	63.68	ref.
Removal hurts		
`medical_knowledge`	63.55	−0.13pp
`research_methodology`	63.51	−0.17pp

Table 9.1: **Content-type ablation.** We remove one type and measure Δ relative to the full corpus.

`drug_information`, `background_review`, `policy_administrative`, `patient_case` (36k), and `other`. In total, 2.16 million passages are annotated.

9.4 Content-Type Ablation

We measure the effect of each content type by ablation: we remove one type and compare against the full corpus on three evaluation tasks (DiaMed, FrACCO-30, and EMEA, 9 seeds). Table 9.1 shows that removing `drug_information` *improves* performance by +0.67pp, while removing `medical_knowledge` degrades it by −0.13pp. A random control (removing the same amount of text at random) gives +0.04pp, confirming that the effect is specific to the content type.

We exclude `drug_information` from the final corpus (redundant with the EMEA source already included separately) and `other`. Paragraphs shorter than 50 tokens are also excluded.

9.5 Quality Filtering and Upsampling

We then test the effect of thresholding on the quality signals (Table 9.2). Joint filtering with `edu`≥ 7 and `cont`≥ 7 produces a +0.48pp gain; a stricter threshold (`edu`≥ 8 and `cont`≥ 8) reaches +0.55pp. By contrast, the writing-quality and terminological-precision signals *degrade* performance when used as filters (−0.26 and −0.33pp). This degradation likely comes from the fact that these filters eliminate informal but informative clinical documents, such as case notes and lightly edited reports.

Rather than applying a strict filter, we adopt a quality-weighted upsampling inspired by FineWeb (Penedo et al., 2024). Each article receives a quality ratio, defined as the proportion of its paragraphs satisfying `edu`≥ 7 AND `cont`≥ 7. Articles are

Filter	Avg F_1	Δ
Helpful filters		
`edu`≥ 8 AND `cont`≥ 8	**59.37**	+0.55pp
`edu`≥ 7 AND `cont`≥ 7	<u>59.30</u>	**+0.48pp**
`edu`≥ 6 AND `cont`≥ 6	58.95	+0.13pp
No filter (reference)	58.82	ref.
Harmful filters		
`term`≥ 7	58.57	−0.26pp
`writ`≥ 7	58.49	−0.33pp

Table 9.2: **Quality-signal ablation.** Educational value and content richness improve performance; writing quality and terminological precision degrade it.

Ratio of `edu`≥ 7 AND `cont`≥ 7 paragraphs	Coefficient
0%	excluded
1–25%	4.3×
26–50%	8.5×
51–75%	12.8×
76–99%	21.3×
100%	34.1×

Table 9.3: **Upsampling coefficients** per article-quality bucket.

then upsampled in proportion to this ratio (Table 9.3). Articles with 0% high-quality paragraphs are excluded from the final corpus.

9.6 Final Composition

The final training mixture, after cleaning and quality-weighted upsampling, contains 10 billion tokens (Table 9.4). These 10 billion include the repetitions induced by upsampling (coefficients ranging from 4.3× to 34.1× depending on quality, see Section 9.5); the underlying unique content is more limited. HAL and ISTEX articles (passed through the pipeline of Section 9.3) and FineWiki-bio (integrated as-is) are grouped under "curated biomedical articles." The three other sources (EMEA, E3C, synthetic textbooks) are also integrated as-is.

Conclusion

Two findings emerge from the ablations. First, not all content types contribute equally: removing drug information actually improves downstream performance, while removing medical knowledge degrades it. Second, not all quality signals

Source	Tokens	%	Content
Curated biomedical articles	7B	70%	HAL + ISTEX + FineWiki-bio
Synthetic ontology textbooks	2B	20%	CIM-10, CCAM, ATC
EMEA (drug inserts)	600M	6%	Pharmacology
E3C (clinical sentences)	400M	4%	Patient cases
Total	**10B**	**100%**	

Table 9.4: **Composition of the *biomed-fr-v2* training mixture** after quality-weighted upsampling.

work as filters: educational value and content richness help, but writing-quality and terminological-precision filters *hurt* performance, likely because they remove the informal but informative clinical documents that ultimately matter most. The lesson, perhaps, is that curation should be guided by what is being said, not by how well it is written.

The corpus is now ready. The next two chapters use it to train French biomedical encoders. Chapter 10 starts with the simplest recipe (continual pretraining with MLM on *biomed-fr*) and shows that even a modest public corpus produces a competitive model. Chapter 11 returns to *biomed-fr-v2* with a more ambitious recipe: a temporary detour through causal language modeling that, somewhat unexpectedly, leaves a permanent imprint on the encoder.

Part IV

Pretraining Language Models

10 Encoder Models for French Biomedicine

10.1 Introduction

In Chapter 7, we found that 55% of biomedical subwords are absent from CamemBERT's vocabulary. PubMedBERT (Gu et al., 2022) turned this kind of observation into a principle: for domains with enough unlabeled text, training a language model from scratch with a specialized vocabulary outperforms continual pretraining. The gain over BioBERT (Lee et al., 2020) was modest, but the conclusion was adopted widely.

For French, Labrak et al. (2023) went further. They reported that continual pretraining from CamemBERT led to F_1 losses of up to 20 points, and released DrBERT, trained from scratch on 128 V100 GPUs for 20 hours. The claim was strong, the compute was expensive, and the implication was clear: continual pretraining does not work for French biomedical models. Yet the size of the reported loss is difficult to reconcile with what we know about continual pretraining. A model that has already seen 138 GB of French text should not lose 20 points from seeing additional biomedical text. Something in the evaluation deserves closer examination.

In this chapter, we show that continual pretraining on *biomed-fr* (Chapter 7) yields an average improvement of +2.54 F_1 across five biomedical NER benchmarks, using 2 V100 GPUs for 39 hours. We also identify the methodological differences that led to the contradictory conclusion.

10.2 Pre-training

For the adaptation of CamemBERT to the biomedical domain, we conducted a second phase of pre-training on both versions of the *biomed-fr* corpus, starting from the weights and configuration of the camembert-base model. We applied the Masked Language Modeling (MLM) task with whole-word masking, following the method of Martin et al. (2020). We used the Adam optimizer (Kingma and Ba, 2015) with $\beta_1 = 0.9$ and $\beta_2 = 0.98$, and a learning rate of 5×10^{-5}. We performed 50,000 steps over 39 hours using two Tesla V100 GPUs. A batch size of 8 per GPU and gradient accumulation over 16 steps were used to achieve an effective batch size of 256.

10.3 Fine-tuning and Evaluation

Regarding model evaluation, we collected three named entity recognition evaluation datasets. These datasets cover various styles, allowing us to assess the model's versatility across different subdomains of biomedicine.

QUAERO. The QUAERO corpus (Névéol et al., 2014) consists of two evaluation sets: EMEA, containing drug leaflets, and MEDLINE, containing scientific article titles. The entities are manually annotated following 10 semantic groups from the UMLS (Lindberg et al., 1993). As some of these entities are nested, we kept only the entities with the coarsest granularity.

E3C. For evaluation, unlike the *biomed-fr* corpus, we use layers 1 and 2. These layers contain documents of different types, including clinical cases extracted from scientific articles. Layer 2 is semi-annotated, and it is used as the training set for fine-tuning, with 10% dedicated to the validation set. We evaluate on layer 1, which is fully manually annotated. There is only one class, and the objective is to find clinical entities in the text, regardless of their type.

CAS. The CAS corpus (Grouin et al., 2019) also consists of clinical cases from scientific articles. We focus on task 3 of DEFT 2020 (Cardon et al., 2020), which is an information extraction task based on CAS. It includes two subtasks, and thus two sets of annotations. In the first subtask, two classes need to be identified: *pathology* and *signs or symptoms*. The second subtask concerns associated information, including *anatomy*, *dose*, *examination*, *mode*, *timing*, *substance*, *treatment*, and *value*. These two tasks will be referred to as CAS1 and CAS2, respectively.

Fine-tuning. For fine-tuning, we used Optuna (Akiba et al., 2019) for hyperparameter selection. We set the learning rate to 5×10^{-5}, the warmup ratio to 0.224, and the batch size to 16. We performed 2000 steps. Predictions were made using a simple linear layer on top of the model. None of the CamemBERT layers were frozen.

Evaluation. Scores are measured using the seqeval tool (Nakayama, 2018) in strict mode with micro-average and the IOB2 scheme. For each evaluation, the best fine-tuned model on the validation set is selected to measure the final score on the test set. We average the results over 10 evaluations with different seeds.

10.4 Results and Discussion

CamemBERT vs CamemBERT-bio. We observe a significant performance gain on all evaluation datasets with our new model (Table 10.1). On average, we achieve a 2.54-

Dataset	CamemBERT	CamemBERT-bio biomed-fr-small	biomed-fr
Clinical			
CAS1	70.50	72.94	**73.03**
CAS2	79.02	80.00	**81.66**
E3C	67.63	67.96	**69.85**
Leaflets			
EMEA	74.14	75.93	**76.71**
Scientific			
MEDLINE	65.73	65.48	**68.47**
Average	71.40	72.46	**73.94**

F_1 scores (micro-average, seqeval strict, IOB2). Mean over 10 seeds. biomed-fr-s = biomed-fr-small (10% subset).

Table 10.1: **CamemBERT vs CamemBERT-bio** on five biomedical NER benchmarks. Continual pretraining on *biomed-fr* yields +2.54 F_1 on average.

point improvement in F-score. This gain is observed across all styles, demonstrating the model's versatility for both clinical and scientific domains.

BIOMED-FR-SMALL VS BIOMED-FR. We observe a decrease in performance with the biomed-fr-small dataset, but there is still a significant gain on certain datasets compared to CamemBERT. This confirms that the size of the corpus positively influences the performance, even in a specialized domain like biomedicine.

COMPARISON WITH THE STATE OF THE ART. We compared the performance of CamemBERT-bio with various previously mentioned approaches (Table 10.2). CamemBERT-bio achieves the best results for almost all evaluation datasets. Dura et al. (2022) did not observe improvement on EMEA and MEDLINE compared to CamemBERT because their pre-training corpus consists of documents from APHP, making it a less diverse corpus. However, they gain several points on their evaluation dataset, which is also based on APHP documents. van Mulligen et al. (2016) presents the highest score on MEDLINE and the best recall on EMEA. Their approach is based on a knowledge-based model, allowing them to achieve the best recall on both QUAERO evaluation datasets. Furthermore, their approach is the only one in this table capable of handling nested entities, giving them an advantage.

It is important to note that these different CamemBERT-based approaches have various experimental setups. The presence of CRF layers instead of a simple linear layer after CamemBERT, freezing CamemBERT layers, and variations in hyperpa-

Authors	CAS1	CAS2	EMEA	MEDLINE
seqeval				
Dura et al. (2022)	–	–	72.90	59.70
Ours	**73.03**	**81.66**	**76.71**	**68.47**
BRATeval				
Le Clercq de Lannoy et al. (2022)	–	–	67.40	55.30
van Mulligen et al. (2016)	–	–	74.90	**69.80**
Copara et al. (2020)	61.53	73.70	–	–
Ours	**84.97**	**83.25**	**77.80**	56.16

F_1 scores. BRATeval is the evaluation tool from the CLEF eHealth 2016 campaign (Névéol et al., 2016).

Table 10.2: **Comparison with existing approaches** on four NER tasks, using two evaluation tools.

rameters are examples of differences observed in addition to the pre-training corpus, which makes the comparison more challenging.

10.5 Evaluating Models: Methodology Matters

In a recent work, Labrak et al. (2023) introduced a new French biomedical language model named DrBERT. The authors contend that performing continual-pretraining on biomedical data from CamemBERT leads to reduced performance. They used a different methodology for the evaluation of named entity recognition, prompting us to examine the implications of each approach on the interpretation of the results.

Our evaluation approach is centered around micro F_1, which is measured using seqeval in strict mode with the IOB2 scheme. Their approach is based on weighted F_1 with independent token classification. Every token has a label and the model is evaluated for each of them, whereas with seqeval, the model is evaluated for each entity. Notably, the "O" token, representing non-entity, is the most frequent token and thus holds significant weight in the evaluation process. As all tokens are labeled, the number of predictions depends on the tokenizer, thereby influencing the final score.

In order to explore the impact of token labeling variations on the evaluation, we conducted an additional experiment on EMEA and MEDLINE where we reproduced the DrBERT methodology that we named *token-with-O*, and another one where we excluded the "O" token and only labeled the first token of each entity, referred to as *entity-without-O*.

We observe significant differences in scores between the two methodologies (Table 10.3). The methodology *token-with-O* consistently reports higher scores across all tasks compared to the alternative method, *entity-without-O*. Notably, the best-

Method	Model	EMEA w-F_1	EMEA m-F_1	MEDLINE w-F_1	MEDLINE m-F_1
token-with-O					
	DrBERT-7GB	87.45	34.95	75.52	**15.07**
	CamemBERT-bio	**90.37**	36.27	**77.89**	14.82
	CamemBERT	88.33	**47.45**	76.20	11.92
entity-without-O					
	DrBERT-7GB	66.72	**24.72**	60.70	**10.80**
	CamemBERT-bio	**73.53**	24.15	**62.04**	8.70
	CamemBERT	71.85	22.71	60.95	9.41

w-F_1 = weighted F_1, m-F_1 = macro F_1. Averaged over 10 runs.

Table 10.3: **Impact of evaluation methodology.** The best model varies depending on the metric and the treatment of the "O" token.

Model	Time	Hardware	GPU-h	kg CO_2
DrBERT	20h	128× V100	2,560	26.11
AliBERT	20h	48× A100	960	8.16
CamemBERT-bio	39h	2× V100	78	**0.80**

Table 10.4: **Carbon emissions during training.** CamemBERT-bio emits 32× less CO_2 than DrBERT. Estimation based on 34g CO_2eq. per kWh (France, 2022–2023).

performing model varies depending on the chosen methodology for one metric. CamemBERT achieves the highest macro F_1 score on the EMEA dataset, whereas with *entity-without-O*, DrBERT-7GB emerges as the top model. This observation underscores the influential role of the "O" class in determining the best-performing model.

In terms of macro F_1, DrBERT consistently outperforms CamemBERT-bio when evaluated using the *entity-without-O* methodology. On the other hand, CamemBERT-bio exhibits better performance across all other metrics. This indicates that DrBERT demonstrates a more balanced performance across different classes.

Furthermore, Labrak et al. (2023) suggested that continual-pretraining from a French model on French biomedical data is not effective, as it results in an F_1-score loss of up to 20 points. However, considering our success with continual-pretraining and the points we raised about the methodology, we suggest a reassessment of these findings.

10.6 Environmental Impact

Considering estimated carbon emissions during training, AliBERT emits 10 times the amount of CamemBERT-bio, while DrBERT releases 32 times more (Table 10.4). While a direct comparison with AliBERT was not possible, the minor performance variance with DrBERT, set against the pronounced disparities in computational and environmental costs, leads us to advocate for continual-pretraining as the preferred adaptation method for biomedical language models.

10.7 Limitations

Our evaluation focused on one task, named entity recognition. This might mean our understanding of how well our model performs is restricted. Future studies should aim to assess our model across a wider range of tasks and datasets to get a clearer picture of its strengths and weaknesses.

Additionally, the corpus limitations discussed in Chapter 7 apply here as well: *biomed-fr* is dominated by scientific literature and does not contain genuine clinical notes. Models pretrained on this corpus may still underperform on real hospital data compared to models trained on private clinical corpora (Dura et al., 2022).

Conclusion

The vocabulary mismatch identified in Chapter 7 is real, but it does not prevent effective adaptation: +2.54 F_1 at 32× less CO_2 than the from-scratch alternative. The 20-point loss reported by Labrak et al. (2023) turns out to be an artifact of the evaluation protocol, not a property of continual pretraining itself.

Continual pretraining works for encoders. The bidirectional nature of MLM may make these models particularly robust to domain shifts. But the field has moved toward autoregressive decoders, where the dynamics of continual pretraining are different, and the results, as the next chapter discusses, are less encouraging.

11 Beyond Masked Language Modeling

11.1 Introduction

Masked language modeling sees only 15% of its input. At every training step, 85% of the tokens receive no gradient signal. Causal language modeling, by contrast, predicts every token from its predecessors: every position contributes, every position learns. The price is that it can only look backward. The question is whether an encoder can benefit from this dense training signal without permanently losing its ability to look in both directions.

In this chapter, we show that it can. We introduce a two-phase pretraining strategy for ModernCamemBERT-bio: first, continual pretraining with a causal language modeling objective, then a short "decay" phase that converts the model back to masked language modeling. The resulting encoder outperforms the standard MLM-only approach by +2.9 F_1 on average across 8 French biomedical tasks (Base) and +1.5 F_1 (Large), winning on all 8. The effect transfers to English at three scales, with gains of +0.3 to +0.8 F_1 depending on model size.

The CLM phase leaves a lasting imprint on low transformer layers. CKA analysis shows that layers 0–7 diverge 5–44× more between CLM and MLM models than seed noise alone, and this divergence persists through extended MLM decay. Freeze interventions establish a double dissociation: freezing low layers during CLM eliminates the downstream benefit; freezing mid layers preserves it. The advantage cannot be localized post hoc: transplanting layer groups from CLM to MLM recovers at most 35% of the gap. The effect is emergent, requiring the coordinated interaction between all layers.

11.2 Motivation

11.2.1 When Decoder Continue-Pretraining Stops Working

11.2.2 Architecture and Objective

CamemBERT-bio (Chapter 10) demonstrated that continual pretraining works for biomedical encoders. However, it used the original CamemBERT architecture: 512-token context, absolute position embeddings, no Flash Attention. Clinical documents

Figure 11.1: **The CLM detour pipeline.** Starting from a pretrained MLM encoder, we continue pretraining with a causal objective (Phase 1), then decay back to MLM (Phase 2). The model does not return to an MLM-like state.

are often longer than 512 tokens, and the architecture has not kept pace with recent advances.

ModernBERT (Warner et al., 2025) addresses these limitations with Flash Attention, RoPE position embeddings, alternating local/global attention, and an 8,192-token context window. For biomedical adaptation, we start from ModernCamemBERT (French) and ModernBERT (English), in both Base (150M) and Large (350M) sizes.

The standard approach would be to continue MLM pretraining on biomedical data, as we did for CamemBERT-bio. But MLM has a structural limitation: only 15% of tokens are masked and receive gradient signal at each step. Causal language modeling does not have this limitation: every token predicts the next, and every position contributes to the loss. The idea is to use CLM for the compute-intensive pretraining phase, and switch back to MLM only at the end, to recover bidirectional attention. The key question is whether this switch erases the benefits of CLM or preserves them.

Related approaches. A handful of recent works explore objective switching for encoder pretraining. Gisserot-Boukhlef et al. (2025) pretrain encoders *from scratch* (210M–1B parameters, 100B tokens) and find that a biphasic CLM-then-MLM schedule outperforms pure MLM under fixed compute. Weller et al. (2025) train matched encoder/decoder pairs (up to 1B parameters, 1.7T tokens) and show that continued pretraining on the reverse objective does not bridge the encoder-decoder performance gap. AntLM (Team, 2024) alternates between CLM and MLM epochs at small scale (10M words). Our approach differs in three ways: we study the round-trip MLM→CLM→MLM with continued pretraining (not from scratch); we focus on the lasting effect of the CLM phase after returning to MLM; and we provide causal evidence via freeze interventions of where the benefit comes from. As Gisserot-Boukhlef et al. (2025) note, understanding how training objectives transfer to downstream tasks "remains largely unexplored."

11.3 Method

11.3.1 The CLM Detour

Our approach consists of two phases, illustrated in Figure 11.1.

Phase 1: CLM pretraining. Starting from a pretrained ModernBERT encoder, we replace the bidirectional attention mask with a causal mask and train with next-token prediction. The model architecture is unchanged: only the attention mask and loss computation differ. This forces the model to compress information into each token representation, since it cannot look ahead.

Phase 2: MLM decay. After CLM pretraining, we restore bidirectional attention and train with MLM at 15% masking for approximately 10% of the Phase 1 budget. The optimizer state is kept between phases; only the learning rate scheduler resets. Phase 2 decays the learning rate from peak to 10% of peak.

Controlled comparison. The MLM baseline follows the same two-phase structure with 30% masking in Phase 1 and 15% in Phase 2, identical schedule and optimizer. Both trajectories use identical architecture, data, and total compute:

$$\begin{aligned} \theta_A &= T_{\text{MLM}}(T_{\text{CLM}}(\theta_0, n), 0.1n) && \text{(CLM detour)} \\ \theta_B &= T_{\text{MLM}}(\theta_0, 1.1n) && \text{(MLM baseline)} \end{aligned}$$

The only difference is the Phase 1 objective. Any performance gap is attributable to the training objective alone.

11.3.2 Freeze Interventions

To test which layers carry the CLM benefit, we run three freeze experiments on the 22-layer French Base model, where the CLM-MLM gap is largest (+2.9pp):

- **Experiment 1** (low freeze, CLM phase): Layers 0–7 frozen during CLM, then normal decay. Tests whether low-layer modification during CLM is necessary.
- **Experiment 2** (low freeze, decay phase): Normal CLM, then layers 0–7 frozen during decay. Tests whether low-layer CLM changes persist without further updates.
- **Experiment 3** (mid freeze, CLM phase): Layers 8–14 frozen during CLM, then normal decay. Together with Experiment 1, tests selectivity: if freezing low layers eliminates the benefit while freezing mid layers preserves it, this constitutes a double dissociation.

11.3.3 Data

French. We compile 10B tokens (Base) or 25B tokens (Large) from four sources. The main source (7B tokens) is French biomedical literature (scientific articles, clinical guidelines, medical theses), where each paragraph is scored for educational value using Qwen3-235B and articles are upsampled based on their proportion of high-scoring paragraphs, following the Biomed-Enriched approach (Chapter 8). The remaining sources are synthetic medical QA from French coding systems (2B), clinical cases from the European Clinical Case Corpus (E3C, 400M), and drug package inserts from the European Medicines Agency (600M).

English. For the 50B scale, we mix three components: 60% biomedical literature from Biomed-Enriched (PMC Open Access articles filtered by educational value $\geq$ 3.0), 20% medical instruction-following datasets (MedS-Ins, ReasonMed, MediFlow, UltraMedical, EHR-Ins-Reasoning), and 20% MIMIC-III discharge summaries and clinical notes. The smaller 10B variant uses Biomed-Enriched with clinical case paragraphs upsampled 100× and other clinical documents 10× (80%), plus MediFlow medical instructions (20%), without MIMIC.

11.3.4 Training Details

All runs use decoupled AdamW with peak lr 2×10^{-4}, $\beta_1 = 0.9$, $\beta_2 = 0.98$, weight decay 10^{-5}, and a global batch size of 384 sequences (~3.1M tokens). Phase 1 uses linear warmup over 100M tokens then constant learning rate. Documents are packed into 8,192-token sequences with end-of-sequence tokens between documents. Training uses bf16 mixed precision on 4×H100 GPUs.

11.3.5 Fine-tuning Protocol

French tasks. Classification (DiaMED) and multilabel tasks (MedDialog, FrACCO, CANTEMIST, DisTEMIST) use a linear head on the CLS token, trained for 15 epochs with lr $= 2 \times 10^{-5}$, batch size 4, and 10% warmup. Multilabel tasks use max sequence length 4096 with weighted BCEWithLogitsLoss; classification uses 2048. NER tasks (EMEA, Medline) use a span-based architecture with BiLSTM, CRF, and biaffine scorer, trained for 4000 steps with lr $= 5 \times 10^{-5}$ for the encoder and 10^{-3} for the head, batch size 16, max length 1024. All tasks use AdamW with weight decay 0.01 and select the best checkpoint by validation F_1.

English tasks. We follow the BioClinical-ModernBERT evaluation protocol (Sounack et al., 2025): lr $= 5 \times 10^{-5}$, weight decay 0.01, batch size 16, 10 epochs for most tasks (20 for NER). All models are fine-tuned with the same hyperparameters per task.

11.3.6 Evaluation

We evaluate on 8 French and 11 English biomedical tasks (Table 11.1), using 9 seeds for French and 5 for English. French baselines include ModernCamemBERT, DrBERT (Labrak et al., 2023), CamemBERT-bio, and CamemBERT (Martin et al.,

Table 11.1: **Evaluation tasks.** Max sequence length reflects the fine-tuning configuration; longer values indicate tasks with longer input documents.

Task	Type	Lang	Max len.
French			
DiaMED	Classification	FR	2048
MedDialog-FR	Multilabel	FR	4096
FrACCO (30/100)	Multilabel	FR	4096
CANTEMIST	Multilabel	FR	4096
DisTEMIST	Multilabel	FR	4096
QUAERO (EMEA/Medline)	NER	FR	1024
English			
ChemProt	Relation Extr.	EN	512
Phenotyping	Classification	EN	8192
COS	NER	EN	512
Social History	NER	EN	512
De-identification	NER	EN	8192
AnatEM	NER	EN	512
BC5CDR	NER	EN	512
JNLPBA	NER	EN	512
NCBI Disease	NER	EN	512
GAD	Classification	EN	512
HoC	Classification	EN	512

2020). English baselines include ModernBERT, PubMedBERT (Gu et al., 2022), BioBERT (Lee et al., 2020), and BioClinical-ModernBERT (Sounack et al., 2025).

11.4 Results

11.4.1 French Biomedical Tasks

Table 11.2 presents the French results. For Base, the CLM detour achieves 60.8% average F_1, outperforming the MLM baseline on all 8 tasks (+2.9pp). For Large, CLM reaches 63.3% versus 61.8% for MLM (+1.5pp). CamemBERT-bio and CamemBERT average 36% and 35% overall, limited by their 512-token context on long clinical documents. We release the CLM-detour models as ModernCamemBERT-bio (Base and Large).

11.4.2 English Biomedical Tasks

Table 11.3 shows the English results at three scales. CLM outperforms MLM on average at all scales, with the gap widening at Large scale (+0.8pp, 7/11 task wins). The smaller English effect is expected: ModernBERT was pretrained on web data that commonly includes biomedical corpora such as PubMed, so its low layers

Table 11.2: **French biomedical results** (macro F_1, 9 seeds). Bold: best per column; underline: second best.

		Multilabel Classification					CLS	NER		
	Ctx	FrACCO-30	FrACCO-100	CANTEMIST	DISTEMIST	MedDialog	DiaMED	EMEA	Medline	Avg
Baselines										
ModernCamemBERT	8192	70.1	55.3	63.3	20.2	60.6	56.4	63.4	55.3	55.6
DrBERT	512	53.0	35.6	37.9	21.4	63.6	57.0	**68.0**	**62.3**	49.9
CamemBERT-bio	512	41.9	20.1	12.8	9.6	38.6	47.7	61.6	56.6	36.1
CamemBERT	512	40.8	19.4	11.9	9.5	37.4	40.6	61.5	54.7	34.5
Our Models (Base, 150M)										
MLM baseline	8192	69.9	56.8	64.9	23.5	62.5	63.4	65.4	56.8	57.9
CLM detour	8192	74.8	60.1	71.0	25.5	63.6	**67.4**	65.9	58.2	60.8
Our Models (Large, 350M)										
MLM baseline	8192	79.4	63.3	72.6	29.1	**64.5**	61.2	66.1	58.0	61.8
CLM detour	8192	**80.7**	**65.4**	**74.4**	**30.4**	**64.5**	64.8	67.0	59.5	**63.3**

Table 11.3: **English biomedical results** across 11 benchmarks (5 seeds). Base at 10B and 50B scales; Large at 50B. Bold: best; underline: second best.

		Clinical Tasks					BigBIO Tasks						
	Ctx	ChemPr	Pheno	COS	Social	DEID	AnatEM	BC5CDR	JNLPBA	NCBI	GAD	HoC	Avg
Baselines													
ModernBERT-base	8192	89.5	48.4	94.0	53.1	78.3	77.2	87.9	74.3	77.7	76.8	66.6	74.9
PubMedBERT	512	90.2	52.0	95.0	48.7	80.4	**83.3**	89.7	74.9	**82.1**	79.3	**71.0**	77.0
BioClinical-MdnBERT	8192	90.0	60.7	94.8	56.0	81.8	79.2	88.7	74.8	78.7	75.8	67.0	77.0
Base 150M (10B tokens)													
MLM baseline	8192	90.4	61.3	94.4	55.5	79.3	78.9	88.3	74.9	79.2	78.6	68.9	77.3
CLM detour	8192	90.3	61.0	94.3	55.7	82.6	79.9	89.1	74.2	80.4	79.3	69.4	77.8
Base 150M (50B tokens)													
MLM baseline	8192	**90.5**	61.6	95.1	55.7	81.6	80.0	88.7	74.9	80.3	78.7	67.1	77.7
CLM detour	8192	90.1	**61.9**	**95.2**	54.2	83.2	81.0	89.1	74.5	80.1	78.8	70.0	78.0
Large 350M (50B tokens)													
MLM baseline	8192	**90.5**	61.0	94.9	55.0	82.3	82.0	89.4	**75.5**	81.8	76.4	67.8	77.9
CLM detour	8192	90.4	61.3	94.7	**56.5**	**84.2**	83.2	**89.8**	75.3	81.7	**79.7**	69.3	**78.7**

already partially encode biomedical features. ModernCamemBERT, by contrast, was pretrained on general French web text without biomedical sources, leaving a larger domain gap and correspondingly stronger CLM benefit (+2.9pp vs +0.3pp). We release the English models as ModernBERT-bio.

11.4.3 Optimal Decay Ratio

We sweep the decay length from 2.5% to 50% of the CLM phase at two scales, evaluating on 3 French tasks (DiaMED, FrACCO-30, FrACCO-100; Table 11.4). At both scales, 10% decay gives the best performance; shorter decay (2.5–4%) scores about 1.0pp below optimal, and longer decay (20–50%) provides no additional benefit. This ratio matches the cooldown lengths used in language model pretraining.

11.4.4 Needle-in-a-Haystack

Long-context encoding is critical in biomedical NLP, where clinical documents routinely span thousands of tokens and key information (diagnoses, coding labels) can

Table 11.4: **Decay ratio sweep** on 3 French tasks (DiaMED, FrACCO-30, FrACCO-100; 9 seeds each). 10% decay is optimal at both scales.

Decay tokens	Decay / Phase 1	CLM+decay F_1	MLM+decay F_1
10B Phase 1 (Base, 150M)			
0.25B	2.5%	71.6%	69.1%
1.0B	**10%**	**72.6%**	70.4%
2.0B	20%	71.9%	70.4%
3.0B	30%	72.1%	69.7%
5.0B	50%	71.7%	70.4%
25B Phase 1 (Large, 350M)			
1.0B	4%	73.2%	70.1%
2.5B	**10%**	**73.5%**	70.5%
5.0B	20%	72.7%	70.1%

appear anywhere in the document. We design a synthetic needle-in-a-haystack task in French to evaluate long-context retrieval without entity-type confounds.

Protocol. A medical fact (the "needle") is inserted into a clinical document (the "haystack") at a controlled position, and the model must determine whether a given query fact is present (binary classification). Negative examples use the same template with different slot values, so the task requires precise retrieval rather than shallow pattern matching. We define 8 French medical fact templates covering drugs, vital signs, lab results, diagnoses, procedures, allergies, treatment duration, and symptoms. Each template has 2–4 slots filled from small lexicons:

- *Le patient a reçu 200 mg de morphine par voie intraveineuse.*
  Distractor: *Le patient a reçu 500 mg de tramadol par voie orale.*
- *La tension artérielle mesurée était de 160/90 mmHg.*
- *Le diagnostic retenu est cirrhose hépatique de stade III.*
- *Le patient rapporte une dyspnée évoluant depuis une semaine.*

We generate 1,500 balanced positive/negative pairs across 5 lengths (512–8,192 tokens) and 3 positions (start, middle, end), split 70/15/15 for train/validation/test. We freeze each encoder and train a 2-layer MLP probe (Dropout → Linear → GELU → Dropout → Linear) on CLS representations for 3 epochs (lr = 2×10^{-5}, batch size 4, AdamW), selecting the best checkpoint by validation accuracy.

Results. CLM outperforms MLM at every context length and position (+10.7pp overall: 62.2% vs 51.6%). MLM accuracy degrades monotonically with document length (57% at 512 tokens to 43% at 8,192), while CLM degrades more slowly and remains above MLM throughout. The CLM advantage is largest at mid-document positions, where the inserted fact is farthest from both sequence boundaries. This is consistent with the downstream pattern: the French tasks with the largest CLM gains (FrACCO, CANTEMIST) use sequence lengths of 4,096 tokens during fine-tuning.

11.5 Analysis: Why Does the CLM Detour Work?

11.5.1 CKA Methodology

We measure representational similarity with linear Centered Kernel Alignment (Kornblith et al., 2019). Given two representation matrices $X \in \mathbb{R}^{n\times p}$ and $Y \in \mathbb{R}^{n\times q}$ (one per model, n samples, mean-pooled over tokens), we center each column, compute Gram matrices $K = XX^\top$ and $L = YY^\top$, and define:

$$\mathrm{CKA}(X, Y) = \frac{\mathrm{tr}(KL)}{\sqrt{\mathrm{tr}(K^2)\,\mathrm{tr}(L^2)}}$$

We report divergence $d = 1 - \mathrm{CKA}$, so higher values indicate greater representational difference. CKA equals 1 when representations are identical up to invertible linear transformation, 0 when there is no linear relationship. All computations use float64 arithmetic. For French we use 500 held-out texts from DiaMED and FrACCO; for English, PubMed abstracts. Results are averaged over 3 seeds.

11.5.2 The CLM Imprint Persists in Low Layers

Comparing layer-by-layer CKA between CLM-detour and MLM-only models, the CLM phase modifies low-layer representations in a way that subsequent MLM training does not undo, even when the decay budget matches the CLM phase. CKA divergence reaches approximately 56.5% after 1.5B tokens of MLM decay and remains stable, with only 0.6pp variation over 8B additional tokens.

To isolate where CLM differs from MLM *specifically* rather than just from training noise, we normalize each layer's CLM-MLM divergence by a seed-noise control (two MLM models trained with different random seeds on identical data):

$$r_\ell = \frac{1 - \mathrm{CKA}(\mathrm{CLM}_\ell,\ \mathrm{MLM}_\ell^{s_1})}{1 - \mathrm{CKA}(\mathrm{MLM}_\ell^{s_2},\ \mathrm{MLM}_\ell^{s_1})}$$

A ratio of 1 means no CLM-specific effect. Low layers (0–7) show ratios of 5–44×: CLM changes them far more than random seed variation does. Mid and deep layers are near 1×: both objectives modify them similarly, so CKA cannot distinguish CLM from MLM there.

Table 11.5: **Freeze intervention results** (French Base, 8 tasks, 9 seeds). Freezing low layers during CLM eliminates the benefit ($p = 0.36$ vs MLM baseline). Freezing mid layers preserves it. Freezing low layers during decay has negligible effect.

Condition	**Avg F_1**	**Δ vs MLM**
	References	
MLM baseline	57.9	—
CLM detour	**60.8**	+2.9
	Freeze interventions	
Freeze L0–7 (CLM phase)	58.4	+0.5
Freeze L8–14 (CLM phase)	60.3	+2.4
Freeze L0–7 (decay phase)	60.3	+2.4

Table 11.6: **Layer transplantation** on DiaMED (3 seeds, linear probing). No transplant recovers more than 35% of the 7.1pp gap.

Transplant	F_1 (%)	Δ	**Recovery**
	References		
MLM baseline	32.8	—	0%
CLM detour	39.9	+7.1	100%
	Layer group transplants		
Low (0–7)	31.9	−0.9	−12%
Mid (8–14)	35.3	+2.5	+35%
Late (15–21)	32.4	−0.4	−6%

11.5.3 Causal Evidence: Freeze Interventions

Do these low-layer changes actually drive the downstream improvement? We answer with the freeze interventions described in §11.3.2, summarized in Table 11.5.

Freezing low layers (0–7) during CLM drops F_1 from 60.8% to 58.4%. We cannot reject the null hypothesis that the resulting model matches the MLM baseline ($p = 0.36$, paired bootstrap across 8 tasks × 9 seeds). Freezing mid layers (8–14) during CLM preserves the full benefit (60.3%). Together, these two experiments establish a double dissociation: low layers are necessary and mid layers are not. Freezing low layers during decay has negligible effect (−0.5pp, 60.3%), confirming that MLM decay preserves the CLM imprint in low layers.

11.5.4 Non-Localizability

The freeze experiments show that low layers are necessary *during training*. We now ask whether the resulting benefit can be localized *after training*, by copying full parameter blocks from the CLM model into the MLM model.

Table 11.7: **Component-level transplants** on DiaMED (3 seeds, linear probing). Neither attention nor MLP modules alone recover the full gap.

Transplant	F_1 (%)	Δ	Recovery
All attention	33.9	+1.1	+15%
All MLP	35.3	+2.5	+35%

Table 11.8: **CKA divergence scales with model size and is asymmetric.** The encoder-decoder comparison uses different model families (see text).

Model	Params	Div.	CI
Encoders (MLM→CLM→MLM)			
ModernCamemBERT Base	150M	56.5%	±0.7%
ModernCamemBERT Large	350M	67.2%	±0.3%
Decoder (CLM→MLM→CLM)			
Gemma-3	270M	26.3%	±1.0%

No transplant recovers more than 35% of the gap (Table 11.6). Low and late layer transplants hurt performance. Inserting CLM low layers into an MLM model creates a mismatch with the mid and deep layers that co-adapted under MLM, yielding representations worse than either pure model.

Component-level transplants (Table 11.7) confirm the same pattern. Replacing all attention modules from CLM into MLM recovers only 15% of the gap; replacing all MLP blocks recovers 35%. No single component type captures the full benefit.

Unlike freeze interventions, which act during training, transplants replace already-trained weights and cannot reconstruct the cross-layer co-adaptation that developed during CLM. The freeze experiments show low layers are necessary during training; the transplant experiments show the resulting benefit cannot be isolated post hoc.

11.5.5 Scaling and Architectural Asymmetry

The CLM imprint increases with model capacity (Table 11.8): Large (350M) reaches 67.2% $\pm$ 0.3% divergence versus 56.5% $\pm$ 0.7% for Base (150M). Testing the reverse on Gemma-3 (270M decoder), MLM-then-CLM retains only 26.3% $\pm$ 1.0% divergence versus 67.2% for an encoder of similar scale. One explanation is the density of the decay signal: MLM decay at 15% masking provides weak erasure, while CLM decay at 100% provides strong erasure. This comparison is exploratory: it is confounded by model family and language, and a controlled comparison using encoder and decoder variants of the same family would be needed to confirm the asymmetry.

11.6 Limitations

Our freeze interventions use French Base models only. The CKA patterns that motivate them (low-layer divergence well above seed noise) are consistent across French Base, French Large, and English Base, but replication of the freeze interventions on English and Large models would strengthen the causal claim. The encoder-decoder asymmetry comparison is confounded by model family. The position trade-off at 2048+ tokens means that the approach may not be suitable for tasks requiring very long context, such as full discharge summary coding.

Conclusion

Training an encoder like a decoder, then converting it back, produces stronger representations than training with MLM alone. The CLM detour reshapes low transformer layers in ways that subsequent MLM decay does not undo: a lasting imprint that scales with model capacity and produces consistent downstream gains across 8 French and 11 English biomedical tasks.

The practical implication is straightforward: for biomedical encoder pretraining, the CLM detour is a free lunch. Same data, same compute, better results. MLM decay of roughly 10% of the detour length suffices, and the effect is strongest when the domain gap between pretraining and target data is large.

Parts 1 and 2 of this thesis have focused on data and pretraining: how to collect biomedical text, how to filter it, and how to train encoder models on it. But a pretrained model is only useful if it can be deployed on real tasks. In the next part, we turn to adaptation: how to extract clinical information when labeled data is scarce, labels number in the thousands, and the only available text is public.

12 Knowledge-Enriched Pretraining

12.1 Introduction

Medical coding is the task of assigning standardized codes from medical ontologies to clinical documents. It is central to hospital billing, epidemiological surveillance, and clinical research. In France, over 30 million hospital stays per year require accurate coding using the CIM-10 classification (the French adaptation of ICD-10), CCAM procedure codes, and ATC medication codes. This task requires understanding not only medical vocabulary but also the relational structure of medical ontologies: hierarchical links between codes, causal associations, differential diagnoses, and exclusion rules. These relationships are explicitly encoded in ontology graphs but are absent from clinical text corpora.

Pretrained biomedical language models such as BioBERT (Lee et al., 2020), PubMedBERT (Gu et al., 2022), and CamemBERT-bio (Touchent et al., 2023) learn medical vocabulary from large text corpora. However, they do not capture the relational structure of medical ontologies, since this structure is not expressed in running text. Knowledge graph approaches such as RDF2Vec (Ristoski and Paulheim, 2016) and Snomed2Vec (Agarwal et al., 2019) encode ontology structure through random walks, but produce static, non-contextualized embeddings that cannot benefit from transformer pretraining. Knowledge-enhanced transformers such as DRAGON (Yasunaga et al., 2022a) and KEPLER (Wang et al., 2021) integrate knowledge graph structure into language model pretraining, but they rely on existing text-graph alignment rather than generating new training data from ontology structure alone.

We propose OntoBook, a pretraining method that converts medical ontology structure into training signal for encoder language models. Our approach proceeds in three stages. First, we generate random walks through ontology graphs to produce sequences that capture hierarchical, causal, and differential relationships between medical codes. Second, we reformulate these walks into fluent medical prose using a large language model, creating synthetic textbooks grounded in ontology structure. Third, we train an encoder with two objectives on the same textbook data: masked language modeling and relation prediction between code pairs. We refer to this co-training on shared data as *alignment*, and our experiments show that it is a necessary condition for the method to work.

We evaluate OntoBook on three French medical coding benchmarks and present ablation studies on the contribution of each component. We release 1.3 million LLM-

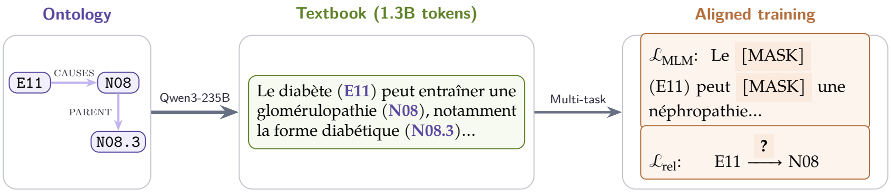

Figure 12.1: **OntoBook pipeline.** An ontology subgraph is converted into a random walk, then reformulated into textbook prose by Qwen3-235B. The encoder trains on this text with two aligned objectives: masked language modeling and relation prediction between code pairs.

reformulated medical textbooks across three French ontologies (CIM-10, CCAM, ATC) and all pretrained model checkpoints.

12.2 Related Work

This section reviews three research directions: biomedical language model pretraining, knowledge-enhanced language models, and automatic medical coding.

12.2.1 Biomedical Pretrained Language Models

Domain-specific pretraining has proven essential for biomedical NLP. BioBERT (Lee et al., 2020) further pretrained BERT on PubMed abstracts and PMC full-text articles, improving named entity recognition and relation extraction. PubMedBERT (Gu et al., 2022) demonstrated that pretraining from scratch on in-domain text outperforms mixed-domain initialization. ClinicalBERT (Huang et al., 2019) targeted clinical notes from MIMIC-III. For French, CamemBERT-bio (Touchent et al., 2023) adapted CamemBERT (Martin et al., 2020) to biomedical text, while DrBERT (Labrak et al., 2023) was trained on French biomedical and clinical text from the NACHOS corpus. These models learn from flat text corpora and do not capture the hierarchical structure of medical ontologies.

12.2.2 Knowledge-Enhanced Language Models

Random walks on knowledge graphs generate sequences capturing relational structure. RDF2Vec (Ristoski and Paulheim, 2016) adapted the walk-then-embed paradigm to RDF graphs. In the biomedical domain, Snomed2Vec (Agarwal et al., 2019) applied random walk and Poincaré embeddings to SNOMED-CT for clinical prediction tasks. OWL2Vec* (Chen et al., 2021) combined random walk sequences with lexical and logical features from OWL ontologies. These methods produce static embeddings (Word2Vec-style) and cannot be used for transformer pretraining.

Several approaches inject knowledge graph structure into transformers. ERNIE (Zhang et al., 2019) aligns entity embeddings with text representations. K-BERT (Liu et al., 2020) injects knowledge triples into the input sequence using soft-position indices and a visibility mask. For multi-task pretraining, KEPLER (Wang et al., 2021) combines MLM with TransE-style knowledge embedding on Wikidata. DRAGON (Yasunaga et al., 2022a) jointly trains MLM and knowledge graph link prediction, with a biomedical variant on UMLS achieving +3% on MedQA. CODER (Yuan et al., 2022c) uses UMLS relation triplets for contrastive pretraining, and SAPBERT (Liu et al., 2021) uses contrastive learning on UMLS synonyms for biomedical entity linking. Closest to the present work, BioOntoBERT (Shashikumar et al., 2023) generates sentences from biomedical ontologies using Onto2Sen templates and pretrains BERT with MLM on the resulting corpus. However, the template-based generation produces formulaic sentences (e.g., "X is a type of Y") rather than fluent prose, and the method uses only MLM without multi-task relation prediction.

Recent work has explored synthetic data generation from knowledge sources. The Phi series (Gunasekar et al., 2023) demonstrated that LLM-generated textbook-quality data enables strong small-model performance. EntiGraph (Yang et al., 2025b) generates entity-centric synthetic data from knowledge sources but targets decoder language models rather than encoder pretraining.

12.2.3 Automatic Medical Coding

Medical coding assigns ICD, procedure, or medication codes to clinical text. CAML (Mullenbach et al., 2018) introduced label-wise attention for explainable predictions. PLM-ICD (Huang et al., 2022) showed that fine-tuning pretrained encoders with label attention directly improves coding performance. Kirchler et al. (2026) recently demonstrated that LLM embeddings improve transferability of EHR-based predictions across countries and coding systems, highlighting the importance of encoding medical knowledge structure for cross-system generalization. These methods improve the classification head or the input representations but rely on generic encoders pretrained on flat text corpora.

Table 12.1 summarizes how OntoBook combines components not jointly present in prior work.

12.3 Method

OntoBook converts medical ontology structure into pretraining signal through three stages: (1) generating random walks through ontology graphs, (2) reformulating these walks into fluent textbook prose using a large language model, and (3) training an encoder with multi-task learning combining masked language modeling and

Method	Walks	LLM	MLM+Rel
Prior work			
Snomed2Vec	✓		
BioOntoBERT	✓		
DRAGON			✓
KEPLER			✓
Phi-1		✓	
Ours			
OntoBook	✓	✓	✓

Table 12.1: **Comparison with related approaches.** OntoBook uniquely combines all three components for biomedical encoder pretraining.

relation prediction. We first describe the ontologies used, then detail each stage. Figure 12.1 illustrates the full pipeline.

12.3.1 Medical Ontologies

We use three French medical ontologies published by the Agence du Numérique en Santé (ANS) through its terminology server (SMT) in RDF/OWL format. CIM-10 FR PMSI is the French adaptation of the WHO's ICD-10, enriched by the Agence Technique de l'Information sur l'Hospitalisation (ATIH) for the French hospital information system (Programme de Médicalisation des Systèmes d'Information, PMSI). It contains 19,161 diagnostic codes organized in a 5-level hierarchy (chapters, blocks, categories, subcategories), with textual annotations including 23,282 synonyms, 8,171 inclusion notes, and 381 definitions. CIM-10 is the only ontology with semantic edges beyond the hierarchy: 1,317 causal edges (e.g., E11 type 2 diabetes *causes* N08.3 diabetic glomerulopathy), 5,739 exclusion edges, and 341 manifestation edges, yielding a mean node degree of 2.77 compared to 2.00 for the purely hierarchical ontologies.

CCAM (Classification Commune des Actes Médicaux) is the French procedure classification maintained by the Caisse Nationale d'Assurance Maladie (CNAM), containing 38,191 procedure codes in a 4-level hierarchy with 8,991 synonyms and 1,188 disjointness constraints between mutually exclusive procedures. ATC (Anatomical Therapeutic Chemical) is the WHO medication classification with 6,950 substance codes in a strict 5-level hierarchy from anatomical group to chemical substance, with no semantic edges, no synonyms, and no definitions beyond labels. This contrast in relational richness directly shapes the walk generation strategy: CIM-10 walks can follow causal and differential diagnosis paths, whereas CCAM and ATC walks are limited to hierarchical and sibling traversals. Table 12.2 summarizes the structural properties of the three ontologies.

	CIM-10	CCAM	ATC
Graph structure			
Codes	19,161	38,191	6,950
Hierarchy depth	5	4	5
Hierarchical edges	19,139	38,191	6,950
Causal edges	1,317	—	—
Manifestation edges	341	—	—
Exclusion edges	5,739	—	—
Disjoint edges	—	1,188	—
Mean degree	2.77	2.06	2.00
Textual attributes			
Synonyms	23,282	8,991	0
Inclusion notes	8,171	—	—
Definitions	381	151	0

Table 12.2: **Structural properties** of the three French medical ontologies used for walk generation. CIM-10 is the only ontology with semantic edges beyond the hierarchy.

12.3.2 Ontology Walk Generation

We generate random walks through ontology graphs to capture relational structure in a sequential format suitable for language model pretraining. We sample walks starting from each code in the ontology. At each step, the next node is selected via weighted sampling where edge type weights depend on the walk type. This biased sampling is necessary because semantic edges are rare (Table 12.2), and a uniform walk would rarely traverse them. Each walk records the traversed codes along with their labels and the relation types connecting them. Walk length is sampled uniformly between 8 and 20 steps. We track visited nodes to avoid cycles, with a fallback that allows revisiting nodes when all neighbors have been explored.

For CIM-10, we define five walk types, each emphasizing different aspects of medical knowledge: *Etiologie* (causal chains), *Diagnostic Différentiel* (codes to distinguish clinically), *Codage Double* (mandatory code pairs linking etiology to manifestation), *Syndrome* (multi-system manifestations), and *Cross-Chapter* (connections between different ICD chapters, e.g., diabetes E11 to its renal complication N08). We generate 402,328 walks for CIM-10 (average 4,830 characters), 762,958 for CCAM, and 138,980 for ATC, totaling 1,304,266 walks.

12.3.3 Textbook Generation

Raw walks contain structured markers (e.g., "`>> À distinguer de:`") that are useful for parsing but not natural for language model pretraining. We use a large language model to reformulate walks into fluent medical prose resembling textbook paragraphs.

Structured Walk

```
[F02.0] Démence de la maladie de Pick
>> Cette pathologie peut provoquer:
[G31.0] Atrophie cérébrale circonscrite
>> Parmi les causes possibles:
[F02.02] ...avec symptômes hallucinatoires
```

↓ *Qwen3-235B*

Medical Textbook

Le code **F02.0**, désignant la démence de la maladie de Pick, s'inscrit dans un réseau de connexions inter-systèmes entre les troubles mentaux et les affections neurologiques. Ce code entretient une relation étroite avec **G31.0** (atrophie cérébrale circonscrite), établissant un lien étiologique réciproque...

Figure 12.2: **Walk-to-textbook transformation.** The structured walk is reformulated into fluent medical prose while preserving all codes and relations.

We prompt Qwen3-235B-A22B-Instruct (Yang et al., 2025a) with FP8 quantization to transform each walk into a coherent paragraph. The prompt instructs the model to: (1) preserve all medical codes and their relationships exactly as stated, (2) use natural medical language without lists or bullet points, (3) not add any information beyond what appears in the walk. This ensures the textbook content remains grounded in the ontology structure. We apply two filters: we discard reformulations shorter than 50 characters or that fail to mention the source codes. The reformulation runs on 4 NVIDIA H100 GPUs using vLLM (Kwon et al., 2023) with guided decoding to ensure valid JSON output, taking approximately 20 hours for all 1.3 million walks. The resulting textbooks contain approximately 1.3 billion tokens across all three ontologies. Figure 12.2 shows an example transformation.

Figure 12.3 shows the system prompt used for CIM-10 walks (translated from French for readability; prompts for CCAM and ATC follow the same structure with ontology-specific terminology).

12.3.4 OntoBook Training

We train a ModernCamemBERT-base encoder (Antoun et al., 2025), initialized from a French checkpoint trained on 1 trillion tokens.

The model optimizes two objectives jointly. The first, $\mathcal{L}_{\text{MLM}}$, follows standard masked language modeling (Devlin et al., 2019): we mask 15% of tokens from textbook paragraphs and train to predict them. The second, $\mathcal{L}_{\text{rel}}$, classifies the relationship between pairs of medical codes. Our main model uses CIM-10; we explore transfer from CCAM and ATC in Section 12.4.5. We extract 282,907 code pairs from the CIM-10 ontology across 6 relation types: Parent, Child, Sibling,

Reformulation Prompt (CIM-10, translated)

You are an expert medical writer specializing in high-quality educational material.

Reformulate the provided CIM-10 walk into fluent, textbook-quality medical prose.

Strict rules:
(1) **No invention**: never add information not in the source.
(2) **Total fidelity**: preserve all definitions, codes, clinical notes, exclusions.
(3) **Completeness**: do not omit any source information.
(4) **Ignore metadata**: never mention walk types or structural headers.

Style: Continuous narrative prose, no bullet points or markdown. Integrate CIM-10 codes naturally. Preserve exact medical terminology.

Goal: Transform the walk into text suitable for a medical nosology textbook chapter while remaining 100% faithful to the original content.

Figure 12.3: System prompt for LLM-based walk reformulation. Temperature is set to 0 for deterministic, faithful output.

Differential Diagnosis, Causes, and Caused_By. For each pair, we retrieve the corresponding textbook descriptions generated in Section 12.3.3. The input format is "`[CLS]` text$_A$ `[SEP]` text$_B$ `[SEP]`", where text$_A$ and text$_B$ are the textbook descriptions of the two codes, and a separate linear classification head on the `[CLS]` token predicts the relation type, trained with cross-entropy loss over the 6 classes.

The combined loss is:

$$\mathcal{L} = \mathcal{L}_{\text{MLM}} + \lambda \mathcal{L}_{\text{rel}} \tag{12.1}$$

where $\lambda = 1.0$. We train for 4 epochs with learning rate 2×10^{-5}, batch size 128, and AdamW optimizer with weight decay 0.01 and 1,000 warmup steps. Training takes approximately 4 hours per epoch on one NVIDIA H100 GPU. The key design choice is that both objectives operate on the *same textbook data*. We call this property *alignment*, and we show in Section 12.4.3 that it is essential for performance.

12.4 Experiments

12.4.1 Experimental Setup

Evaluation Benchmarks. We evaluate on the standard French medical coding evaluation suite:

- **FRACCO** (Pignat et al., 2025): a native French oncology corpus with 1,301 clinical cases annotated with ICD-O-3.1 codes, framed as multi-label classification over the top 100 codes.
- **Cantemist-FR** (Zaghir et al., 2024): the French adaptation of the Spanish tumor coding task (Miranda-Escalada et al., 2020), with 2,051 clinical documents.

Model	**FRACCO**	**Cant.**	**Dist.**	**Avg.**
External baselines				
CamemBERT-bio	20.2$_{\pm 0.2}$	12.1$_{\pm 0.4}$	9.0$_{\pm 0.2}$	13.8
DrBERT	36.3$_{\pm 0.7}$	37.7$_{\pm 1.0}$	22.5$_{\pm 0.7}$	32.1
ModernCamemBERT	56.4$_{\pm 1.0}$	63.5$_{\pm 1.3}$	23.4$_{\pm 1.7}$	47.7
Our models				
CodeInfill+MLM	56.5$_{\pm 0.2}$	66.4$_{\pm 1.9}$	20.0$_{\pm 0.9}$	47.6
MLM-only	55.8$_{\pm 0.4}$	66.0$_{\pm 1.1}$	24.2$_{\pm 5.8}$	48.7
OntoBook	**58.3**$_{\pm 0.3}$	**67.1**$_{\pm 1.1}$	**32.2**$_{\pm 1.1}$	**52.5**

Table 12.3: **Main results** on French medical coding (micro-F_1 %). External baselines averaged over 9 seeds; our models over 3 seeds. Best results in **bold**.

- **Distemist-FR** (Zaghir et al., 2024): the French adaptation of the disease mention coding task (Miranda-Escalada et al., 2022).

Cantemist-FR and Distemist-FR are cross-lingual projections from Spanish. FRACCO is the only native French benchmark. All results are micro-F1 scores averaged over 3 random seeds.

Baselines. We compare against French biomedical language models: CamemBERT-bio (Touchent et al., 2023), DrBERT (Labrak et al., 2023), and ModernCamemBERT (Antoun et al., 2025) (which serves as our initialization checkpoint). We also report results for three ablations of our method: MLM-only (trained with $\mathcal{L}_{\text{MLM}}$ only on our textbooks), Rel-only (trained with $\mathcal{L}_{\text{rel}}$ only), and CodeInfill+MLM, which replaces relation prediction with a code infilling objective that masks medical code tokens (e.g., "E11", "N08.3") in textbook text and trains the model to predict them. We do not compare against knowledge-enhanced models such as DRAGON (Yasunaga et al., 2022a) or SAPBERT (Liu et al., 2021) as they are English encoders that cannot be applied to French text.

Fine-tuning. For downstream evaluation, each pretrained encoder receives a clinical document as input and predicts a set of medical codes (multi-label classification over the top 100 codes per benchmark). We fine-tune with a linear classification head for 10 epochs, learning rate 2×10^{-5}, and batch size 16. External baselines (CamemBERT-bio, DrBERT, ModernCamemBERT) are averaged over 9 seeds; our models over 3 seeds due to the large number of configurations evaluated.

12.4.2 Main Results

Table 12.3 compares OntoBook against French biomedical baselines.

Configuration	FRACCO	Cant.	Dist.	Avg.	Δ
Loss ablations					
OntoBook (aligned)	**58.33**	**67.06**	**32.24**	**52.54**	—
− $\mathcal{L}_{\text{rel}}$ (MLM-only)	55.81	66.01	24.23	48.68	−3.86
− $\mathcal{L}_{\text{MLM}}$ (Rel-only)	48.24	51.15	20.18	39.86	−12.68
− Alignment (misaligned)	33.39	20.11	12.63	22.04	−30.50
Data ablation					
MLM-only (raw walks)	55.51	66.00	22.76	48.09	−4.45

Table 12.4: **Ablation study** (micro-F_1 %). Misalignment degrades all benchmarks, with the largest drop on Cantemist (−46.95). All variants use the same base model and training budget.

OntoBook outperforms all baselines on all three benchmarks. The improvement over MLM-only pretraining is significant on FRACCO (+2.5 micro-F1, $p < 0.001$) and Distemist (+8.0 micro-F1, $p < 0.001$), with consistent but not individually significant gains on Cantemist ($p = 0.11$). The gain is largest on Distemist, suggesting that ontology-aware pretraining particularly helps with fine-grained disease coding. OntoBook also reduces variance on Distemist from ±5.8 (MLM-only) to ±1.1.

12.4.3 Ablation Study

Table 12.4 ablates the key components of OntoBook.

Alignment is the most critical factor. Training $\mathcal{L}_{\text{MLM}}$ and $\mathcal{L}_{\text{rel}}$ on different data causes catastrophic degradation across all benchmarks, with Cantemist dropping from 67.06 to 20.11 (−46.95). Relation prediction alone also fails (−12.68 average F1), confirming that $\mathcal{L}_{\text{rel}}$ cannot serve as a standalone pretraining objective. The last row trains MLM-only on raw structured walks, removing both reformulation and relation prediction. The raw walk model (48.09) is close to the textbook MLM-only model (48.68), indicating that reformulation alone contributes little to MLM pretraining (+0.59 F1). However, the gap to full OntoBook is 4.45 points, reflecting the combined effect of adding both reformulation and relation prediction. This suggests that reformulation primarily benefits the relation prediction objective: on raw walks with structural markers (e.g., "`>> À distinguer de:`"), the classifier can exploit formatting shortcuts rather than learning medical semantics, whereas fluent textbook prose forces the model to learn from content rather than format.

12.4.4 Training Dynamics and Hyperparameter Sensitivity

Figure 12.4 shows training dynamics and hyperparameter sensitivity. Epoch 1 (51.98) and epoch 2 (52.54) both outperform MLM-only pretraining (dashed line), but epoch 3 (49.43) degrades below it, possibly due to overfitting on the relation prediction

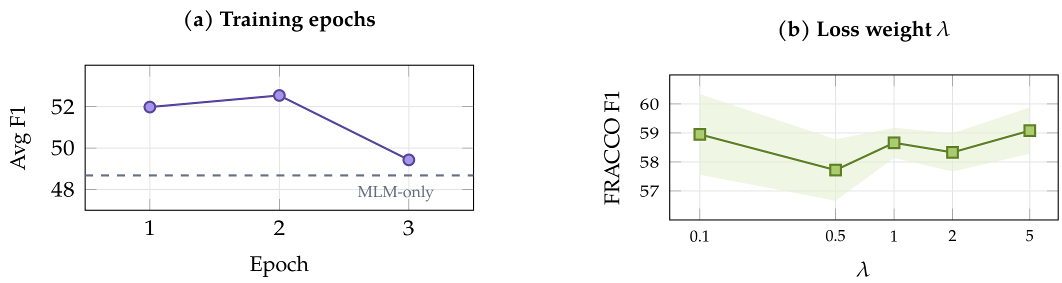

Figure 12.4: **Training dynamics.** (a) Average F_1 peaks at epoch 2 then degrades. (b) Performance is stable across λ values (shaded: ± 1 std).

Ontology source	FRACCO	Cant.	Dist.	Avg.
Main configuration				
CIM-10 (OntoBook)	58.33	67.06	32.24	52.54
Other ontology sources				
All (CIM-10+CCAM+ATC)	59.31	**67.64**	31.44	52.80
CCAM only	58.59	66.54	34.26	**53.13**
ATC only	**59.66**	63.03	**36.05**	52.91

Table 12.5: **Effect of ontology source** on downstream performance (micro-F_1 %). Each row trains a separate model using walks and relations from only that ontology. All models use the same training procedure and base checkpoint.

task. All results in this paper use the epoch 2 checkpoint. The loss weight λ is robust across $\{0.1, 0.5, 1.0, 2.0, 5.0\}$: all values achieve 57.7–59.1% F1 on FRACCO. We use $\lambda = 1.0$ as it achieves the lowest variance ($\pm 0.52\%$). This robustness suggests that the exact balance between objectives matters less than ensuring both are present and aligned.

12.4.5 Multi-Ontology Transfer

The main OntoBook model uses CIM-10 walks (Table 12.3). To test whether other ontologies also provide useful pretraining signal, we train separate models on walks from each ontology individually and on all three combined.

All three individual ontologies improve over the MLM-only baseline, indicating that the pipeline generalizes across different medical knowledge structures. ATC (medications) yields the best Distemist score (+3.81 over CIM-10), despite Distemist being a disease coding task, suggesting that drug-disease relationships transfer well to disease understanding. CCAM (procedures) achieves the best average micro-F1 (53.13). Combining all three ontologies does not outperform the best individual ontologies. We hypothesize that this is because the relation prediction head must accommodate heterogeneous edge types across ontologies.

Model	Chapter	Depth	Dist. ρ
Baselines			
Base	97.5	99.3	0.195
MLM-only	98.9	99.4	0.287
CodeInfill+MLM	**99.0**	**99.8**	**0.397**
Rel-only	98.0	98.2	0.203
Ours			
OntoBook	98.7	99.7	0.073

Table 12.6: **Probing analysis** on CIM-10 code embeddings (% accuracy for Chapter and Depth, Spearman ρ for Distance). CodeInfill+MLM achieves the best geometric organization but OntoBook achieves the best downstream performance (Table 12.3).

12.4.6 Probing Analysis

To understand how OntoBook organizes code representations, we evaluate with three probing tasks on CIM-10 code embeddings (mean-pooled over code descriptions): chapter classification (26 classes), hierarchy depth prediction (4 levels), and distance correlation (Spearman ρ between cosine distances and ontology tree distances). We compare against a code infilling variant (CodeInfill+MLM) that masks and predicts code tokens during pretraining.

CodeInfill+MLM achieves the best probing metrics ($\rho = 0.397$) but the worst downstream performance among our models (47.6 average micro-F1, Table 12.3), while OntoBook has lower distance correlation ($\rho = 0.073$) despite achieving the best downstream performance (52.5 micro-F1). This inverse relationship suggests that for encoder pretraining, representational flexibility is more valuable than strict geometric preservation of ontology structure.

12.5 Discussion

The 30.50-point gap between aligned and misaligned training highlights the central role of objective alignment. When $\mathcal{L}_{\text{MLM}}$ trains on textbooks while $\mathcal{L}_{\text{rel}}$ trains on unrelated data, the model receives conflicting gradient signals: the MLM objective pushes representations toward textbook language patterns, while the relation objective pushes them toward a different data distribution. Aligned training avoids this conflict by ensuring both objectives reinforce the same semantic structure. Relation prediction guides attention toward ontologically meaningful patterns in the textbook text, while MLM ensures the model builds coherent representations of that text. This explains why relation prediction alone fails (−12.68 F1): it is a discriminative task that can exploit surface-level shortcuts, as demonstrated by the small gap between raw-walk and textbook MLM-only models (+0.59 F1). Combined with aligned MLM, however, it provides a complementary signal that improves downstream coding performance.

The cross-ontology transfer results suggest that our approach generalizes beyond CIM-10: training on CCAM or ATC alone also improves over the baseline despite targeting different medical domains. However, combining all ontologies does not outperform individual ones, possibly because the relation prediction head must accommodate heterogeneous edge types.

The probing analysis reveals a tension between geometric organization and downstream performance. CodeInfill+MLM achieves the best probing metrics ($\rho = 0.397$) but the worst downstream scores among our models (47.6 micro-F1), while OntoBook achieves the opposite ($\rho = 0.073$, 52.5 micro-F1). This suggests that for downstream coding tasks, flexible representations adapt better than rigid geometric structure that must be partially unlearned during fine-tuning.

LLM reformulation plays a dual role. For MLM alone, the effect is modest (+0.59 F1), but reformulation is essential for relation prediction: on raw walks, structural markers provide formatting shortcuts that the classifier can exploit without learning medical semantics, whereas fluent prose forces the model to learn from content rather than format. The full gap between OntoBook and MLM-only on raw walks is 4.45 points.

Limitations. All experiments use French medical coding. Extending to English ontologies such as ICD-11 and SNOMED-CT is left for future work. The reformulation step requires significant compute for LLM inference, and sensitivity to the choice of LLM has not been evaluated. Cantemist-FR and Distemist-FR are cross-lingual projections from Spanish (Zaghir et al., 2024), which may introduce translation artifacts; FRACCO is the only native French benchmark. All experiments use ModernCamemBERT; generalization to other architectures remains untested.

Future Work. Future work will extend OntoBook to English medical coding using UMLS and SNOMED-CT ontologies. A promising direction is cross-ontology walks that traverse edges between different ontologies (e.g., from a CIM-10 diagnosis to its ATC treatment to the CCAM procedure), generating richer walks that capture inter-ontology relationships in a single sequence.

12.6 Conclusion

We presented OntoBook, a method that converts medical ontology structure into pretraining signal for encoder language models through random walks, LLM-based textbook reformulation, and multi-task learning combining masked language modeling and relation prediction. Our key finding is that alignment between objectives is essential: both tasks must train on the same data, as misalignment causes a 30-point degradation. On French medical coding benchmarks, OntoBook improves over MLM-only pretraining by +2.5 micro-F1 on FRACCO and +8.0 on Distemist. We release 1.3 million LLM-reformulated medical textbooks across three French ontologies (CIM-10, CCAM, ATC) and pretrained model checkpoints.

Part V

Adapting to Clinical Tasks

13 Limits of Direct Fine-Tuning

14 Architectures for Low-Resource Extraction

15 Synthetic Data for Task Adaptation

15.1 Introduction

Clinical text is scarce, but structured medical knowledge is not. France maintains medical ontologies that span tens of thousands of codes (CIM-10 for diagnoses, CCAM for procedures, ATC for medications), and PubMed Central contains millions of clinical case descriptions embedded in scientific articles. Both are publicly available, both are richly structured, and neither can be used directly to train a language model. The question is whether we can convert them into synthetic training data that is closer to what clinical applications actually need.

This chapter presents two instances of this idea. The first, OntoBook, generates synthetic medical textbooks from ontology graphs by reformulating random walks with a large language model. The second, currently in progress, reformulates the clinical case paragraphs extracted in Chapter 8 into synthetic clinical reports. Both share the same recipe: take a structured source, hand it to an LLM with a tight prompt, and get back fluent medical text that preserves the structure. They differ in input (graph vs. paragraph) and target (encoder pretraining vs. task adaptation). The full method and downstream results for OntoBook are presented in Chapter 12; here we focus on the data artifacts themselves.

15.2 Synthetic Textbooks from Medical Ontologies

15.2.1 Walk Generation

We generate random walks through ontology graphs to capture relational structure in a sequential format suitable for language model pretraining. We sample walks starting from each code in the ontology. At each step, the next node is selected via weighted sampling where edge type weights depend on the walk type. This biased sampling is necessary because semantic edges are rare (Table 12.2 in Chapter 12), and a uniform walk would rarely traverse them. Each walk records the traversed codes along with their labels and the relation types connecting them. Walk length is sampled uniformly between 8 and 20 steps. We track visited nodes to avoid cycles, with a fallback that allows revisiting nodes when all neighbors have been explored.

For CIM-10, we define five walk types, each emphasizing different aspects of medical knowledge:

- **Étiologie** (causal chains): traverses CAUSES and CAUSED_BY edges to follow disease etiology.
- **Diagnostic Différentiel**: traverses EXCLUSION edges to enumerate codes to distinguish clinically.
- **Codage Double**: follows mandatory code pairs linking etiology to manifestation, encoding the dual-coding rules of CIM-10.
- **Syndrome**: traverses MANIFESTATION edges to follow multi-system manifestations of a single underlying condition.
- **Cross-Chapter**: connects different ICD chapters, e.g., a metabolic diagnosis (E11, type 2 diabetes) and its renal complication (N08.3, diabetic glomerulopathy).

We generate 402,328 walks for CIM-10 (average 4,830 characters), 762,958 for CCAM, and 138,980 for ATC, totaling 1,304,266 walks across all three ontologies.

15.2.2 Walk-to-Textbook Reformulation

Raw walks contain structured markers (e.g., `>> À distinguer de:`) that are useful for parsing but not natural for language model pretraining. We use a large language model to reformulate walks into fluent medical prose resembling textbook paragraphs.

We prompt Qwen3-235B-A22B-Instruct (Yang et al., 2025a) with FP8 quantization to transform each walk into a coherent paragraph. The prompt instructs the model to: (1) preserve all medical codes and their relationships exactly as stated, (2) use natural medical language without lists or bullet points, (3) not add any information beyond what appears in the walk. This ensures the textbook content remains grounded in the ontology structure. We apply two filters: we discard reformulations shorter than 50 characters or that fail to mention the source codes. The reformulation runs on 4 NVIDIA H100 GPUs using vLLM (Kwon et al., 2023) with guided decoding to ensure valid JSON output, taking approximately 20 hours for all 1.3 million walks. The resulting textbooks contain approximately 1.3 billion tokens across all three ontologies.

Chapter 12 shows one walk-to-textbook transformation in detail (Figure 12.2). The same LLM prompt structure is used for CCAM and ATC, with ontology-specific terminology substituted for medical-coding domain terms.

15.2.3 The Role of Alignment

The key result of OntoBook, presented in Chapter 12, is that aligned multi-task training (MLM and relation prediction on the same textbook data) produces a +3.86 F_1 improvement on French medical coding, while misaligned training (different data per task) causes a −30.50 point degradation. The synthetic textbooks themselves are necessary but not sufficient: training MLM on raw structured walks (without LLM reformulation) yields nearly the same downstream performance as training on the reformulated textbooks (+0.59 F_1). The reformulation matters specifically for the relation prediction objective, where structural markers in raw walks would let the classifier exploit formatting shortcuts rather than medical semantics.

15.3 Synthetic Clinical Reports from Case Paragraphs

This section is a placeholder. The project that motivates it exists, but the paper is still in progress.

15.4 Two Instances, One Pattern

Conclusion

This is the last chapter of Part 3. The thesis conclusion revisits the research questions raised in the introduction and discusses what remains open.

16 Conclusion

Acronyms

PCA	Principal component analysis
SNF	Smith normal form
TDA	Topological data analysis

Glossary

LaTeX	A document preparation system
$\mathbb{R}$	The set of real numbers

Bibliography

Khushbu Agarwal, Tome Eftimov, Raghavendra Addanki, Sutanay Choudhury, Suzanne Tamang, and Robert Rallo. 2019. Snomed2vec: Random walk and poincaré embeddings of a clinical knowledge base for healthcare analytics. In *Proceedings of the KDD Workshop on Applied Data Science for Healthcare (DSHealth)*.

Monica Agrawal, Stefan Hegselmann, Hunter Lang, Yoon Kim, and David Sontag. 2022. Large language models are few-shot clinical information extractors. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 1998–2022. Association for Computational Linguistics.

Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebron, and Sumit Sanghai. 2023. GQA: Training generalized multi-query transformer models from multi-head checkpoints. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 4895–4901, Singapore. Association for Computational Linguistics.

Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. 2019. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of KDD*, pages 2623–2631.

Emily Alsentzer, John Murphy, William Boag, Wei-Hung Weng, Di Jindi, Tristan Naumann, and Matthew McDermott. 2019. Publicly available clinical BERT embeddings. In *Proceedings of the 2nd Clinical Natural Language Processing Workshop*, pages 72–78, Minneapolis, Minnesota, USA. Association for Computational Linguistics.

Bruce Anderson, Irwin D. J. Bross, and Naomi Sager. 1975. Grammatical compression in notes and records: Analysis and computation. *American Journal of Computational Linguistics*, 2(4):68–81.

Wissam Antoun, Francis Kulumba, Rian Touchent, Éric de la Clergerie, Benoît Sagot, and Djamé Seddah. 2024. CamemBERT 2.0: A smarter French language model aged to perfection.

Wissam Antoun, Benoît Sagot, and Djamé Seddah. 2023. Data-efficient French language modeling with CamemBERTa. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 5174–5185, Toronto, Canada. Association for Computational Linguistics.

Wissam Antoun, Benoît Sagot, and Djamé Seddah. 2025. ModernBERT or DeBERTaV3? Examining architecture and data influence on transformer encoder models performance. In *Proceedings of the International Joint Conference on Natural Language Processing and the Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP-AACL 2025)*.

Amanda Askell, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, Nicholas Joseph, Ben Mann, Nova DasSarma, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Jackson Kernion, Kamal Ndousse, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, and Jared Kaplan. 2021. A general language assistant as a laboratory for alignment.

John L. Austin. 1962. *How to Do Things with Words*. Oxford University Press, Oxford.

Dzmitry Bahdanau, Kyung Hyun Cho, and Yoshua Bengio. 2015. Neural machine translation by jointly learning to align and translate. 3rd International Conference on Learning Representations, ICLR 2015 ; Conference date: 07-05-2015 Through 09-05-2015.

Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion, Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Tristan Hume, Scott Johnston, Shauna Kravec, Liane Lovitt, Neel Nanda, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, Ben Mann, and Jared Kaplan. 2022a. Training a helpful and harmless assistant with reinforcement learning from human feedback.

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Catherine Olsson, Christopher Olah, Danny Hernandez, Dawn Drain, Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Kamile Lukosuite, Liane Lovitt, Michael Sellitto, Nelson Elhage, Nicholas Schiefer, Noemi Mercado, Nova DasSarma, Robert Lasenby, Robin Larson, Sam Ringer, Scott Johnston, Shauna Kravec, Sheer El Showk, Stanislav Fort, Tamera Lanham, Timothy Telleen-Lawton, Tom Conerly, Tom Henighan, Tristan Hume, Samuel R. Bowman, Zac Hatfield-Dodds, Ben Mann, Dario Amodei, Nicholas Joseph, Sam McCandlish, Tom Brown, and Jared Kaplan. 2022b. Constitutional AI: Harmlessness from AI feedback.

Simon Baker, Ilona Silins, Yufan Guo, Imran Ali, Johan Högberg, Ulla Stenius, and Anna Korhonen. 2016. Automatic semantic classification of scientific literature according to the hallmarks of cancer. *Bioinformatics*, 32(3):432–440.

Iz Beltagy, Kyle Lo, and Arman Cohan. 2019. SciBERT: A pretrained language model for scientific text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3615–3620, Hong Kong, China. Association for Computational Linguistics.

Iz Beltagy, Matthew E Peters, and Arman Cohan. 2020. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*.

Yoshua Bengio, Réjean Ducharme, Pascal Vincent, and Christian Jauvin. 2003. A neural probabilistic language model. *Journal of Machine Learning Research*, 3:1137–1155.

Yoshua Bengio, Patrice Simard, and Paolo Frasconi. 1994. Learning long-term dependencies with gradient descent is difficult. *IEEE Transactions on Neural Networks*, 5(2):157–166.

Aman Berhe, Guillaume Draznieks, Vincent Martenot, Valentin Masdeu, Lucas Davy, and Jean-Daniel Zucker. 2023. AliBERT: A pre-trained language model for French biomedical text. In *Proceedings of the 22nd Workshop on Biomedical Natural Language Processing and BioNLP Shared Tasks*, pages 223–236, Toronto, Canada. Association for Computational Linguistics.

Douglas Biber. 1988. *Variation across Speech and Writing*. Cambridge University Press, Cambridge.

Douglas Biber and Edward Finegan. 1994. Intra-textual variation within medical research articles. In Nelleke Oostdijk and Pieter de Haan, editors, *Corpus-based Research into Language*, pages 201–221. Rodopi, Amsterdam.

Olivier Bodenreider. 2004. The Unified Medical Language System (UMLS): integrating biomedical terminology. *Nucleic Acids Research*, 32(Database issue):D267–D270.

Sergei Bogdanov, Alexandre Constantin, Timothée Bernard, Benoit Crabbé, and Etienne P Bernard. 2024. NuNER: Entity recognition encoder pre-training via LLM-annotated data. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 11829–11841. Association for Computational Linguistics.

Àlex Bravo, Janet Piñero, Núria Queralt-Rosinach, Michael Rautschka, and Laura I. Furlong. 2015. Extraction of relations between genes and diseases from text and large-scale data analysis: implications for translational research. *BMC Bioinformatics*, 16(1):55.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901. Curran Associates, Inc.

M. Teresa Cabré. 2003. Theories of terminology: Their description, prescription and explanation. *Terminology*, 9(2):163–199.

Pengfei Cao, Yubo Chen, Kang Liu, Jun Zhao, Shengping Liu, and Weifeng Chong. 2020. HyperCore: Hyperbolic and co-graph representation for automatic ICD coding. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 3105–3114. Association for Computational Linguistics.

Rémi Cardon, Natalia Grabar, Cyril Grouin, and Thierry Hamon. 2020. Présentation de la campagne d'évaluation DEFT 2020. In *Actes de TALN 2020, Atelier DÉfi Fouille de Textes*, pages 1–13. ATALA et AFCP.

Wendy W. Chapman, Will Bridewell, Paul Hanbury, Gregory F. Cooper, and Bruce G. Buchanan. 2001. A simple algorithm for identifying negated findings and diseases in discharge summaries. *Journal of Biomedical Informatics*, 34(5):301–310.

Jiaoyan Chen, Pan Hu, Ernesto Jimenez-Ruiz, Ole Magnus Holter, Denvar Antonyrajah, and Ian Horrocks. 2021. OWL2Vec*: Embedding of OWL ontologies. *Machine Learning*, 110(7):1813–1845.

Zeming Chen, Alejandro Hernández Cano, Angelika Romanou, Antoine Bonnet, Kyle Matoba, Francesco Salvi, Matteo Pagliardini, Simin Fan, Andreas Köpf, Amirkeivan Mohtashami, Alexandre Sallinen, Alireza Sakhaeirad, Vinitra Swamy, Igor Krawczuk, Deniz Bayazit, Axel Marmet, Syrielle Montariol, Mary-Anne Hartley, Martin Jaggi, and Antoine Bosselut. 2023. MEDITRON-70B: Scaling medical pretraining for large language models.

Bharath Chintagunta, Namit Katariya, Xavier Amatriain, and Anitha Kannan. 2021. Medically aware GPT-3 as a data generator for medical dialogue summarization. In *Proceedings of the Second Workshop on Natural Language Processing for Medical Conversations*, pages 66–76. Association for Computational Linguistics.

Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, pages 4299–4307.

Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, Albert Webson, Shixiang Shane Gu, Zhuyun Dai, Mirac Suzgun, Xinyun Chen, Aakanksha Chowdhery, Alex Castro-Ros, Marie Pellat, Kevin Robinson, Dasha Valter, Sharan Narang, Gaurav Mishra, Adams Yu, Vincent Zhao, Yanping Huang, Andrew Dai, Hongkun Yu, Slav Petrov, Ed H. Chi, Jeff Dean, Jacob Devlin, Adam Roberts, Denny Zhou, Quoc V. Le, and Jason Wei. 2024. Scaling instruction-finetuned language models. *Journal of Machine Learning Research*, 25(70):1–53.

Junyoung Chung, Caglar Gulcehre, Kyunghyun Cho, and Yoshua Bengio. 2014. Empirical evaluation of gated recurrent neural networks on sequence modeling. In *NIPS 2014 Workshop on Deep Learning, December 2014*.

Kevin Clark, Minh-Thang Luong, Quoc V. Le, and Christopher D. Manning. 2020. Electra: Pre-training text encoders as discriminators rather than generators. In *International Conference on Learning Representations*.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

Nigel Collier, Tomoko Ohta, Yoshimasa Tsuruoka, Yuka Tateisi, and Jin-Dong Kim. 2004. Introduction to the bio-entity recognition task at JNLPBA. In *Proceedings of the International Joint Workshop on Natural Language Processing in Biomedicine and its Applications (NLPBA/BioNLP)*, pages 73–78. COLING.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451. Association for Computational Linguistics.

Jenny Copara, Julien Knafou, Nona Naderi, Claudia Moro, Patrick Ruch, and Douglas Teodoro. 2020. Contextualized French Language Models for Biomedical Named Entity Recognition. In *Actes de TALN 2020, Atelier DÉfi Fouille de Textes*, pages 36–48. ATALA.

Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. 2022. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*.

Chunyuan Deng, Yilun Zhao, Xiangru Tang, Mark Gerstein, and Arman Cohan. 2024. Investigating data contamination in modern benchmarks for large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 8706–8719. Association for Computational Linguistics.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4171–4186. Association for Computational Linguistics.

Jesse Dodge, Maarten Sap, Ana Marasović, William Agnew, Gabriel Ilharco, Dirk Groeneveld, Margaret Mitchell, and Matt Gardner. 2021. Documenting large webtext corpora: A case study on the colossal clean crawled corpus. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 1286–1305. Association for Computational Linguistics.

Rezarta Islamaj Doğan, Robert Leaman, and Zhiyong Lu. 2014. NCBI disease corpus: A resource for disease name recognition and concept normalization. *Journal of Biomedical Informatics*, 47:1–10.

Felix J. Dorfner et al. 2024. Biomedical large languages models seem not to be superior to generalist models on unseen medical data.

Basile Dura, Charline Jean, Xavier Tannier, Alice Calliger, Romain Bey, Antoine Neuraz, and Rémi Flicoteaux. 2022. Learning structures of the French clinical language. Technical report, arXiv. ArXiv:2207.12940.

European Medicines Agency. 2025. Product-information (QRD) templates — human. European Medicines Agency. Quality Review of Documents; human product-information template.

European Parliament and Council of the European Union. 2001. Directive 2001/83/EC on the community code relating to medicinal products for human use. Official Journal of the European Union L 311, 28.11.2001, pp. 67–128. CELEX 32001L0083; consolidated text; articles 59, 61, 63.

Ludwik Fleck. 1979. *Genesis and Development of a Scientific Fact*. University of Chicago Press, Chicago. German original 1935.

Michel Foucault. 1963. *Naissance de la clinique : une archéologie du regard médical*. Presses Universitaires de France, Paris.

Susannah Fox and Maeve Duggan. 2013. Health online 2013. Pew Research Center, Washington, DC.

Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, et al. 2020. The Pile: An 800GB dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*.

François Gaudin. 2003. *Socioterminologie : une approche sociolinguistique de la terminologie*. De Boeck/Duculot, Bruxelles.

Hippolyte Gisserot-Boukhlef et al. 2025. Should we still pretrain encoders with masked language modeling? *arXiv preprint arXiv:2507.00994*.

Nathan Godey, Wissam Antoun, Rian Touchent, Rachel Bawden, Éric de la Clergerie, Benoît Sagot, and Djamé Seddah. 2026. Gaperon: A peppered English-French generative language model suite. *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*. To appear.

Natalia Grabar and Rémi Cardon. 2018. CLEAR – Simple Corpus for Medical French. In *Proceedings of the 1st Workshop on Automatic Text Adaptation*, pages 3–9. Association for Computational Linguistics.

Natalia Grabar, Clément Dalloux, and Vincent Claveau. 2020. CAS: corpus of clinical cases in French. *Journal of Biomedical Semantics*, 11(1):7.

Aaron Grattafiori, Abhimanyu Dubey, et al. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

H. Paul Grice. 1975. Logic and conversation. In Peter Cole and Jerry L. Morgan, editors, *Syntax and Semantics, Vol. 3: Speech Acts*, pages 41–58. Academic Press, New York.

Cyril Grouin, Natalia Grabar, Vincent Claveau, and Thierry Hamon. 2019. Clinical case reports for NLP. In *Proceedings of the 18th BioNLP Workshop and Shared Task*, pages 273–282. Association for Computational Linguistics.

Aditya Grover and Jure Leskovec. 2016. node2vec: Scalable feature learning for networks. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 855–864. ACM.

Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. 2022. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1):1–23.

Suriya Gunasekar, Yi Zhang, Jyoti Aneja, Caio César Teodoro Mendes, Allie Del Giorno, Sivakanth Gopi, Mojan Javaheripi, Piero Kauffmann, Gustavo de Rosa, Olli Saarikivi, et al. 2023. Textbooks are all you need. *CoRR*, abs/2306.11644.

Daya Guo et al. 2025a. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645(8081):633–638.

Yiduo Guo, Jie Fu, Huishuai Zhang, and Dongyan Zhao. 2025b. Efficient domain continual pretraining by mitigating the stability gap. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 32850–32870, Vienna, Austria. Association for Computational Linguistics.

Suchin Gururangan, Ana Marasović, Swabha Swayamdipta, Kyle Lo, Iz Beltagy, Doug Downey, and Noah A. Smith. 2020. Don't stop pretraining: Adapt language models to domains and tasks. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8342–8360, Online. Association for Computational Linguistics.

Zellig S. Harris. 1968. *Mathematical Structures of Language*. Interscience Tracts in Pure and Applied Mathematics. Wiley-Interscience, New York.

Zellig S. Harris, Michael Gottfried, Thomas Ryckman, Paul Mattick, Anne Daladier, T. N. Harris, and S. Harris. 1989. *The Form of Information in Science: Analysis of an Immunology Sublanguage*, volume 104 of *Boston Studies in the Philosophy of Science*. Kluwer Academic, Dordrecht.

Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. 2021. {DEBERTA}: {DECODING}-{enhanced} {bert} {with} {disentangled} {attention}. In *International Conference on Learning Representations*.

Sam Henry, Kevin Buchan, Michele Filannino, Amber Stubbs, and Ozlem Uzuner. 2020. 2018 n2c2 shared task on adverse drug events and medication extraction in electronic health records. *Journal of the American Medical Informatics Association*, 27(1):3–12.

Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long short-term memory. *Neural computation*, 9(8):1735–1780.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. 2022. Training compute-optimal large language models.

Chao-Wei Huang, Shang-Chi Tsai, and Yun-Nung Chen. 2022. PLM-ICD: Automatic ICD coding with pretrained language models. In *Proceedings of the 4th Clinical Natural Language Processing Workshop*, pages 10–20. ACL.

Kexin Huang, Jaan Altosaar, and Rajesh Ranganath. 2019. Clinicalbert: Modeling clinical notes and predicting hospital readmission. *arXiv preprint arXiv:1904.05342*.

Ken Hyland. 2005. *Metadiscourse: Exploring Interaction in Writing*. Continuum, London.

Ken Hyland and Polly Tse. 2004. Metadiscourse in academic writing: A reappraisal. *Applied Linguistics*, 25(2):156–177.

Adam Ibrahim, Benjamin Thérien, Kshitij Gupta, Mats L. Richter, Quentin Anthony, Timothée Lesort, Eugene Belilovsky, and Irina Rish. 2024. Simple and scalable strategies to continually pre-train large language models. *Transactions on Machine Learning Research*, 2024. ArXiv:2403.08763.

Julia Ive, Natalia Viani, Joyce Kam, Lucia Yin, Somain Verma, Stephen Puntis, Rudolf N. Cardinal, Angus Roberts, Robert Stewart, and Sumithra Velupillai. 2020. Generation and evaluation of artificial mental health records for natural language processing. *npj Digital Medicine*, 3(1):69.

Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. 2021. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. In *Applied Sciences*.

Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. 2019. PubMedQA: A dataset for biomedical research question answering. In *Proceedings* *of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2567–2577, Hong Kong, China. Association for Computational Linguistics.

Alistair E. W. Johnson, Lucas Bulgarelli, Lu Shen, Alvin Gayles, Ayad Shammout, Steven Horng, Tom J. Pollard, Sicheng Hao, Benjamin Moody, Brian Gow, Li-wei H. Lehman, Leo Anthony Celi, and Roger G. Mark. 2023. Mimic-iv, a freely accessible electronic health record dataset. *Scientific Data*, 10(1):1.

Alistair E. W. Johnson, Tom J. Pollard, Lu Shen, Li-wei H. Lehman, Mengling Feng, Mohammad Ghassemi, Benjamin Moody, Peter Szolovits, Leo Anthony Celi, and Roger G. Mark. 2016. Mimic-iii, a freely accessible critical care database. *Scientific Data*, 3(1):160035.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeff Wu, and Dario Amodei. 2020. Scaling laws for neural language models. *ArXiv*, abs/2001.08361.

Diederik P. Kingma and Jimmy Ba. 2015. Adam: A method for stochastic optimization. ICLR 2015. arXiv:1412.6980.

Matthias Kirchler, Marcelo Ferro, Valentina Lorenzini, et al. 2026. Large language models improve transferability of electronic health record-based predictions across countries and coding systems. *npj Digital Medicine*.

James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. 2017. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526.

Richard Kittredge. 1982. Sublanguages. *American Journal of Computational Linguistics*, 8(2):79–84.

Reinhard Kneser and Hermann Ney. 1995. Improved backing-off for M-gram language modeling. In *1995 International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, volume 1, pages 181–184.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. Similarity of neural network representations revisited. In *ICML*.

Martin Krallinger, Obdulia Rabal, Saber A. Akhondi, et al. 2017. Overview of the BioCreative VI chemical-protein interaction track. In *Proceedings of the Sixth BioCreative Challenge Evaluation Workshop*, volume 1, pages 141–146. BioCreative VI workshop proceedings.

Taku Kudo. 2018. Subword regularization: Improving neural network translation models with multiple subword candidates. In *Proceedings of the 56th Annual Meeting* *of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 66–75, Melbourne, Australia. Association for Computational Linguistics.

Taku Kudo and John Richardson. 2018. SentencePiece: A simple and language independent subword tokenizer and detokenizer for neural text processing. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 66–71. Association for Computational Linguistics.

Francis Kulumba, Wissam Antoun, Guillaume Vimont, and Laurent Romary. 2024. Harvesting textual and structured data from the HAL publication repository. *CoRR*, abs/2407.20595.

Sunjun Kweon, Junu Kim, Jiyoun Kim, Sujeong Im, Eunbyeol Cho, Seongsu Bae, Jungwoo Oh, Gyubok Lee, Jong Hak Moon, Seng Chan You, Seungjin Baek, Chang Hoon Han, Yoon Bin Jung, Yohan Jo, and Edward Choi. 2024. Publicly shareable clinical large language model built on synthetic clinical notes. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 5148–5168. Association for Computational Linguistics.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th ACM Symposium on Operating Systems Principles*, pages 611–626. ACM.

Yanis Labrak, Adrien Bazoge, Richard Dufour, Béatrice Daille, Pierre-Antoine Gourraud, Emmanuel Morin, and Mickael Rouvier. 2022. FrenchMedMCQA: A French multiple-choice question answering dataset for medical domain. In *Proceedings of the LREC 2022 Workshop on Multilingual Biomedical Text Processing*.

Yanis Labrak, Adrien Bazoge, Richard Dufour, Mickael Rouvier, Emmanuel Morin, Béatrice Daille, and Pierre-Antoine Gourraud. 2023. DrBERT: A robust pre-trained model in French for biomedical and clinical domains. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16207–16221, Toronto, Canada. Association for Computational Linguistics.

Yanis Labrak, Adrien Bazoge, Oumaima El Khettari, Mickael Rouvier, Pacome Constant Dit Beaufils, Natalia Grabar, Béatrice Daille, Solen Quiniou, Emmanuel Morin, Pierre-Antoine Gourraud, and Richard Dufour. 2024a. DrBenchmark: A large language understanding evaluation benchmark for French biomedical domain. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 5376–5390. ELRA and ICCL.

Yanis Labrak, Adrien Bazoge, Emmanuel Morin, Pierre-Antoine Gourraud, Mickael Rouvier, and Richard Dufour. 2024b. BioMistral: A collection of open-source pretrained large language models for medical domains. In *Findings of the Association* *for Computational Linguistics: ACL 2024*, pages 5848–5864, Bangkok, Thailand. Association for Computational Linguistics.

John D. Lafferty, Andrew McCallum, and Fernando C. N. Pereira. 2001. Conditional random fields: Probabilistic models for segmenting and labeling sequence data. In *Proceedings of the Eighteenth International Conference on Machine Learning (ICML 2001)*, pages 282–289. Morgan Kaufmann.

Nathan Lambert, Jacob Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James V. Miranda, Alisa Liu, et al. 2024. Tulu 3: Pushing frontiers in open language model post-training.

Guillaume Lample, Miguel Ballesteros, Sandeep Subramanian, Kazuya Kawakami, and Chris Dyer. 2016. Neural architectures for named entity recognition. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 260–270. Association for Computational Linguistics.

Bruno Latour and Steve Woolgar. 1979. *Laboratory Life: The Construction of Scientific Facts*. Sage, Beverly Hills, CA. 2nd ed., Princeton University Press, 1986.

Hugo Laurençon, Lucile Saulnier, Thomas Wang, Christopher Akiki, Albert Villanova del Moral, Teven Le Scao, Leandro von Werra, Chenghao Mou, Eduardo González Ponferrada, Huu Nguyen, Jörg Frohberg, Mario Sasko, Quentin Lhoest, Angelina McMillan-Major, Gérard Dupont, Stella Biderman, Anna Rogers, Loubna Ben Allal, Francesco De Toni, Giada Pistilli, Olivier Nguyen, Somaieh Nikpoor, Maraim Masoud, Pierre Colombo, Javier de la Rosa, Paulo Villegas, Tristan Thrush, Shayne Longpre, Sebastian Nagel, Leon Weber, Manuel Muñoz, Jian Zhu, Daniel van Strien, Zaid Alyafeai, Khalid Almubarak, Minh Chien Vu, Itziar Gonzalez-Dios, Aitor Soroa, Kyle Lo, Manan Dey, Pedro Ortiz Suarez, Aaron Gokaslan, Shamik Bose, David Ifeoluwa Adelani, Long Phan, Hieu Tran, Ian Yu, Suhas Pai, Jenny Chim, Violette Lepercq, Suzana Ilic, Margaret Mitchell, Alexandra Sasha Luccioni, and Yacine Jernite. 2022. The bigscience ROOTS corpus: A 1.6tb composite multilingual dataset. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022) Datasets and Benchmarks Track*.

Hang Le, Loïc Vial, Jibril Frej, Vincent Segonne, Maximin Coavoux, Benjamin Lecouteux, Alexandre Allauzen, Benoit Crabbé, Laurent Besacier, and Didier Schwab. 2020. FlauBERT: Unsupervised language model pre-training for French. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 2479–2490, Marseille, France. European Language Resources Association.

Tiphaine Le Clercq de Lannoy, Romaric Besançon, Olivier Ferret, Julien Tourille, Frédérique Brin-Henry, and Bianca Vieru. 2022. Stratégies d'adaptation pour la reconnaissance d'entités médicales en français. In *Actes de la 29e Conférence sur le Traitement Automatique des Langues Naturelles*, pages 215–225. ATALA.

Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. 2020. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240.

Eric Lehman, Evan Hernandez, Diwakar Mahajan, Jonas Wulff, Micah J Smith, Zachary Ziegler, Daniel Nadler, Peter Szolovits, Alistair Johnson, and Emily Alsentzer. 2023. Do we still need clinical language models? In *Proceedings of the Conference on Health, Inference, and Learning*, pages 578–597.

Jeffrey Li, Alex Fang, Georgios Smyrnis, et al. 2024. DataComp-LM: In search of the next generation of training sets for language models. *arXiv preprint arXiv:2406.11794*.

Jiao Li, Yueping Sun, Robin J. Johnson, Daniela Sciaky, Chih-Hsuan Wei, Robert Leaman, Allan Peter Davis, Carolyn J. Mattingly, Thomas C. Wiegers, and Zhiyong Lu. 2016. BioCreative V CDR task corpus: a resource for chemical disease relation extraction. *Database*, 2016:baw068.

D. A. Lindberg, B. L. Humphreys, and A. T. McCray. 1993. The Unified Medical Language System. *Methods of Information in Medicine*, 32(4):281–291.

Fangyu Liu, Ehsan Shareghi, Zaiqiao Meng, Marco Basaldella, and Nigel Collier. 2021. Self-alignment pretraining for biomedical entity representations. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4228–4238. ACL.

Jiacheng Liu, Sewon Min, Luke Zettlemoyer, Yejin Choi, and Hannaneh Hajishirzi. 2024. Infini-gram: Scaling unbounded n-gram language models to a trillion tokens. *CoRR*, abs/2401.17377. Presented at COLM 2024.

Weijie Liu, Peng Zhou, Zhe Zhao, Zhiruo Wang, Qi Ju, Haotang Deng, and Ping Wang. 2020. K-BERT: Enabling language representation with knowledge graph. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 2901–2908.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach.

LLM360. 2024. TxT360: A top-quality LLM pre-training dataset requires the perfect blend. Hugging Face dataset.

María José Luzón Marco. 2000. Collocational frameworks in medical research papers: a genre-based study. *English for Specific Purposes*, 19(1):63–86.

Bernardo Magnini, Begoña Altuna, Alberto Lavelli, Manuela Speranza, and Roberto Zanoli. 2020. The E3C project: Collection and annotation of a multilingual corpus of clinical cases. *Proceedings of the Seventh Italian Conference on Computational Linguistics CLiC-it 2020*.

Andrei A. Markov. 2006. An example of statistical investigation of the text Eugene Onegin concerning the connection of samples in chains. *Science in Context*, 19(4):591–600. English translation of the 1913 lecture.

Louis Martin, Benjamin Muller, Pedro Javier Ortiz Suárez, Yoann Dupont, Laurent Romary, Éric de la Clergerie, Djamé Seddah, and Benoît Sagot. 2020. CamemBERT: a tasty French language model. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7203–7219, Online. Association for Computational Linguistics.

Michael McCloskey and Neal J. Cohen. 1989. Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of Learning and Motivation*, volume 24, pages 109–165. Elsevier.

Tomas Mikolov, Wen-tau Yih, and Geoffrey Zweig. 2013. Linguistic regularities in continuous space word representations. In *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 746–751, Atlanta, Georgia. Association for Computational Linguistics.

Tomáš Mikolov, Martin Karafiát, Lukáš Burget, Jan Černocký, and Sanjeev Khudanpur. 2010. Recurrent neural network based language model. In *Proc. Interspeech 2010*, pages 1045–1048.

Carolyn R. Miller. 1984. Genre as social action. *Quarterly Journal of Speech*, 70(2):151–167.

Antonio Miranda-Escalada, Eulàlia Farré-Maduell, and Martin Krallinger. 2020. Named entity recognition, concept normalization and clinical coding: Overview of the Cantemist track for cancer text mining in Spanish, corpus, guidelines, methods and results. In *Proceedings of the Iberian Languages Evaluation Forum (IberLEF 2020), CEUR Workshop Proceedings*, volume 2664, pages 303–323.

Antonio Miranda-Escalada, Luis Gascó, Salvador Lima-López, Eulàlia Farré-Maduell, Darryl Estrada, Anastasios Nentidis, Anastasia Krithara, Georgios Katsimpras, Georgios Paliouras, and Martin Krallinger. 2022. Overview of DisTEMIST at BioASQ: Automatic detection and normalization of diseases from clinical texts: Results, methods, evaluation and multilingual resources. In *Working Notes of the Conference and Labs of the Evaluation Forum (CLEF 2022), CEUR Workshop Proceedings*, volume 3180, pages 179–203.

James Mullenbach, Sarah Wiegreffe, Jon Duke, Jimeng Sun, and Jacob Eisenstein. 2018. Explainable prediction of medical codes from clinical text. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 1101–1111. Association for Computational Linguistics.

Hiroki Nakayama. 2018. seqeval: A Python framework for sequence labeling evaluation. Https://github.com/chakki-works/seqeval.

Aurélie Névéol, K. Bretonnel Cohen, Cyril Grouin, Thierry Hamon, Thomas Lavergne, Liadh Kelly, Lorraine Goeuriot, Grégoire Rey, Aude Robert, Xavier Tannier, and Pierre Zweigenbaum. 2016. Clinical information extraction at the CLEF eHealth evaluation lab 2016. *CEUR Workshop Proceedings*, 1609:28–42.

Aurélie Névéol, Hercules Dalianis, Sumithra Velupillai, Guergana Savova, and Pierre Zweigenbaum. 2018. Clinical natural language processing in languages other than English: opportunities and challenges. *Journal of Biomedical Semantics*, 9(1):12.

Aurélie Névéol, Cyril Grouin, Jeremy Leixa, Sophie Rosset, and Pierre Zweigenbaum. 2014. The QUAERO French medical corpus: A resource for medical entity recognition and normalization. In *Proc. of BioTextMining Workshop*, pages 24–30.

Kevin Ngozi Nwogu. 1997. The medical research paper: Structure and functions. *English for Specific Purposes*, 16(2):119–138.

Pedro Javier Ortiz Suárez, Benoît Sagot, and Laurent Romary. 2019. Asynchronous pipelines for processing huge corpora on medium to low resource infrastructures. In *Proceedings of the Workshop on Challenges in the Management of Large Corpora (CMLC-7)*, pages 9–16. Leibniz-Institut für Deutsche Sprache.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke E. Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, pages 27730–27744.

Ankit Pal, Logesh Kumar Umapathi, and Malaikannan Sankarasubbu. 2022. MedMCQA: A large-scale multi-subject multi-choice dataset for medical domain question answering. In *Proceedings of the Conference on Health, Inference, and Learning*.

Razvan Pascanu, Tomas Mikolov, and Yoshua Bengio. 2013. On the difficulty of training recurrent neural networks. In *Proceedings of the 30th International Conference on Machine Learning (ICML)*, volume 28 of *PMLR*, pages 1310–1318.

Guilherme Penedo, Hynek Kydlíček, Loubna Cappelli, Hailey Saez de Ocáriz Borde, et al. 2024. FineWeb: decanting the web for the finest text data at scale. *arXiv preprint arXiv:2406.17557*.

Matthew E. Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, and Luke Zettlemoyer. 2018. Deep contextualized word representations. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1*

(*Long Papers*), pages 2227–2237, New Orleans, Louisiana. Association for Computational Linguistics.

Matthew E. Peters, Mark Neumann, Robert Logan, Roy Schwartz, Vidur Joshi, Sameer Singh, and Noah A. Smith. 2019. Knowledge enhanced contextual word representations. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 43–54. Association for Computational Linguistics.

Johann Pignat, Milena Vucetic, Christophe Gaudet-Blavignac, Jamil Zaghir, Amandine Stettler, Fanny Amrein, Jonatan Bonjour, Jean-Philippe Goldman, Olivier Michielin, Christian Lovis, and Mina Bjelogrlic. 2025. FRACCO: A gold-standard annotated corpus of oncological entities with ICD-O-3.1 normalisation.

Jacob Portes, Alexander Trott, Sam Havens, Daniel King, Abhinav Venigalla, Moin Nadeem, Nikhil Sardana, Daya Khudia, and Jonathan Frankle. 2023. MosaicBERT: A bidirectional encoder optimized for fast pretraining. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*.

Ofir Press, Noah Smith, and Mike Lewis. 2022. Train short, test long: Attention with linear biases enables input length extrapolation. In *International Conference on Learning Representations*.

Sampo Pyysalo and Sophia Ananiadou. 2014. Anatomical entity mention recognition at literature scale. *Bioinformatics*, 30(6):868–875.

Alec Radford and Karthik Narasimhan. 2018. Improving language understanding by generative pre-training.

Jack W. Rae, Sebastian Borgeaud, Trevor Cai, Katie Millican, Jordan Hoffmann, et al. 2021. Scaling language models: Methods, analysis & insights from training gopher. *CoRR*, abs/2112.11446.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67.

Preethi Raghavan, James L. Chen, Eric Fosler-Lussier, and Albert M. Lai. 2014. How essential are unstructured clinical narratives and information fusion to clinical trial recruitment? *AMIA Summits on Translational Science Proceedings*, 2014:218–223.

David Rein, Betty Li Hou, Asa Cooper Stickland, et al. 2023. GPQA: A graduate-level google-proof q&a benchmark. *arXiv preprint arXiv:2311.12022*.

Petar Ristoski and Heiko Paulheim. 2016. RDF2Vec: RDF graph embeddings for data mining. In *International Semantic Web Conference*, pages 498–514. Springer.

David E. Rumelhart and James L. McClelland. 1987. *Learning Internal Representations by Error Propagation*, pages 318–362.

Naomi Sager. 1981. *Natural Language Information Processing: A Computer Grammar of English and Its Applications*. Addison-Wesley, Reading, MA.

Naomi Sager, Carol Friedman, and Margaret S. Lyman. 1987. *Medical Language Processing: Computer Management of Narrative Data*. Addison-Wesley, Reading, MA.

Oscar Sainz, Iker García-Ferrero, Rodrigo Agerri, Oier Lopez de Lacalle, German Rigau, and Eneko Agirre. 2024. GoLLIE: Annotation guidelines improve zero-shot information-extraction. In *The Twelfth International Conference on Learning Representations (ICLR 2024)*. OpenReview.net.

Victor Sanh, Albert Webson, Colin Raffel, Stephen H. Bach, Lintang Sutawika, Zaid Alyafeai, Antoine Chaffin, Arnaud Stiegler, Teven Le Scao, Arun Raja, Manan Dey, M Saiful Bari, Canwen Xu, Urmish Thakker, Shanya Sharma Sharma, Eliza Szczechla, Taewoon Kim, Gunjan Chhablani, Nihal Nayak, Debajyoti Datta, Jonathan Chang, Mike Tian-Jian Jiang, Han Wang, Matteo Manica, Sheng Shen, Zheng Xin Yong, Harshit Pandey, Rachel Bawden, Thomas Wang, Trishala Neeraj, Jos Rozen, Abheesht Sharma, Andrea Santilli, Thibault Fevry, Jason Alan Fries, Ryan Teehan, Tali Bers, Stella Biderman, Leo Gao, Thomas Wolf, and Alexander M. Rush. 2022. Multitask prompted training enables zero-shot task generalization. In *The Tenth International Conference on Learning Representations (ICLR 2022)*.

John R. Searle. 1969. *Speech Acts: An Essay in the Philosophy of Language*. Cambridge University Press, Cambridge.

Rico Sennrich, Barry Haddow, and Alexandra Birch. 2016. Neural machine translation of rare words with subword units. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1715–1725, Berlin, Germany. Association for Computational Linguistics.

Claude E. Shannon. 1951. Prediction and entropy of printed English. *The Bell System Technical Journal*, 30(1):50–64.

Claude Elwood Shannon. 1948. A mathematical theory of communication. *The Bell System Technical Journal*, 27:379–423.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Yu Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models.

Supreeth P Shashikumar, Fatemeh Amrollahi, and Shamim Nemati. 2023. Leveraging biomedical ontologies to boost pre-training for language models. In *Proceedings of the International Conference on Biomedical Ontologies (ICBO)*, volume 3603 of *CEUR Workshop Proceedings*.

Haizhou Shi, Zihao Xu, Hengyi Wang, Weiyi Qin, Wenyuan Wang, Yibin Wang, Zifeng Wang, Sayna Ebrahimi, and Hao Wang. 2025. Continual learning of large language models: A comprehensive survey. *ACM Computing Surveys*, 58(5):1–42.

Luca Soldaini, Rodney Kinney, Akshita Bhagia, Dustin Schwenk, David Atkinson, Russell Authur, Ben Bogin, Khyathi Chandu, Jennifer Dumas, Yanai Elazar, Valentin Hofmann, Ananya Jha, Sachin Kumar, Li Lucy, Xinxi Lyu, Nathan Lambert, Ian Magnusson, Jacob Morrison, Niklas Muennighoff, Aakanksha Naik, Crystal Nam, Matthew Peters, Abhilasha Ravichander, Kyle Richardson, Zejiang Shen, Emma Strubell, Nishant Subramani, Oyvind Tafjord, Evan Walsh, Luke Zettlemoyer, Noah Smith, Hannaneh Hajishirzi, Iz Beltagy, Dirk Groeneveld, Jesse Dodge, and Kyle Lo. 2024. Dolma: an open corpus of three trillion tokens for language model pretraining research. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15725–15788. Association for Computational Linguistics.

Luciana B. Sollaci and Mauricio G. Pereira. 2004. The introduction, methods, results, and discussion (IMRAD) structure: a fifty-year survey. *Journal of the Medical Library Association*, 92(3):364–371.

Ali Soroush, Benjamin S. Glicksberg, Eyal Zimlichman, Yiftach Barash, Robert Freeman, Alexander W. Charney, Girish N. Nadkarni, and Eyal Klang. 2024. Large language models are poor medical coders — benchmarking of medical code querying. *NEJM AI*, 1(5).

Thomas Sounack, Joshua Davis, Brigitte Durieux, Antoine Chaffin, Tom J. Pollard, Eric Lehman, Alistair E. W. Johnson, Matthew McDermott, Tristan Naumann, and Charlotta Lindvall. 2025. BioClinical ModernBERT: A state-of-the-art long-context encoder for biomedical and clinical NLP.

Dan Sperber and Deirdre Wilson. 1995. *Relevance: Communication and Cognition*, 2nd edition. Blackwell, Oxford. 1st ed. 1986.

Robert Stalnaker. 2002. Common ground. *Linguistics and Philosophy*, 25(5–6):701–721.

Dan Su, Kezhi Kong, Ying Lin, Joseph Jennings, Brandon Norick, Markus Kliegl, Mostofa Patwary, Mohammad Shoeybi, and Bryan Catanzaro. 2025. Nemotron-cc: Transforming common crawl into a refined long-horizon pretraining dataset. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2459–2475. Association for Computational Linguistics.

Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. 2024. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063.

Tianxiang Sun, Yunfan Shao, Xipeng Qiu, Qipeng Guo, Yaru Hu, Xuanjing Huang, and Zheng Zhang. 2020. CoLAKE: Contextualized language and knowledge embedding. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 3660–3670. International Committee on Computational Linguistics.

Mujeen Sung, Hwisang Jeon, Jinhyuk Lee, and Jaewoo Kang. 2020. Biomedical entity representations with synonym marginalization. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 3641–3650. Association for Computational Linguistics.

John M. Swales. 1990. *Genre Analysis: English in Academic and Research Settings*. Cambridge University Press, Cambridge.

Xavier Tannier, Salam Abbara, Rémi Flicoteaux, Youness Khalil, Aurélie Névéol, Pierre Zweigenbaum, and Emmanuel Bacry. 2026. PARHAF: a human-authored corpus of clinical reports for fictitious patients in French. *arXiv preprint arXiv:2603.20494*.

AntLM Team. 2024. AntLM: Bridging causal and masked language models. In *CoNLL*.

OLMo Team, Dirk Groeneveld, et al. 2025. OLMo 2: Furious. *arXiv preprint arXiv:2501.00656*.

Rita Temmerman. 2000. *Towards New Ways of Terminology Description: The Sociocognitive Approach*, volume 3 of *Terminology and Lexicography Research and Practice*. John Benjamins, Amsterdam.

Rian Touchent, Laurent Romary, and Éric De La Clergerie. 2023. CamemBERT-bio: a tasty French language model better for your health. In *Actes de la 30ème Conférence sur le Traitement Automatique des Langues Naturelles (TALN)*, pages 323–334.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. Llama: Open and efficient foundation language models.

Özlem Uzuner, Brett R. South, Shuying Shen, and Scott L. DuVall. 2011. 2010 i2b2/VA challenge on concepts, assertions, and relations in clinical text. *Journal of the American Medical Informatics Association*, 18(5):552–556.

Erik M. van Mulligen, Zubair Afzal, Saber A. Akhondi, Dang Vo, and Jan A. Kors. 2016. Erasmus MC at CLEF eHealth 2016: Concept recognition and coding in French texts. In *Working Notes of CLEF 2016*.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Ł ukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.

Xiaozhi Wang, Tianyu Gao, Zhaocheng Zhu, Zhengyan Zhang, Zhiyuan Liu, Juanzi Li, and Jian Tang. 2021. KEPLER: A unified model for knowledge embedding and pre-trained language representation. *Transactions of the Association for Computational Linguistics*, 9:176–194.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023a. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations (ICLR 2023)*.

Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. 2023b. Self-instruct: Aligning language models with self-generated instructions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13484–13508, Toronto, Canada. Association for Computational Linguistics.

Yizhong Wang, Swaroop Mishra, Pegah Alipoormolabashi, Yeganeh Kordi, Amirreza Mirzaei, Atharva Naik, Arjun Ashok, Arut Selvan Dhanasekaran, Anjana Arunkumar, David Stap, Eshaan Pathak, Giannis Karamanolakis, Haizhi Lai, Ishan Purohit, Ishani Mondal, Jacob Anderson, Kirby Kuznia, Krima Doshi, Kuntal Kumar Pal, Maitreya Patel, Mehrad Moradshahi, Mihir Parmar, Mirali Purohit, Neeraj Varshney, Phani Rohitha Kaza, Pulkit Verma, Ravsehaj Singh Puri, Rushang Karia, Savan Doshi, Shailaja Keyur Sampat, Siddhartha Mishra, Sujan Reddy A, Sumanta Patro, Tanay Dixit, and Xudong Shen. 2022. Super-NaturalInstructions: Generalization via declarative instructions on 1600+ NLP tasks. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 5085–5109, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Griffin Thomas Adams, Jeremy Howard, and Iacopo Poli. 2025. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2526–2547, Vienna, Austria. Association for Computational Linguistics.

Maurice Weber, Daniel Y. Fu, Quentin Anthony, Yonatan Oren, Shane Adams, Anton Alexandrov, Xiaozhong Lyu, Huu Nguyen, Xiaozhe Yao, Virginia Adams, Ben Athiwaratkun, Rahul Chalamala, Kezhen Chen, Max Ryabinin, Tri Dao, Percy Liang, Christopher Ré, Irina Rish, and Ce Zhang. 2024. Redpajama: an open dataset for training large language models. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024) Datasets and Benchmarks Track*.

Lawrence L. Weed. 1968. Medical records that guide and teach. *New England Journal of Medicine*, 278(11):593–600.

Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V. Le. 2022a. Finetuned language models are zero-shot learners. In *The Tenth International Conference on Learning Representations (ICLR 2022)*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, brian ichter, Fei Xia, Ed H. Chi, Quoc V Le, and Denny Zhou. 2022b. Chain of thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*.

Orion Weller, Kathryn Ricci, Marc Marone, Antoine Chaffin, Dawn Lawrie, and Benjamin Van Durme. 2025. Seq vs seq: An open suite of paired encoders and decoders. *arXiv preprint arXiv:2507.11412*.

Alexander Wettig, Kyle Lo, Sewon Min, Hannaneh Hajishirzi, Danqi Chen, and Luca Soldaini. 2025. Organize the web: Constructing domains enhances pre-training data curation. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, Proceedings of Machine Learning Research. PMLR.

Chaoyi Wu, Weixiong Lin, Xiaoman Zhang, Ya Zhang, Weidi Xie, and Yanfeng Wang. 2024. PMC-LLaMA: toward building open-source language models for medicine. *Journal of the American Medical Informatics Association*, 31(9):1833–1843.

Eugen Wüster. 1979. *Einführung in die allgemeine Terminologielehre und terminologische Lexikographie*. Springer, Wien.

Yong Xie, Karan Aggarwal, and Aitzaz Ahmad. 2024. Efficient continual pre-training for building domain specific large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 10184–10201, Bangkok, Thailand. Association for Computational Linguistics.

Cheng Xu, Shuhao Guan, Derek Greene, and M. Tahar Kechadi. 2024. Benchmark data contamination of large language models: A survey. *CoRR*, abs/2406.04244.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Wang, Bowen Zheng, Bowen Yu, et al. 2025a. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Zhichao Yang, Avijit Mitra, Sunjae Kwon, and Hong Yu. 2024. ClinicalMamba: A generative clinical language model on longitudinal clinical notes. In *Proceedings of the 6th Clinical Natural Language Processing Workshop*, pages 54–63, Mexico City, Mexico. Association for Computational Linguistics.

Zitong Yang, Neil Band, Shuangping Li, Emmanuel Candès, and Tatsunori Hashimoto. 2025b. Synthetic continued pretraining. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Michihiro Yasunaga, Antoine Bosselut, Hongyu Ren, Xikun Zhang, Christopher D Manning, Percy Liang, and Jure Leskovec. 2022a. Deep bidirectional language-knowledge graph pretraining. In *Advances in Neural Information Processing Systems*, volume 35, pages 37309–37323.

Michihiro Yasunaga, Jure Leskovec, and Percy Liang. 2022b. LinkBERT: Pretraining language models with document links. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8003–8016, Dublin, Ireland. Association for Computational Linguistics.

Anthony Yazdani, Ihor Stepanov, and Douglas Teodoro. 2026. GLiNER-BioMed: A suite of efficient models for open biomedical named entity recognition. *Bioinformatics*.

Çağatay Yıldız, Nishaanth Kanna Ravichandran, Nitin Sharma, Matthias Bethge, and Beyza Ermis. 2024. Investigating continual pretraining in large language models: Insights and implications.

Zheng Yuan, Chuanqi Tan, and Songfang Huang. 2022a. Code synonyms do matter: Multiple synonyms matching network for automatic ICD coding. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 808–814. Association for Computational Linguistics.

Zheng Yuan, Zhengyun Zhao, Haixia Sun, Jiao Li, Fei Wang, and Sheng Yu. 2022b. Coder: Knowledge-infused cross-lingual medical term embedding for term normalization. *Journal of Biomedical Informatics*, 126:103983.

Zheng Yuan, Zhengyun Zhao, Haixia Sun, Jiao Li, Fei Wang, and Sheng Yu. 2022c. CODER: Knowledge-infused cross-lingual medical term embedding for term normalization. *Journal of Biomedical Informatics*, 126:103983.

Jamil Zaghir, Mina Bjelogrlic, Jean-Philippe Goldman, Soukaïna Aananou, Christophe Gaudet-Blavignac, and Christian Lovis. 2024. FRASIMED: A clinical French annotated resource produced through crosslingual BERT-based annotation projection. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 7450–7460, Torino, Italia. ELRA and ICCL.

Manzil Zaheer, Guru Guruganesh, Kumar Avinava Dubey, Joshua Ainslie, Chris Alberti, Santiago Ontañón, Philip Pham, Anirudh Ravula, Qifan Wang, Li Yang, and Amr Ahmed. 2020. Big bird: Transformers for longer sequences. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*.

Urchade Zaratiana, Gil Pasternak, Oliver Boyd, George Hurn-Maloney, and Ash Lewis. 2025. GLiNER2: Schema-driven multi-task learning for structured information extraction. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025), System Demonstrations*, pages 130–140. Association for Computational Linguistics.

Urchade Zaratiana, Nadi Tomeh, Pierre Holat, and Thierry Charnois. 2024. GLiNER: Generalist model for named entity recognition using bidirectional transformer. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5364–5376. Association for Computational Linguistics.

Biao Zhang and Rico Sennrich. 2019. Root mean square layer normalization. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, pages 12360–12371.

Zhengyan Zhang, Xu Han, Zhiyuan Liu, Xin Jiang, Maosong Sun, and Qun Liu. 2019. ERNIE: Enhanced language representation with informative entities. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 1441–1451. ACL.

Junhao Zheng, Xidi Cai, Shengjie Qiu, and Qianli Ma. 2025. Spurious forgetting in continual learning of language models. In *The Thirteenth International Conference on Learning Representations (ICLR 2025)*.

Wenxuan Zhou, Sheng Zhang, Yu Gu, Muhao Chen, and Hoifung Poon. 2024. UniversalNER: Targeted distillation from large language models for open named entity recognition. In *The Twelfth International Conference on Learning Representations (ICLR 2024)*. OpenReview.net.

Appendix